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1 Introduction

1.1 Motivation and Research Problem

Treatment effects need not be uniform across an outcome distribution. An additional year of education may affect wages differently near the lower and upper parts of the wage distribution, just as job training may have different consequences for workers with low and high earnings potential. Mean effects can conceal such differences because they summarize the response at only one location. Quantile methods provide a complementary view by examining lower, middle, and upper parts of the conditional outcome distribution (Koenker and Bassett 1978). Giving these differences a structural treatment-effect interpretation is difficult, however, when treatment is endogenous. Education may depend on unobserved ability, program participation on unobserved motivation, and prices on factors jointly determined with demand. Conventional quantile regression then describes conditional associations but does not generally recover the effect of externally changing treatment. Instrumental variable quantile regression (IVQR) addresses this problem by combining instruments with structural quantile restrictions that allow treatment to depend on unobserved outcome determinants (Chernozhukov and Hansen 2005).

Modern applications add a second difficulty. Researchers may observe many controls, transformations, and interactions that could matter for the outcome or the relationship between instruments and controls. Estimating all resulting nuisance coefficients without structure can be noisy or infeasible, even though these coefficients are not the primary objects of interest. Regularization makes high-dimensional nuisance estimation manageable by controlling model complexity, but shrinkage and selection also introduce estimation error. The double machine learning framework uses Neyman-orthogonal scores to reduce the first-order sensitivity of a target moment to sufficiently small nuisance-estimation errors (Chernozhukov et al. 2018). In the IVQR setting, Chen et al. (2021) combine this orthogonal construction with regularized nuisance estimation. DML-IVQR thus provides a way to handle many controls while keeping inference focused on a low-dimensional structural quantile effect; it does not imply that the nuisance quantities are known or estimated without error.

Instrument relevance remains a separate concern. DML changes how nuisance-estimation errors enter the structural moment, but it does not create information in the instruments. A valid instrument may still contain little information about the endogenous treatment. When excluded-instrument relevance is weaker, changing a candidate treatment-effect value may produce only a small change in the population moment, making competing values difficult to distinguish. This problem concerns the identifying content of the instruments rather than the dimension of the controls. Conventional inference centered on an approximately normal point estimator can become unreliable when identification is weak. The candidate-specific approach of Chernozhukov and Hansen (2008) instead tests the IVQR restrictions at each hypothesized value and obtains a confidence region by retaining the values that are not rejected. Chen et al. (2021) carry this weak-identification-robust inference logic into their high-dimensional DML-IVQR framework.

Weak-identification robustness is important, but it is not the end of the practical question. Asymptotic validity means that a procedure has the intended limiting rejection behavior under its maintained assumptions without requiring strong identification. Informativeness asks whether it can exclude false treatment-effect values: a confidence region may be valid yet very wide, and a test may retain the truth frequently while also retaining many alternatives. Finite-sample calibration asks whether the asymptotic reference approximation produces rejection probabilities close to their nominal level in the sample sizes researchers actually have. An asymptotically valid test can still be poorly calibrated in a moderate sample if that approximation is inaccurate. Point-estimation performance, inferential informativeness, and finite-sample calibration must therefore be assessed separately. Their behavior may also vary across quantiles because the structural effect, the quantile score, information near the quantile boundary, and nuisance-estimation difficulty need not be uniform over the outcome distribution. Studying only the median could consequently miss relevant patterns, although no particular ordering between central and tail quantiles should be presumed.

Simulation provides a controlled way to examine these finite-sample issues. The practical value of DML-IVQR depends not only on its asymptotic foundations but also on how point accuracy, accepted-region informativeness, power against false values, and coverage behave as excluded-instrument relevance is progressively reduced. This thesis studies those questions within the Monte Carlo framework of Chen et al. (2021), systematically weakening excluded-instrument relevance while holding the main structural features of the design fixed. The resulting research problem is not whether the literature supplies a DML-IVQR procedure with asymptotically weak-identification-robust inference; it already does. The issue is how informative and well calibrated that procedure is in finite samples as the instruments contain progressively less information about treatment, and whether this behavior differs across quantiles. Section 1.2 states the research question and the contribution of the controlled simulation exercise more precisely.

1.2 Research Question and Contribution

The research question is: How does weakening instrument strength affect the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how do these effects vary across quantiles? The experiment operationalizes this question by weakening excluded-instrument relevance in the Monte Carlo design of Chen et al. (2021). Its sole structural modification to the baseline DGP is the treatment equation, in which the parameter κ controls the loading on the excluded-instrument primitives. Across values of κ, the design preserves the marginal variance and distribution of the latent treatment index, the endogeneity channel, the structural outcome equation, the structural quantile treatment effect, the control distribution, and the observed instrument construction. This produces a controlled simulation comparison in which relevance is reduced without simultaneously changing these other DGP components. The parameter κ is therefore an experimental control for excluded-instrument relevance, not a formal or universal statistic for IVQR identification strength.

The main design uses $n \in \{500, 1000\}$, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, and $\kappa \in \{1, 0.50, 0.25, 0.10\}$, with 500 Monte Carlo replications in each design cell. Three estimators are evaluated on the same simulated samples. Oracle-GMM is an infeasible benchmark that is given the designated relevant control set. Full-GMM uses the complete high-dimensional control set without the DML regularization strategy. DML-IVQR-BC uses the preserved high-dimensional procedure with regularized nuisance estimation, a Neyman-orthogonal score, and the BC pivotal penalty. These comparisons show how the three implemented procedures behave under the same structural design; they do not make Oracle-GMM an upper bound or isolate the causal effect of regularization, because the estimators also differ in aspects of their nuisance and instrument-residualization procedures.

The evaluation separates three dimensions of finite-sample performance. Point estimation is summarized by Bias, MAE, and RMSE. Inferential informativeness is assessed through power at prespecified false alternatives around $\alpha_0(\tau)$ and a grid-based accepted-set measure within the prespecified primary computational domain. Because that measure concerns only a finite numerical domain, it is not interpreted as the unrestricted length of a theoretical confidence region; a separate domain-expansion diagnostic checks whether acceptance continues beyond the primary domain. Finite-sample calibration is evaluated through coverage at the exact true parameter and the corresponding empirical size. Coverage is computed by evaluating the test directly at $\alpha_0(\tau)$, rather than by asking whether the numerical confidence-region grid happens to contain the truth, so it is independent of that grid. Power is likewise evaluated directly at the prespecified false values. All three dimensions are needed: coverage alone does not show whether false candidates can be excluded, accepted-set behavior and power do not establish correct calibration, and point-estimation accuracy is distinct from test validity. Evaluating five quantiles allows the analysis to ask whether the response to weaker relevance differs across the conditional outcome distribution rather than averaging over τ or considering only the median.

Relative to Chen et al. (2021), the thesis contributes a controlled finite-sample stress test of their preserved estimator and inference architecture under systematically weaker excluded-instrument relevance. Chen, Huang, and Tien already provide the structural DML-IVQR framework, high-dimensional regularization, the orthogonal score, GMM estimation, candidate-specific weak-identification-robust inference, estimator comparisons, and evidence across several quantiles. The extension studies controlled relevance variation, repeated-sampling accepted-set informativeness, power against false alternatives, coverage and size across relevance settings, and heterogeneity in these responses across quantiles. Preserving the authors' executable estimator and inference architecture makes the exercise an extension of their baseline rather than a redesign of the method. It does not propose a new structural IVQR model, identification theorem, orthogonal score, DML estimator, weak-identification-robust test, or finite-sample correction. Its contribution is empirical Monte Carlo evidence on an existing methodology under a deliberately modified instrument-relevance design. Having defined the research question and contribution, Section 1.3 explains how the methodology, simulation design, results, and discussion are organized to address them.

1.3 Structure of the Thesis

Chapter 2 establishes the conceptual basis for the analysis by reviewing quantile treatment effects under endogeneity, the structural IVQR framework, and the distinction between identification and weak identification. It then connects weak-identification-robust inference with the separate problem of high-dimensional nuisance estimation, introduces the role of double machine learning, and locates the thesis question within the DML-IVQR literature. Building on this background, Chapter 3 formalizes the methodology used in the simulation. It presents the structural quantile model, profiled nuisance quantities, Neyman-orthogonal score, and regularized nuisance estimators before defining Oracle-GMM, Full-GMM, and DML-IVQR and the common GMM criterion and candidate-specific robust inference procedure.

The analysis then moves from methodology to experimental design and evidence. Chapter 4 specifies the baseline data-generating process and the instrument-relevance extension governed by κ. It describes the design dimensions and performance measures, the finite numerical construction used to summarize accepted sets, and the post-main diagnostics that examine sensitivity to the chosen computational domain. Chapter 5 reports the resulting Monte Carlo evidence. Its presentation proceeds from calibration of the instrument-relevance settings to point-estimation performance, accepted-set informativeness, power, and coverage and size. It then examines heterogeneity across quantiles and draws together the comparisons among the three estimators. This sequence keeps accuracy, informativeness, calibration, and cross-quantile comparison distinct within a common experimental framework.

Chapter 6 interprets this evidence through the distinctions developed earlier. It relates weaker relevance to identifying information, separates asymptotic validity from finite-sample calibration and informativeness, considers what DML can and cannot address, and states the limitations of the exercise. Chapter 7 concludes by summarizing the answer to the research question, bringing together the main implications, and briefly indicating directions for further research. The appendices support the main discussion with complete Monte Carlo tables, detailed power results, boundary and grid-domain diagnostics, coverage forensic checks, additional $W_N(a)$ profiles, and implementation and reproducibility material.

2 Econometric Background and Related Literature

2.1 Quantile Treatment Effects and Endogeneity

Mean regression summarizes how covariates are related to the conditional mean of an outcome. Quantile methods ask a different distributional question: whether those relationships vary across lower, central, and upper parts of the conditional outcome distribution. A treatment may, for example, have a different effect for observations near the lower part of an earnings or wealth distribution than around its median or upper tail. A mean effect alone does not describe such variation unless the treatment shifts the distribution uniformly. Quantiles are not inherently superior to means; they reveal features that a mean can conceal and may be the relevant objects when distributional heterogeneity matters. The regression-quantile framework introduced by Koenker and Bassett (1978) provides the statistical foundation for estimating conditional quantile relationships.

A causal distributional object can be defined using potential outcomes. Let $Y_i(d)$ denote the outcome that observation i would realize if treatment were fixed at state d. Conditional on controls $X_i = x$, let $q(\tau, d, x)$ be the τ-th quantile of the potential-outcome distribution $Y_i(d) \mid X_i = x$. For two treatment states d_1 and d_0, the conditional quantile treatment effect (QTE) is

$$\mathrm{QTE}(\tau, x) = q(\tau, d_1, x) - q(\tau, d_0, x). \tag{1}$$

The QTE compares the same quantile index of two potential-outcome distributions at common controls and is therefore a distributional treatment-effect object (Chernozhukov and Hansen 2005). In general, it is not an individual treatment effect: the observation occupying rank τ under d_1 need not be the same observation occupying that rank under d_0. A stronger individual-rank interpretation requires additional restrictions, such as rank invariance. Without such restrictions, the QTE remains a comparison between marginal potential-outcome quantiles.

This structural quantile differs from an observed conditional quantile. The object $q(\tau, d, x)$ describes the potential-outcome distribution when treatment is conceptually fixed at d. By contrast, $Q_{Y|D,X}(\tau \mid d, x)$ is the τ-th quantile among observations whose realized treatment is $D_i = d$ and whose controls equal x. These objects need not coincide. Treatment is endogenous when it is statistically related to unobserved determinants of the outcome that matter for the structural relationship. Program participation may depend on unobserved motivation, for example, or education may be related to unobserved ability. People observed at a particular treatment value can then have a different distribution of latent characteristics from people observed at other values. Conditioning on realized treatment consequently changes which unobserved ranks are represented. Conventional quantile regression generally does not recover the structural quantile effect of an endogenous treatment (Chernozhukov and Hansen 2005, 2013). It can still describe the observed conditional association $Q_{Y|D,X}(\tau \mid D, X)$, but a causal or structural interpretation requires further assumptions. Treatment-effect heterogeneity and endogeneity are therefore separate issues: quantile methods allow relationships to vary across the outcome distribution, whereas endogeneity concerns selection through unobserved outcome determinants. Allowing heterogeneity alone does not solve the selection problem.

Chernozhukov and Hansen (2005) develop an instrumental-variable model for quantile treatment effects under endogeneity. Their framework combines potential outcomes and structural rank variables with instruments and restrictions on how ranks behave across treatment states. Under rank invariance, an individual's latent rank is unchanged when treatment changes. Rank similarity relaxes that exact equality: ranks may change across potential treatment states, but, conditional on the information governing treatment selection, the rank variables must have the same distribution across those states. Rank similarity is therefore a substantive identifying restriction, not merely a technical convenience (Chernozhukov and Hansen

2013; Chernozhukov et al. 2017). These restrictions make it possible to use exogenous instrument variation while retaining heterogeneous structural effects. The framework can accommodate binary, discrete, or continuous endogenous treatments under its maintained assumptions. Other IV approaches target related but distinct distributional effects. For example, Abadie et al. (2002) study quantile treatment effects for a binary treatment within an IV framework tied to the complier population. That approach and the structural IVQR framework rely on different assumptions and answer different questions; neither is generally superior to the other (Chernozhukov and Hansen 2013).

The distinction between heterogeneity and endogeneity motivates instrumental variable quantile regression (IVQR). A suitable instrument supplies variation in treatment that is separated, under the maintained model, from the unobserved determinants of potential outcomes. IVQR uses that variation to learn about structural quantiles rather than treating the observed conditional quantile among self-selected treatment values as causal. This thesis studies a continuously valued endogenous treatment and quantile-specific structural effects, so conventional quantile regression is not the relevant inferential framework. An IVQR structure is needed to connect instrument-induced treatment variation to the structural quantile effect. Section 2.2 introduces that framework and explains its central conditional quantile restriction.

2.2 Instrumental Variable Quantile Regression

Instrumental variable quantile regression (IVQR) addresses the gap between observed and structural quantiles described in Section 2.1. The model begins from potential outcomes represented conceptually as $Y_i(d) = q(U_{di}, d, X_i)$, where U_{di} is the rank associated with treatment state d. It combines this structure with instrument exogeneity, a treatment-selection mechanism, and rank invariance or the more general rank-similarity condition (Chernozhukov and Hansen 2005; Chernozhukov et al. 2017). Instrument exogeneity requires the potential-outcome ranks to be independent of Z_i conditional on X_i. Rank similarity allows realized ranks to differ across treatment states while requiring their distributions to be the same after conditioning on the information governing selection. The selection mechanism may nevertheless allow D_i to depend on latent variables related to the outcome ranks. IVQR therefore does not assume that treatment itself is exogenous. Instead, a valid instrument provides externally generated variation that is separated from the structural ranks under the maintained assumptions. Validity and relevance have different roles: validity concerns the instrument's relationship with structural unobservables, whereas relevance concerns whether the instrument changes the distribution of treatment. An instrument can satisfy the validity condition without generating useful treatment variation, while relevance alone does not establish exogeneity.

The main testable implication derived by Chernozhukov and Hansen (2005) is the conditional quantile restriction

$$\Pr[Y_i \leq q(\tau, D_i, X_i) \mid X_i, Z_i] = \tau. \tag{2}$$

At the true structural quantile function, each observation is evaluated at its realized, possibly endogenous treatment. Within cells defined by the controls and instruments, the probability that the observed outcome lies at or below the resulting structural τ-quantile is τ. The presence of Z_i in the conditioning set is essential: it carries the exogenous variation used to distinguish the structural relationship from selection into D_i. The same implication can be written as the conditional moment

$$\mathrm{E}\big[\tau - \mathbb{1}\{Y_i \leq q(\tau, D_i, X_i)\} \bigm| X_i, Z_i\big] = 0. \tag{3}$$

Thus, at the correct structural quantile, the indicator-based quantile residual has conditional mean zero given controls and instruments. This result is the bridge from the economic model of potential outcomes and selection to a statistical condition usable for estimation. It follows from the maintained IVQR rank and exogeneity assumptions; it is not a generic property of every quantile model that happens to include an instrument (Chernozhukov et al. 2017).

A widely used specialization makes the structural quantile function linear in parameters:

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau). \tag{4}$$

Here $\alpha_0(\tau)$ is the low-dimensional structural treatment effect of interest, $\beta_0(\tau)$ contains the control coefficients, and $c_0(\tau)$ is an intercept. All may vary across quantiles. This form connects the IVQR literature to the model used later in the thesis. It also makes the difference from ordinary quantile regression explicit. Ordinary quantile regression models $Q_{Y|D,X}(\tau \mid D, X)$ by conditioning directly on treatment and controls. IVQR instead seeks the structural function $q(\tau, D, X)$ by requiring its quantile residual to satisfy restrictions conditional on X and Z. The procedure does not simply add Z as another explanatory variable. The instrument enters through the model's identifying restriction, while the structural treatment coefficient remains attached to the endogenous D.

Estimation must consequently choose $\alpha_0(\tau)$ so that the instrument-based restrictions are satisfied, rather than obtain it from an ordinary quantile regression of Y on D and X. The inverse quantile regression logic of Chernozhukov and Hansen (2008) makes this candidate-based problem tractable. For a candidate a, subtract $D_i'a$ from Y_i and estimate the remaining quantile relationship using X_i and Z_i. If a equals the true $\alpha_0(\tau)$, the population coefficient on Z_i in this auxiliary quantile regression is zero. One can therefore examine how compatible each candidate is with a zero instrument coefficient and search over a. The search remains generally nonconvex in the endogenous coefficient, even though each auxiliary quantile regression is a familiar problem (Chernozhukov et al. 2017). More generally, the conditional restriction can be converted into unconditional moments by multiplying the quantile residual by suitable functions of X_i and Z_i and averaging. GMM-type criteria then summarize the distance of those sample moments from zero: candidates producing smaller moments are more compatible with the model. This is the conceptual link to the candidate-specific criterion defined formally in Section 3.6, without fixing a particular weighting or computational algorithm here.

Chen et al. (2021) retain this structural IVQR restriction and adapt estimation to settings with a low-dimensional treatment effect and high-dimensional controls. Their contribution is discussed in Sections 2.5–2.6 and developed formally in Chapter 3. For the present discussion, the important point is that IVQR gives structural meaning to the instrument-based quantile condition only under its maintained potential-outcome, exogeneity, selection, and rank assumptions. Instrument validity makes the restriction meaningful, while relevance determines whether instrument variation can distinguish competing treatment effects. Even when the restriction is valid, it need not uniquely or sharply determine $\alpha_0(\tau)$. Moment validity and point identification are separate properties. Section 2.3 therefore turns from the construction of the IVQR restriction to the conditions under which it identifies the structural quantile effect.

2.3 Identification and Weak Identification in IVQR

The validity of the IVQR restriction does not by itself establish identification. Moment validity means that the true structural parameter satisfies the population restriction implied by the model. Identification asks whether the restriction distinguishes that parameter from alternatives. If only $\alpha_0(\tau)$ satisfies the relevant population restrictions within the maintained parameter space, it is point identified. If several values satisfy them, the restrictions identify a set rather than a unique value, producing partial identification. At the limit, the restrictions may fail to distinguish among some or all candidate values. Chernozhukov and Hansen (2005) make this distinction by first deriving the model's conditional quantile restriction and then imposing additional conditions for uniqueness. The set of functions or parameter values satisfying the restriction is therefore conceptually prior to any claim of point identification. A correct model can supply a valid moment while leaving its parameter only partially identified or unidentified.

Instrument relevance is central to whether the restriction separates competing treatment effects. Validity permits Z_i to enter the structural moment because it restricts the relationship between the instrument and the potential-outcome ranks. Relevance asks whether changes in Z_i induce enough treatment variation to reveal how the structural quantile changes with D_i. If treatment responds little to the instrument, changing a candidate value a may change the population moment only slightly. The true value can remain the unique zero while nearby candidates produce moments that are almost indistinguishable from zero. This is weak identification in a local, informational sense: uniqueness may hold, but the restriction separates the truth from alternatives only weakly. Under stronger identification, the population moment moves more sharply away from zero as the candidate moves away from $\alpha_0(\tau)$. Relevant instrument-induced treatment variation supports this separation, but descriptive relevance alone is not identification. The instrument must also be valid, and the full nonlinear population mapping must rule out other solutions.

The identification analysis in Chernozhukov and Hansen (2005) formalizes this idea through conditions on the mapping from candidate structural quantiles to the IVQR restrictions. For discrete treatments, their sufficient conditions have a full-rank character: instrument variation must affect the joint distribution of outcome and treatment richly enough for the relevant population mapping to have a unique solution. For continuously valued treatments, the corresponding analysis uses completeness-type conditions rather than a simple linear first-stage rank condition. The later review by Chernozhukov et al. (2017) emphasizes that IVQR identification is quantile-specific. The structural coefficient $\alpha_0(\tau)$ can vary with τ, and the sensitivity of the moment depends on both instrument-induced treatment variation and the outcome distribution near the quantile boundary. The relevant relationship is consequently density weighted and may differ across quantiles. An instrument can be informative about one structural quantile and less informative about another without any universal ordering across the center and tails. This is why evidence from a mean first stage does not completely characterize IVQR identification at a particular τ.

Limited identifying information affects both estimation and conventional inference. When population restrictions distinguish candidates only weakly, sample criteria constructed from those restrictions may also be less discriminating. Sampling variation can then make several candidate values appear similarly compatible with the data. A point minimizer may be dispersed or sensitive to small sample changes, and sets of non-rejected values may be broad; tests may also have low power against some false alternatives. These are possible consequences rather than deterministic outcomes in every sample. The terminology also requires care. Weak identification means that the restrictions separate parameter values only weakly, even if a unique population solution exists. Partial identification means that the maintained restrictions determine an identified set with more than one value. Failed or no identification means that the restrictions contain essentially no information for distinguishing at least some directions or candidates. Chernozhukov and Hansen (2008) design inference to accommodate weak, partial, and failed identification, but these concepts are not interchangeable. Conventional inference based on a strongly identified, approximately normal point estimator can be unreliable when the population criterion supplies limited curvature or separation. The estimator may then have nonstandard behavior, substantial dispersion, and a poor normal approximation.

The later Monte Carlo design operationalizes only one component of this problem. Building on the IVQR setting of Chen et al. (2021), the thesis varies excluded-instrument relevance while holding the remaining features of the data-generating process fixed. The parameter κ controls that relevance by changing the amount of instrument-induced treatment variation; it is not itself a complete measure of IVQR identification. Likewise, the first-stage F-statistic and partial R^2 reported later are useful descriptive summaries of relevance, not quantile-specific IVQR identification statistics. The design therefore does not assign any value of κ a formal label of weak, partial, or failed identification. Its purpose is to study how finite-sample estimation and inference respond as excluded-instrument relevance, and hence the potential identifying information in the moments, is reduced. Because standard point-estimator approximations may become unreliable when identification is limited, inference should not depend on strong identification alone. Section 2.4 explains the Chernozhukov–Hansen approach that addresses this problem by testing candidate values directly.

2.4 Weak-Identification-Robust Inference

Conventional inference for a point estimator is most straightforward when the structural parameter is uniquely and strongly identified. The direct approach discussed by Chernozhukov and Hansen (2008) uses local full-rank conditions and global one-to-one behavior of the population moment mapping to obtain an approximately normal IVQR estimator. These conditions ensure both local sensitivity and a unique population solution. When they fail, an estimate-centered Wald interval can inherit the irregular behavior described in Section 2.3. Chernozhukov and Hansen therefore develop a robust, or dual, approach that reverses the inferential question. Instead of asking whether a candidate a is far from an approximately normal point estimate, it asks whether the IVQR restrictions are compatible with a itself. The candidate is treated as fixed under its null hypothesis, nuisance quantities are profiled or estimated conditional on it, and the resulting instrument-based quantile moment is tested directly. This approach avoids requiring strong identification of $\alpha_0(\tau)$ for asymptotic validity.

Let $\widehat{g}_N(a)$ denote the sample IVQR moment evaluated at candidate a, and let $\widehat{\Sigma}(a)$ be an estimated covariance-type matrix that records the moment's sampling variation. A generic candidate-specific statistic is

$$W_N(a) = N\widehat{g}_N(a)'\widehat{\Sigma}(a)^{-1}\widehat{g}_N(a). \tag{5}$$

The quadratic form standardizes the distance of the candidate's sample moments from zero. A small value means that the moments are relatively compatible with the restriction at a, whereas a large value provides stronger evidence against that candidate. Under the null hypothesis $a = \alpha_0(\tau)$ and the maintained regularity conditions, Chernozhukov and Hansen (2008) establish an asymptotic chi-square reference distribution for the corresponding candidate-specific statistic. Crucially, this null result does not require the additional full-rank and global uniqueness conditions used for direct inference. The logic is analogous to Anderson–Rubin-type inference: the test evaluates restrictions imposed under the hypothesized structural value rather than relying on a regular limiting distribution for the structural point estimator. It does not first require the researcher to determine how precisely a unique minimizer of the sample criterion can be located.

Repeating this test over the maintained parameter space and retaining every candidate that is not rejected yields the confidence region. For significance level p, let c_{1-p} denote the appropriate asymptotic chi-square critical value. The inverted region is represented generically as

$$C_{1-p}(\tau) = \left\{a : W_N(a) \leq c_{1-p}\right\}. \tag{6}$$

This construction need not produce a symmetric interval around the point estimate. Because it retains candidates according to their own tests, the region may be asymmetric, disconnected, or wide. If the restrictions contain almost no information about the structural parameter, the region may approach or equal the whole maintained parameter space. These shapes are informative about what the model and data can exclude. The dual procedure remains asymptotically applicable when identification is weak, partial, or fails because its null distribution is obtained candidate by candidate without assuming that the population moment selects one value sharply (Chernozhukov et al. 2017). This is the sense in which the procedure is weak-identification robust; it does not make the underlying identification problem disappear.

Robustness, informativeness, and finite-sample calibration are separate properties. Weak-identification robustness concerns asymptotic null validity without a strong-identification assumption. Informativeness concerns the ability to exclude false candidates. When identifying information is limited, many candidates may satisfy the sample restrictions closely enough to remain in the inverted region, and power against false alternatives may be low. A wide or disconnected region can therefore be an honest indication that the data reveal little about the parameter rather than a failure of the robust logic. Calibration asks a third question: whether finite-sample rejection probabilities are close to their nominal level. The Chernozhukov–Hansen procedure used later in this thesis employs an asymptotic chi-square approximation, so robustness does not guarantee exact finite-sample coverage or size control. Its finite-sample calibration depends on how well that approximation works at the available sample size and remains an empirical question. Thus, a procedure can be asymptotically robust yet uninformative, imperfectly calibrated in finite samples, or both.

A different route is provided by Chernozhukov et al. (2009), who exploit the conditional pivotal property of quantile estimating equations to construct exact finite-sample inference under their maintained conditions. Their reference distributions are obtained through a separate conditional simulation procedure and are not the asymptotic chi-square reference distribution described above. That method is not the procedure implemented in this thesis. Chen et al. (2021) instead retain the candidate-specific, weak-identification-robust IVQR inference logic while adapting nuisance estimation to a high-dimensional control setting through an orthogonalized construction. This separates two problems: conducting inference on the structural coefficient when instrument information may be limited, and estimating many nuisance parameters accurately enough to form the structural moment. The present section addresses the first problem. Section 2.5 turns to the second by reviewing high-dimensional nuisance estimation and double machine learning before Section 2.6 brings the two strands together.

2.5 High-Dimensional Nuisance Estimation and Double Machine Learning

Many econometric problems combine a low-dimensional parameter of interest with nuisance parameters that are needed for estimation but are not themselves the main object of study. In the IVQR setting, the structural treatment effect at a given quantile is the target, while the coefficients on controls, conditional density features, and instrument-residualization coefficients are nuisance quantities. When the number of available controls is large relative to the sample size, estimating all of these quantities without structure can be unstable or impossible. Regularization makes the problem manageable by shrinking or selecting coefficients. Its usual justification is approximate sparsity: although many variables may be available, a comparatively small subset provides most of the predictive content, while the remaining effects are zero or small. Approximate sparsity does not require every omitted coefficient to be exactly zero. It instead permits a low-complexity approximation whose error is sufficiently limited. Regularization therefore trades variance reduction and computational feasibility against shrinkage and selection bias. If a structural estimator simply substitutes regularized nuisance estimates into an ordinary moment condition, that bias can enter the target estimate at first order and invalidate standard inference, even when the nuisance estimates predict reasonably well (Chernozhukov et al. 2018).

Double machine learning addresses this problem by constructing a score whose expectation is locally insensitive to small errors in the nuisance estimates. Let W_i collect the observed data, let θ_0 denote a low-dimensional target, and let η_0 collect the nuisance functions or parameters. A generic identifying condition is

$$\mathrm{E}[\psi(W_i;\theta_0,\eta_0)] = 0. \tag{7}$$

The score is Neyman orthogonal when the derivative of its population expectation with respect to perturbations of the nuisance value vanishes at the truth:

$$\partial_\eta \, \mathrm{E}[\psi(W_i;\theta_0,\eta)]\big|_{\eta=\eta_0} = 0. \tag{8}$$

This condition means that a small nuisance-estimation error has no first-order effect on the population score at θ_0. It does not eliminate nuisance error, guarantee that a learner is accurate, or make the score globally insensitive to poor estimates. Remaining higher-order terms must still be controlled through suitable convergence rates, moment bounds, and other regularity conditions. Orthogonality is therefore a bias-reduction device that makes valid inference possible under weaker nuisance-estimation rates than a non-orthogonal plug-in construction would generally allow (Chernozhukov et al. 2018).

The term double or debiased machine learning refers to this combination of orthogonal scores with flexible, regularized nuisance estimation and an appropriate data-use scheme. The "debiased" description concerns the removal of the leading influence of nuisance-estimation error; it does not imply that all finite-sample bias disappears. The machine-learning component may include Lasso or other prediction methods, but the theory does not license an unrestricted choice of learner. The nuisance procedure must satisfy the rates and regularity conditions required for the selected score and model. In the general DML framework, sample splitting estimates the nuisance quantities on one part of the data and evaluates the score on another. This separation limits overfitting bias and avoids restrictive empirical-process conditions that can arise when the same observations perform both tasks. Cross-fitting rotates the training and evaluation roles across several folds and then combines the fold-specific scores, allowing each observation to contribute to score evaluation while using most of the sample for nuisance training. Orthogonality and cross-fitting should not be conflated: orthogonality is a property of the score, whereas cross-fitting is a strategy for organizing estimation and evaluation (Chernozhukov et al. 2018).

These ideas are especially relevant to IVQR with many controls because nuisance estimation must be repeated for candidate treatment effects. For each candidate, the procedure needs a high-dimensional quantile regression for the control component and an estimate of the density-weighted relationship used to residualize the instruments. Estimation errors in either step can contaminate the IVQR moment if it reacts strongly to the nuisance values. An orthogonal score is designed to reduce that sensitivity, while regularization uses approximate sparsity to make the two nuisance steps feasible. This construction addresses a high-dimensional estimation problem, not the separate question of whether the excluded instruments identify the structural effect. It neither validates an invalid instrument nor strengthens an instrument that contains little information about treatment. Consequently, DML does not create identification. Weak-identification-robust testing and orthogonal high-dimensional nuisance estimation solve distinct problems and can be combined: the former avoids relying on strong identification for the candidate-specific reference distribution, while the latter limits the first-order impact of estimating many nuisance quantities.

Chen et al. (2021) adapt this logic to IVQR by forming a score that is locally insensitive to errors in the candidate-specific quantile-regression and instrument-residualization nuisances and by estimating those nuisances with regularized methods. Their analysis connects high-dimensional control adjustment to the candidate-testing logic reviewed in Section 2.4 and provides the methodological basis for the DML-IVQR estimator studied in this thesis. Cross-fitting is part of the general DML framework, and Chen, Huang, and Tien also study a cross-fitted construction. The specific DML-IVQR implementation preserved for the Monte Carlo comparison in this thesis instead estimates the nuisance quantities and evaluates the orthogonal score on the same sample; it does not use sample splitting or cross-fitting. This implementation choice should be kept separate from the definition of orthogonality and is not presented here as a correction to, or improvement on, the general framework. The next section brings these high-dimensional nuisance methods together with IVQR identification-robust inference and states the research gap addressed by the simulation study.

2.6 DML-IVQR and the Research Gap

The literature reviewed in this chapter supplies two foundations for the analysis. The structural IVQR framework of Chernozhukov and Hansen (2005) defines quantile treatment effects when treatment is endogenous and establishes the restrictions through which instruments can identify them. Chernozhukov and Hansen (2008) then develop candidate-specific inference that does not require strong identification, while Chernozhukov et al. (2017) place identification, estimation, and weak-identification-robust inference within a common account of the model. This first strand therefore addresses structural quantile effects, the identifying content of the IVQR restrictions, and inference that can remain asymptotically valid when identification is weak, partial, or fails. It also makes clear that validity and informativeness are different: a robust procedure can control rejection of the truth asymptotically without guaranteeing that false values will be excluded sharply.

A second literature strand addresses the separate difficulty created by high-dimensional nuisance estimation. Chernozhukov et al. (2018) show how Neyman-orthogonal scores, sufficiently accurate nuisance learners, and generally sample splitting or cross-fitting can support inference on a low-dimensional target. Orthogonality reduces the first-order sensitivity of the target moment to nuisance-estimation error, while cross-fitting limits dependence between nuisance training and score evaluation. These tools do not alter the information supplied by an instrument. Weak-identification robustness concerns inference when structural moments distinguish candidate values only weakly; DML nuisance robustness concerns contamination of those moments by estimating many auxiliary quantities. Both problems may occur in the same application, but solving one does not solve the other.

Chen et al. (2021) provide the principal bridge between these strands. They retain the structural IVQR model, estimate high-dimensional quantile-regression and instrument-residualization nuisances with regularization, construct a Neyman-orthogonal score, evaluate the structural parameter with a GMM criterion, and invert candidate-specific tests to obtain weak-identification-robust confidence regions. Their practical DML-IVQR algorithm does not use cross-fitting; under their sparsity-based argument, the remaining nuisance terms are asymptotically negligible. Their high-dimensional Monte Carlo design contains 100 controls, including ten controls designated as relevant, and compares Oracle-GMM, Full-GMM, and DML-IVQR at five quantiles for sample sizes 500 and 1000. They report bias, MAE, and RMSE, compare penalty choices, and display confidence-region profiles. The evidence shows that DML-IVQR can improve substantially on the unregularized full-control estimator and often approach the oracle benchmark in their baseline design. The confidence-region analysis also demonstrates that weak-identification-robust inference is already an integral part of their contribution, rather than a feature introduced by this thesis.

The remaining question is narrower. The principal simulation of Chen et al. (2021) uses the fixed treatment equation $D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i)$; it does not systematically vary excluded-instrument relevance while holding the other elements of the DGP fixed. Their simulations therefore do not trace a controlled response surface showing how progressively weaker relevance changes point-estimation error, the informativeness of inverted confidence regions, or rejection probabilities at false candidate values. Although they present robust confidence-region profiles, their Monte Carlo analysis does not provide a systematic repeated-sampling study of coverage or size across controlled instrument-strength designs, nor does it map power jointly over false alternatives and such designs. They already compare performance across quantiles, but that is distinct from asking whether the effect of systematically weakening relevance differs across quantiles. The literature reviewed here consequently leaves limited systematic evidence on how nuisance-robust and identification-robust IVQR procedures behave jointly as the excluded-instrument signal is weakened within an otherwise fixed high-dimensional design. Later work by Jin and Sun (2025) expands high-dimensional IVQR methodology by developing a Neyman-orthogonal moment for an unconditional quantile treatment effect in a fully nonparametric model and recommending cross-fitting and automatic DML. That contribution broadens the structural and nuisance setting, but it does not study the controlled weak-relevance Monte Carlo question considered here.

The research question of this thesis is: How does weakening instrument strength affect the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how do these effects vary across quantiles? The Monte Carlo design implements this question by weakening excluded-instrument relevance through one experimental parameter while preserving the structural DGP and estimator and inference architecture of Chen et al. (2021) as far as possible. Across relevance settings, it holds fixed the marginal treatment-index variance, the endogeneity channel, the outcome model, the structural quantile treatment effect, the controls, and the instruments. Changing one DGP component in this way provides a controlled simulation comparison: differences across settings can be interpreted as responses to the designed reduction in excluded-instrument relevance without simultaneous changes in the marginal treatment distribution, endogeneity relationship, or structural effect distribution. The parameter governing this change is not treated as a formal measure of IVQR identification.

The contribution is finite-sample evidence about an existing procedure, not a new estimator, identification theorem, score, test, or finite-sample correction. The design compares

Oracle-GMM, Full-GMM, and DML-IVQR for sample sizes 500 and 1000 at five quantiles. It examines three related dimensions: point estimation through bias, MAE, and RMSE; inferential informativeness through accepted-set behavior and power against specified false alternatives; and finite-sample calibration through coverage and size. Studying these outcomes together matters because asymptotic robustness does not by itself reveal whether a confidence region will be informative or well calibrated in a moderate sample when instruments contain little treatment information. The literature provides the structural IVQR model, the orthogonal high-dimensional nuisance construction, and the weak-identification-robust testing framework. Taking those components as given, Chapter 3 formalizes the exact score, nuisance estimators, benchmark estimators, GMM criterion, and inference procedure used in the simulation.

3 DML-IVQR Methodology

3.1 Structural Quantile Model

Following the instrumental-variable quantile regression (IVQR) framework of Chernozhukov and Hansen (2005) and Chernozhukov and Hansen (2008), let Y_i denote a scalar outcome, D_i an endogenous treatment or vector of endogenous variables, X_i a vector of observed controls, and Z_i a vector of instruments. For each potential treatment state d, the potential outcome satisfies

$$Y_{di} = q(U_{di}, d, X_i), \qquad U_{di} \mid X_i \sim \text{Uniform}(0, 1), \tag{9}$$

where U_{di} is the treatment-state-specific structural rank and $q(u, d, x)$ is strictly increasing in u. Thus, $q(\tau, d, x)$ is the conditional τ-quantile of the potential outcome Y_d given $X = x$, and hence the structural quantile obtained by fixing treatment at d. The IVQR assumptions combine conditional instrument exogeneity for the potential-outcome ranks, a treatment-selection mechanism, and rank similarity across treatment states. Define the realized structural rank as $U_i := U_{D_i i}$. Rank similarity does not impose the stronger special case of rank invariance, under which $U_{di} = U_{d'i}$ for all treatment states d and d'. A central implication of these assumptions is the observed-data representation

$$Y_i = q(U_i, D_i, X_i), \qquad U_i \mid X_i, Z_i \sim \text{Uniform}(0, 1). \tag{10}$$

Accordingly, U_{di} indexes the rank associated with potential treatment state d, whereas U_i is the rank realized under the observed treatment D_i; the model does not require an individual to retain an identical rank under every treatment state.

Endogeneity permits D_i to be statistically dependent on U_i. Consequently, the observed conditional quantile $Q_{Y|D,X}(\tau \mid d, x)$ generally differs from the structural quantile $q(\tau, d, x)$: the former reflects selection into the realized treatment, whereas the latter holds treatment fixed. Conditioning directly on $D_i = d$ can change the distribution of latent ranks represented among observations and thus need not recover the corresponding potential-outcome quantile. For two treatment states d_1 and d_0, the structural quantile treatment effect at controls x is $q(\tau, d_1, x) - q(\tau, d_0, x)$. The contrast compares the τ-quantiles of the two conditional potential-outcome distributions at common controls. The distinction between observed and structural quantiles is fundamental to IVQR.

For a fixed $\tau \in (0, 1)$, the DML-IVQR procedure of Chen et al. (2021) adopts the linear-in-parameters structural quantile specification

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau). \tag{11}$$

Here $c_0(\tau)$ is the quantile-specific intercept, $\alpha_0(\tau)$ is the low-dimensional structural parameter of interest, and $\beta_0(\tau)$ contains the quantile-specific slopes on the nonconstant controls. In the high-dimensional setting, $\beta_0(\tau)$ may have many components and is treated as a nuisance parameter. The dimension of $\alpha_0(\tau)$ equals the number of endogenous variables but is assumed small relative to the potentially large control dimension. If D_i is scalar, $\alpha_0(\tau)$ is the quantile-specific structural treatment effect per unit change in D_i under this linear specification.

Strict monotonicity in the rank gives $\{Y_i \le q(\tau, D_i, X_i)\} = \{U_i \le \tau\}$. Together with the conditional uniformity in (10), this yields the key restriction at the true structural parameters,

$$\Pr\big[Y_i \le c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau) \mid X_i, Z_i\big] = \tau. \tag{12}$$

Equivalently, the model implies the conditional moment

$$\mathrm{E}\big[\tau - \mathbb{1}\{Y_i \le c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau)\} \,\big|\, X_i, Z_i\big] = 0. \tag{13}$$

Thus, despite the endogeneity of D_i, the rank and instrument-exogeneity structure supplies a conditional quantile restriction for the structural coefficients: at their true values, the probability of a nonpositive structural residual is τ within cells defined by (X_i, Z_i). This restriction defines the population condition on which the later procedure is based. Directly estimating the low-dimensional target $\alpha_0(\tau)$ jointly with the high-dimensional nuisance $\beta_0(\tau)$ is difficult; the next subsection therefore profiles the nuisance coefficient for each candidate treatment parameter.

3.2 Profiled Nuisance Parameters

Fix $\tau \in (0,1)$ and let a denote a candidate value for the low-dimensional structural coefficient $\alpha_0(\tau)$. Define the augmented control vector $\widetilde{X}_i = (1, X_i')'$, whose first element represents the intercept and whose remaining elements are the nonconstant controls. Profiling temporarily treats a as fixed. After subtracting $D_i'a$ from Y_i, the intercept and control slopes are chosen together to fit the τ-quantile of the remaining outcome according to the quantile check loss

$$\rho_\tau(u) = u\big[\tau - \mathbb{1}\{u < 0\}\big]. \tag{14}$$

The loss gives asymmetric linear weights to positive and negative residuals, with the asymmetry determined by τ. Following Chen et al. (2021), the candidate-specific profiled intercept and slopes are

$$\begin{pmatrix} c(a) \\ \beta(a) \end{pmatrix} = \arg\min_{c,b} \mathrm{E}\big[\rho_\tau\big(Y_i - D_i'a - c - X_i'b\big)\big]\,. \tag{15}$$

Thus, $c(a)$ and $\beta(a)$ are the population intercept and control slopes for the adjusted outcome $Y_i - D_i'a$. Because this outcome changes with a, both quantities can change as well. Under the usual quantile-regression regularity conditions, they have the joint population moment characterization

$$\mathrm{E}\big[\big\{\tau - \mathbb{1}\big[Y_i \le D_i'a + c(a) + X_i'\beta(a)\big]\big\}\,\widetilde{X}_i\big] = 0. \tag{16}$$

The constant element of $\widetilde{X}_i$ gives the intercept condition, while the remaining elements give the slope conditions for the nonconstant controls. Together, these moments define the intercept and control coefficients associated with candidate a. At $a = \alpha_0(\tau)$, the maintained model and uniqueness conditions imply $c(\alpha_0(\tau)) = c_0(\tau)$ and $\beta(\alpha_0(\tau)) = \beta_0(\tau)$. For $a \neq \alpha_0(\tau)$, however, $c(a)$ and $\beta(a)$ are candidate-specific profile nuisances rather than the true structural intercept and control coefficients.

The associated profile residual is

$$\epsilon_i(a) = Y_i - D_i'a - c(a) - X_i'\beta(a). \tag{17}$$

Let $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ denote its conditional density evaluated at zero. Residuals below zero lie on one side of the fitted candidate quantile, while residuals above zero lie on the other. A small change in $c(a)$ or $\beta(a)$ can move observations with residuals close to zero from one side to the other. The density near zero therefore determines how sensitive the quantile moment is to small changes in the fitted intercept and control slopes.

To characterize this sensitivity, suppose $Z_i \in \mathbb{R}^{d_z}$ and $X_i \in \mathbb{R}^p$, so $\widetilde{X}_i \in \mathbb{R}^{p+1}$, and define

$$\begin{aligned} M(a) &= \mathrm{E}\big[Z_i \widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\big], \quad M(a) \in \mathbb{R}^{d_z \times (p+1)}, \\ J(a) &= \mathrm{E}\big[\widetilde{X}_i \widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\big], \quad J(a) \in \mathbb{R}^{(p+1)\times(p+1)}. \end{aligned} \tag{18}$$

The matrices summarize how the quantile moment changes when the fitted intercept or control slopes move slightly. The matrix $J(a)$ describes this relationship using the augmented controls themselves, whereas $M(a)$ describes the corresponding relationship between the instruments and the augmented controls. Both are weighted by the residual density at zero because observations near the fitted quantile are most likely to change sides after such a movement. Assuming that $J(a)$ is nonsingular at the population level, set

$$\delta(a) = M(a)J(a)^{-1}, \qquad \delta(a) \in \mathbb{R}^{d_z \times (p+1)}, \tag{19}$$

Partition this matrix as $\delta(a) = [d_0(a), \varDelta(a)]$, where $d_0(a) \in \mathbb{R}^{d_z}$ contains the residualization intercepts and $\varDelta(a) \in \mathbb{R}^{d_z \times p}$ contains the slopes on the nonconstant controls. The corresponding population residualized instrument is

$$\varPsi_i(a) = Z_i - \delta(a)\widetilde{X}_i = Z_i - d_0(a) - \varDelta(a)X_i. \tag{20}$$

The matrix $\delta(a)$ determines how much of each instrument is associated with the intercept and controls under the density weighting. Hence, $\varPsi_i(a)$ is the part of the instrument left after removing that weighted component. By construction, $\mathrm{E}[(Z_i - \delta(a)\widetilde{X}_i)\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)] = 0$. Because the weighting gives more importance to observations near the candidate quantile boundary, $\varPsi_i(a)$ is not an ordinary unweighted regression residual, and $\delta(a)$ is not a conventional first-stage coefficient.

Collect the candidate-specific nuisance quantities as

$$\eta(a) = \begin{pmatrix} c(a) \\ \beta(a) \\ \mathrm{vec}\{\delta(a)\} \end{pmatrix}, \qquad \eta_0 = \eta(\alpha_0(\tau)) = \begin{pmatrix} c_0(\tau) \\ \beta_0(\tau) \\ \mathrm{vec}\{\delta(\alpha_0(\tau))\} \end{pmatrix}. \tag{21}$$

Here $\mathrm{vec}\{\delta(a)\}$ stacks the entries of the matrix $\delta(a)$ into one vector. Vectorization changes only how the matrix nuisance is stored; it does not alter the residualization itself. These objects are nuisance quantities because the inferential target remains $\alpha_0(\tau)$. Candidate-specific $c(a)$

and $\beta(a)$ handle the structural intercept and high-dimensional control component, while $\delta(a)$ removes the density-weighted component of the instruments associated with $\widetilde{X}_i$. Section 3.3 combines these objects with the IVQR quantile residual to construct a Neyman-orthogonal moment for $\alpha_0(\tau)$.

3.3 Neyman-Orthogonal Score

Following Chen et al. (2021), combine the nuisance quantities from Section 3.2 in the score

$$\begin{aligned} g_\tau(W_i; a, \eta) &= \left[\tau - 1\!\!1\{Y_i \le D_i' a + c + X_i'\beta\}\right]\left(Z_i - \delta \widetilde{X}_i\right), \\ g_\tau(W_i; a, \eta(a)) &= \left[\tau - 1\!\!1\{Y_i \le D_i' a + c(a) + X_i'\beta(a)\}\right]\Psi_i(a), \end{aligned} \tag{22}$$

where $W_i = (Y_i, D_i, X_i, Z_i)$ collects the observed variables and η contains c, β, and δ. The first factor is the quantile residual, or quantile score. It compares the observed outcome with the fitted structural τ-quantile. It equals τ when the outcome is above the fitted quantile and $\tau - 1$ when the outcome is at or below it; it is therefore not an ordinary regression residual. At the true model, its conditional expectation is zero. The second factor is the residualized instrument from Section 3.2. It removes the part of Z_i associated with the intercept and nonconstant controls under the density-weighted projection and leaves the instrument variation used in the moment. Multiplying these two parts gives the IVQR moment used to assess the candidate value a.

At $a = \alpha_0(\tau)$ and $\eta = \eta_0$, the score satisfies

$$\mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta_0)] = 0. \tag{23}$$

Section 3.1 showed that the true quantile residual has conditional mean zero given X_i and Z_i. The residualized instrument is itself a function of X_i and Z_i. Thus, within every value of these variables, multiplying by the residualized instrument still gives conditional mean zero. Averaging over X_i and Z_i gives (23). This moment is valid because of the structural-rank and instrument-exogeneity assumptions in Section 3.1. Neyman orthogonality is a separate property created by the nuisance construction. It requires

$$\partial_\eta \, \mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]\Big|_{\eta=\eta_0} = 0. \tag{24}$$

This condition does not mean that the nuisance parameters are known. It says that small errors around their true values do not change the population moment to first order. Without orthogonality, a small nuisance error can cause a first-order change in the target moment. With it, that first-order change is zero, so the leading effect of a sufficiently small error is of higher order. Orthogonality therefore reduces the effect of nuisance-estimation error; it does not remove that error.

To see the joint cancellation for the intercept and control slopes, write $\epsilon_{0i} = \epsilon_i(\alpha_0(\tau))$. Using the residual density defined in Section 3.2, define

$$\begin{aligned} M_0 &= \mathrm{E}\big[Z_i \widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big], \\ J_0 &= \mathrm{E}\big[\widetilde{X}_i \widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big], \\ \delta_0 &= M_0 J_0^{-1}. \end{aligned} \tag{25}$$

At the truth, differentiation with respect to the fitted intercept and control slopes then gives

$$\begin{aligned} \partial_{(c,\beta')} \mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]\big|_{\eta=\eta_0} &= -\mathrm{E}\big[(Z_i - \delta_0 \widetilde{X}_i)\widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big] \\ &= -[M_0 - \delta_0 J_0] = 0. \end{aligned} \tag{26}$$

A small error in c or β moves the fitted quantile. Observations close to the quantile boundary are the ones most likely to switch sides, which is why the density at zero appears. The choice $\delta_0 = M_0 J_0^{-1}$ makes the residualized instrument density-weighted orthogonal to $\widetilde{X}_i$. Its constant entry handles small errors in the fitted intercept, while its remaining entries handle small errors in the control slopes. The corresponding first-order changes therefore cancel.

Orthogonality also holds in the δ direction. For instrument component j, let δ_j denote the corresponding row of δ. Then

$$\partial_{\delta_j} \mathrm{E}\big[g_{\tau,j}(W_i; \alpha_0(\tau), \eta)\big]\big|_{\eta=\eta_0} = -\mathrm{E}\big[\big\{\tau - \mathbb{1}\big[Y_i \le D_i'\alpha_0(\tau) + c_0(\tau) + X_i'\beta_0(\tau)\big]\big\}\, \widetilde{X}_i'\big] = 0. \tag{27}$$

A small error in δ changes how much of $\widetilde{X}_i$ is removed from an instrument. That error is multiplied by the quantile residual, whose population moment with $\widetilde{X}_i$ is already zero at $c_0(\tau)$ and $\beta_0(\tau)$ by (16). Thus, it has no first-order effect either. The constant entry covers the residualization-intercept direction, while the remaining entries cover the residualization slopes. This protection matters because high-dimensional methods estimate the nuisance quantities imperfectly, and regularization adds estimation and shrinkage error. The orthogonal score allows sufficiently accurate nuisance estimates to be combined later with inference on the low-dimensional target; this is the specific DML-IVQR construction of Chen et al. (2021) and the general principle explained by Chernozhukov et al. (2018).

Orthogonality protects the score against small nuisance-parameter errors, but it does not create identification or add information about the treatment effect. In particular, it does not make a weak instrument strong and does not guarantee that the data contain enough information about $\alpha_0(\tau)$. If the instruments contain little information about the endogenous treatment, the moment may also contain little information for distinguishing nearby values of a. IVQR moment validity comes from the structural assumptions and instrument exogeneity; nuisance robustness comes from orthogonalization. These are different requirements.

3.4 Nuisance Estimation and Regularization

Sections 3.2–3.3 defined $c(a)$, $\beta(a)$, and $\delta(a)$ as population quantities. Their sample counterparts can be difficult to estimate when the number of controls is large relative to the sample size N. Estimating every coefficient without restrictions may then be noisy or unstable. The DML-IVQR procedure therefore uses ℓ_1 regularization. This approach is suited to approximate sparsity: many controls may be available, but only a relatively small group has a large predictive role, while many other coefficients are zero or small. Approximate sparsity does not require every weak or irrelevant coefficient to be exactly zero.

Fix τ and a candidate value a. Nuisance estimation is repeated whenever a relevant candidate is evaluated. In the executable implementation used here, the response for the first nuisance step is $Y_i - D_i'a$. The formula adds an intercept, so let b_0 denote that intercept and let $b = (b_1, \dots, b_p)'$ contain only the slopes on X_i. Thus, $(\widehat{b}_0(a), \widehat{\beta}(a))$ estimates $(c(a), \beta(a))$, while the later pair $(\widehat{d}_0(a), \widehat{\delta}(a))$ corresponds to the intercept-and-slope partition $[d_0(a), \mathit{\Delta}(a)]$ of the population residualization matrix. The candidate-specific quantile-Lasso estimates are

$$\begin{pmatrix} \widehat{b}_0(a) \\ \widehat{\beta}(a) \end{pmatrix} = \operatorname*{arg\,min}_{b_0, b} \left\{ \sum_{i=1}^{N} \rho_\tau \big(Y_i - D_i'a - b_0 - X_i'b\big) + \lambda_0(a)|b_0| + \sum_{k=1}^{p} \lambda_k(a)|b_k| \right\}. \tag{28}$$

The first term measures fit at quantile τ. The second places a separate penalty on each coefficient, including the intercept; there is no post-Lasso refit. The loss is an unscaled sum because this is the scaling used by the fitted quantile-regression routine. Following the pivotal route described by Chen et al. (2021), let $s_k = (N^{-1} \sum_i X_{ik}^2)^{1/2}$ and, for $r = 1, \dots, 1000$, draw independent $U_{ir} \sim \text{Uniform}(0, 1)$. The implemented penalty is

$$\begin{aligned} T_r &= \max_{1 \le k \le p} \left| \frac{\sum_{i=1}^{N} X_{ik} \left[\tau - \mathbb{1}\{U_{ir} < \tau\}\right]}{s_k \sqrt{\tau(1-\tau)}} \right|, \\ \lambda(a) &= 2\, \widehat{Q}_{0.9}\Big(\{T_r\}_{r=1}^{1000}\Big) \sqrt{\tau(1-\tau)}(1, s_1, \dots, s_p)'. \end{aligned} \tag{29}$$

Artificial Uniform variables are useful because the indicator at the correct quantile has a known Bernoulli structure. The simulated statistics approximate how large random correlations between the controls and a quantile score can be without a true signal. The simulated 0.9 quantile sets the reference level for these random correlations, while the factor 2 scales the resulting penalty according to the implemented pivotal rule. A fresh set of Uniform draws is generated every time the penalty routine is called. Hence, candidate evaluations on the same observed sample can use different pivotal draws. This is penalty-construction randomness, not sample splitting. The frozen main simulation uses this rule rather than its older five-fold cross-validation alternative.

The fitted quantile regression produces the candidate residual and density weight

$$\begin{aligned} \widehat{\epsilon}_i(a) &= Y_i - D_i' a - \widehat{b}_0(a) - X_i'\widehat{\beta}(a), \\ \widehat{f}_i(a) &= \phi\Big(\widehat{\epsilon}_i(a); \overline{\widehat{\epsilon}}(a), s^2_{\widehat{\epsilon}(a)}\Big), \end{aligned} \tag{30}$$

where $\phi(e;\mu,\sigma)$ is the normal density with mean μ and standard deviation σ. Thus, the executable simulation forms the weights by evaluating a fitted normal density at the candidate residuals.[1] This fitted-normal quantity is an implementation-specific proxy for the theoretical conditional density $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ from Section 3.2; it is not a direct conditional-density estimate at zero. Zero remains the fitted quantile boundary, and the theoretical reason for density weighting is that observations near this boundary determine local changes in the quantile moment. These proxy weights are then used in the instrument-residualization step, where they play the role assigned to density weights in the theoretical construction.

For each instrument component j, the implementation next fits a separate square-root-density-weighted Lasso. Let $\widehat{d}_{j0}(a)$ denote its intercept and let $\widehat{\delta}_j(a)$ contain only the slopes on the transformed controls. Its transformed regression is

$$\begin{pmatrix} \widehat{d}_{j0}(a) \\ \widehat{\delta}_j(a) \end{pmatrix} = \operatorname*{arg\,min}_{(d_{j0},d_j)} \left\{ \sum_{i=1}^{N} \left[\sqrt{\widehat{f}_i(a)} Z_{ji} - d_{j0} - \sqrt{\widehat{f}_i(a)} X_i' d_j \right]^2 + \mathcal{P}_{j,a}(d_j) \right\}. \tag{31}$$

Here $\mathcal{P}_{j,a}(d_j) = \sum_{k=1}^{p} \lambda^{\delta}_{jk}(a)|d_{jk}|$ is the coefficient-specific ℓ_1 penalty produced by the instrument-regression routine's default iterative heteroskedastic loadings. The intercept is included but not penalized, the same controls are used as in the first nuisance step, and no post-Lasso refit is performed. After fitting the transformed regression, the code retains both the

1. The literal preserved call is `dnorm(e, mean(e), var(e))`. Its third argument is interpreted as a standard deviation, although the code supplies the residual sample variance.

intercept and the slopes. Define $\widehat{d}_0(a) = (\widehat{d}_{10}(a), \ldots, \widehat{d}_{d_z 0}(a))'$ and stack the slope rows in $\widehat{\delta}(a)$. The residualized instrument used in the score is

$$\begin{aligned} \widehat{\Psi}_{ji}(a) &= Z_{ji} - \widehat{d}_{j0}(a) - X_i'\widehat{\delta}_j(a), \\ \widehat{\Psi}_i(a) &= Z_i - \widehat{d}_0(a) - \widehat{\delta}(a)X_i. \end{aligned} \tag{32}$$

Thus, the score residualizes the original, untransformed instrument. The calculation subtracts the retained intercept and the fitted contribution of the original controls. It does not divide by $\sqrt{\widehat{f}_i(a)}$ or use the transformed regression residual directly.

The ℓ_1 penalties place a cost on large collections of nonzero coefficients. Weak coefficients can consequently be shrunk strongly, often to zero, which can stabilize both nuisance steps in a high-dimensional sample. This regularization also creates shrinkage error; the orthogonal construction in Section 3.3 is designed to limit its first-order effect on the target moment, as in the general framework of Chernozhukov et al. (2018). The implementation estimates the nuisance parameters and evaluates the score on the same sample; it does not use sample splitting or cross-fitting. General DML procedures often use cross-fitting, but this specific implementation instead relies on the orthogonal score together with its high-dimensional sparsity and rate conditions.

3.5 Oracle-GMM, Full-GMM, and DML-IVQR

The analysis compares three estimators from the IVQR setting studied by Chen et al. (2021). Oracle-GMM, Full-GMM, and DML-IVQR all target the same low-dimensional parameter $\alpha_0(\tau)$ in the structural quantile model. For any candidate a, they use the same quantile index, outcome, treatment, and instrument vector Z_i. Each estimator forms a structural quantile residual, residualizes the instruments with respect to a chosen control set, and multiplies the resulting instrument vector by the quantile score. Their candidate-specific moments therefore have the same dimension as Z_i. What differs is how the control coefficient and instrument residualization are estimated.

Oracle-GMM is an infeasible benchmark because it is given the identity of the ten controls designated as relevant in the simulation DGP. Let $S_0 = \{1, \ldots, 10\}$ denote this oracle control set and let $X_{S_0,i}$ collect these variables. At each candidate, the executable estimator uses an unpenalized quantile regression with an intercept:

$$\begin{pmatrix} \widehat{b}_{0,\text{orc}}(a) \\ \widehat{\beta}_{\text{orc}}(a) \end{pmatrix} = \underset{b_0, b}{\arg\min} \sum_{i=1}^{N} \rho_\tau\left(Y_i - D_i'a - b_0 - X_{S_0,i}'b\right). \tag{33}$$

Thus, Oracle-GMM first removes $D_i'a$ and then fits the remaining outcome at quantile τ using only this known oracle control set. No coefficient is penalized and there is no later variable-selection or refitting step. The oracle is not a practically available estimator and is not free from sampling error. It provides a benchmark for the cost of having to learn which controls matter.

Full-GMM does not receive this knowledge. It uses all available controls in the same unpenalized, candidate-specific quantile-regression problem:

$$\begin{pmatrix} \widehat{b}_{0,\text{full}}(a) \\ \widehat{\beta}_{\text{full}}(a) \end{pmatrix} = \arg\min_{b_0, b} \sum_{i=1}^{N} \rho_\tau\big(Y_i - D_i'a - b_0 - X_i'b\big)\,. \tag{34}$$

In the executable design, this means that all 100 controls enter freely. Full-GMM is feasible in the sense that it does not use the unknown active set, but estimating many weak or irrelevant coefficients from one finite sample can be difficult. Including many controls does not by itself imply inconsistency.

Oracle-GMM and Full-GMM use the same unregularized instrument-residualization calculation, with their respective control matrices. For $e \in \{\text{orc}, \text{full}\}$, let $X_{\text{orc},i} = X_{S_0,i}$ and $X_{\text{full},i} = X_i$. Let $\widehat{f}_{e,i}(a)$ be the fitted-normal density proxy from the estimator's candidate residual, formed with the literal convention described in Section 3.4. The code computes

$$\begin{aligned} \widehat{M}_e(a) &= \frac{1}{N}\sum_{i=1}^{N} Z_i X_{e,i}' \widehat{f}_{e,i}(a), \qquad & \widehat{J}_e(a) &= \frac{1}{N}\sum_{i=1}^{N} X_{e,i} X_{e,i}' \widehat{f}_{e,i}(a), \\ \widehat{\delta}_e(a) &= \widehat{M}_e(a)\widehat{J}_e(a)^{-1}, & \widehat{\Psi}_{e,i}(a) &= Z_i - \widehat{\delta}_e(a) X_{e,i}. \end{aligned} \tag{35}$$

This is a direct matrix inversion, not a penalized regression. The residualization includes no intercept, even though the preceding quantile regression does. This is an implementation-specific feature of the frozen benchmark estimators; it differs from the augmented population residualization used to describe the orthogonal DML score in Sections 3.2–3.3. Accordingly, the benchmark moments do not implement the full augmented orthogonality calculation. The calculation also uses the original Z_i in the final residual. The implementation has no ridge adjustment or generalized inverse: if $\widehat{J}_e(a)$ is singular, the direct solve fails and the outer evaluation records a numerical failure. Each benchmark then multiplies $\widehat{\Psi}_{e,i}(a)$ by its candidate quantile score.

DML-IVQR instead uses the full high-dimensional control set and regularizes both nuisance-estimation steps, as described in Section 3.4. It estimates the control coefficients by quantile Lasso, forms the density proxy, and estimates the instrument residualization through regularized weighted regressions. These nuisance estimates are then used in the score described in Section 3.3. The implemented version, DML-IVQR-BC, uses the pivotal penalty in (29). Oracle-GMM and Full-GMM also use residualized IVQR moments, but their nuisance parameters are estimated without ℓ_1 regularization. Unlike the benchmark matrix calculation, the DML instrument regression retains an intercept. The comparison therefore covers three cases: the oracle control set is known and estimated without regularization; the full control set is estimated without regularization; or the full control set is handled with regularized nuisance estimation. It does not perfectly separate every source of finite-sample error, but it keeps the structural model, target, and instruments common. Once the estimator-specific moment is constructed, Section 3.6 applies the common criterion and inference procedure.

3.6 GMM Criterion, Point Estimation, and Robust Inference

Let $e \in \{\text{orc}, \text{full}, \text{dml}\}$ index the estimator and let $\widehat{g}_{e,i}(a)$ denote its observation-level IVQR score at candidate a. This score uses the estimator-specific nuisance quantities described above and is a $d_z = \dim(Z_i)$ dimensional vector. Following Chen et al. (2021), its sample average is

$$\widehat{g}_{e,N}(a) = \frac{1}{N} \sum_{i=1}^{N} \widehat{g}_{e,i}(a). \tag{36}$$

For a given candidate, this vector shows how far the sample IVQR conditions are from zero. A compatible candidate should produce relatively small average moments after their sampling variation is taken into account. The frozen code measures that variation with the uncentered second moment

$$\widehat{\Sigma}_e(a) = \frac{1}{N} \sum_{i=1}^{N} \widehat{g}_{e,i}(a)\widehat{g}_{e,i}(a)'. \tag{37}$$

The components can have different scales and can move together, so this matrix standardizes them before they are combined. It uses denominator N, has no finite-sample correction, and does not subtract the sample mean. At the true parameter, where the population moment is zero, this uncentered second moment and the covariance have the same probability limit.

The common candidate-specific GMM criterion is

$$W_{e,N}(a) = N\widehat{g}_{e,N}(a)'\widehat{\Sigma}_e(a)^{-1}\widehat{g}_{e,N}(a). \tag{38}$$

This expression is exactly equivalent to the code, which stores the unnormalized score sum and the unnormalized sum of score outer products. When the second-moment matrix is well behaved, $W_{e,N}(a)$ is nonnegative. A smaller value means that the covariance-weighted moments are closer to zero, while a larger value indicates stronger disagreement between the candidate and the sample restrictions. The criterion combines all instrument components into one number. The code uses a direct matrix inverse without a ridge adjustment or generalized inverse. If inversion fails or the result is non-finite, that candidate is recorded as a numerical failure. When the estimator is clear, the subscript e is suppressed and the statistic is written as $W_N(a)$. Conceptually, the point estimator is

$$\widehat{\alpha}_e(\tau) = \arg\min_{a \in \mathcal{A}} W_{e,N}(a). \tag{39}$$

It selects the candidate that makes the weighted sample moments as close to zero as possible; the minimum need not equal zero. In the Monte Carlo implementation, minimization is numerical over the same prespecified finite profile used for confidence-set evaluation, whose details appear in Section 4.5. The first candidate attaining the minimum is selected if there is an exact tie. An incomplete profile receives no point estimate.

Point estimation and robust inference use the profile differently. Point estimation keeps only its minimizer. Inference instead tests every hypothesized value a separately, so a value need not minimize $W_{e,N}(a)$ to be retained. Under $H_0 : a = \alpha_0(\tau)$, the implemented procedure uses the asymptotic reference result

$$W_{e,N}(a) \xrightarrow{d} \chi^2_{d_z}, \qquad d_z = \dim(Z_i). \tag{40}$$

The degrees of freedom equal the number of instrument components and therefore the number of moment components. Following Chernozhukov and Hansen (2008), this test evaluates the moments directly at the hypothesized value instead of relying on a conventional normal approximation for $\widehat{\alpha}_e(\tau)$. For significance level $q \in (0, 1)$, let c_{1-q,d_z} be the $(1 - q)$ quantile of $\chi^2_{d_z}$. The candidate is rejected when $W_{e,N}(a) > c_{1-q,d_z}$ and is not rejected when the weak inequality holds.

Inverting these candidate-specific tests gives

$$C_{e,1-q}(\tau) = \left\{a \in \mathcal{A} : W_{e,N}(a) \le c_{1-q,d_z}\right\}. \tag{41}$$

Rather than placing one standard-error band around the point estimate, test inversion retains every candidate that is not rejected by its own moment test. Their collection is the confidence region. Because the profile can cross the critical value more than twice, this region need not be one ordinary interval: it may be connected, have separated components, be wide, or be empty. Numerical evaluation over a finite domain describes only the accepted set within that domain.

Acceptance reaching a numerical boundary does not establish that the theoretical region is globally unbounded.

The test-inversion procedure is designed to retain asymptotic validity without requiring strong identification and can accommodate weak, partial, or failed identification under the maintained assumptions. This robustness does not imply that the confidence region will be narrow. With informative instruments, moving a away from the true value tends to move the moments away from zero, so the profile usually rises more clearly. With weak instruments, the moments may change slowly and many candidates may remain below the cutoff. This is the difference between validity and informativeness. Weak-identification robustness concerns asymptotic validity: because the implementation uses an asymptotic chi-square critical value, finite-sample size accuracy remains an empirical question. The method is distinct from the exact finite-sample simulation approach of Chernozhukov et al. (2009).

4 Monte Carlo Design

4.1 Baseline Data-Generating Process

The Monte Carlo experiment builds on the high-dimensional simulation design of Chen et al. (2021), with $p = 100$ observed controls. For each observation i, the structural disturbances satisfy

$$\begin{pmatrix} u_i \\ \epsilon_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}\right). \tag{42}$$

The disturbance ϵ_i enters the treatment equation, whereas u_i enters the outcome through $D_i u_i$. Their positive correlation therefore generates endogeneity.

The remaining primitives are mutually independent and independent of the disturbance pair in Equation (42), and they satisfy

$$x_{ji} \overset{\text{ind}}{\sim} \mathcal{N}(0,1), \quad j = 1, \ldots, 100, \qquad z_{1i}, z_{2i}, v_{1i}, v_{2i} \overset{\text{ind}}{\sim} \mathcal{N}(0,1). \tag{43}$$

Observations are generated independently across i. The observed controls are transformations of the latent Gaussian draws:

$$X_{ji} = \Phi(x_{ji}), \qquad j = 1, \ldots, 100, \tag{44}$$

where Φ denotes the standard-normal cumulative distribution function. By the probability integral transform, $X_{ji} \sim \mathcal{U}(0,1)$. These transformations preserve independence across controls and map the latent Gaussian draws monotonically onto the unit interval.

The executable design constructs two instruments as[2]

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i}, \tag{45}$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}. \tag{46}$$

Here, z_{1i} and z_{2i} are the components of the observed instruments that enter the treatment index directly, while the included X components induce dependence between the instruments and observed controls and v_{1i} and v_{2i} add independent noise. The original baseline treatment equation is

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i). \tag{47}$$

The treatment is continuous and lies in $(0,1)$. The next subsection introduces the instrument-strength extension by modifying this treatment equation while leaving the remaining structural DGP unchanged.

The structural outcome is generated according to

$$Y_i = 1 + D_i + 5\sum_{j=1}^{7} X_{ji} + D_i u_i. \tag{48}$$

Only $X_{1i}, \ldots, X_{7i}$ enter the structural outcome directly, while $X_{8i}, \ldots, X_{10i}$ additionally enter the instrument construction. The union of the controls appearing in the outcome and instrument equations therefore gives $X_{1i}, \ldots, X_{10i}$ as the ten controls relevant to the complete DGP. For a treatment level $d \in (0,1)$, Equation (48) implies the potential outcome

$$Y_i(d) = 1 + 5\sum_{j=1}^{7} X_{ji} + d(1 + u_i). \tag{49}$$

Conditional on X_i, treatment-effect heterogeneity is governed by $1 + u_i$. Because u_i is independent of X_i and standard normal,

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5\sum_{j=1}^{7} X_{ji} + d\left[1 + \Phi^{-1}(\tau)\right]. \tag{50}$$

2. The displayed Monte Carlo equations in Chen et al. (2021) write the corresponding control terms as x_j. The released simulation code instead uses the transformed controls $X_j = \Phi(x_j)$. The simulations in this thesis follow that executable implementation.

The coefficient on d in the structural τ-quantile is therefore

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau). \tag{51}$$

In particular, the median structural quantile treatment effect equals $\alpha_0(0.50) = 1$.

4.2 Instrument-Strength Extension

The thesis-specific extension varies excluded-instrument relevance through a single parameter $\kappa \in [0,1]$ while holding the remaining structural DGP fixed. It adds one primitive, $w_i \sim \mathcal{N}(0,1)$, which is independent of all primitives defined in Section 4.1. The latent treatment index and treatment are

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + \sqrt{2(1-\kappa^2)}\, w_i + \epsilon_i, \tag{52}$$

$$D_i(\kappa) = \Phi\big(d_i^*(\kappa)\big)\,. \tag{53}$$

The common loading κ enters symmetrically on z_{1i} and z_{2i}, so excluded-instrument relevance remains a one-dimensional experimental factor. This treatment equation is the sole structural modification relative to the executable baseline.

At $\kappa = 1$, the coefficient on w_i is $\sqrt{2(1-\kappa^2)} = 0$, and hence

$$D_i(1) = \Phi(z_{1i} + z_{2i} + \epsilon_i), \tag{54}$$

which exactly reproduces Equation (47). The scaling of w_i preserves the variance of the latent treatment index. Independence and the unit variances of z_{1i}, z_{2i}, w_i, and ϵ_i imply

$$\mathrm{Var}\big[d_i^*(\kappa)\big] = 2\kappa^2 + 2(1-\kappa^2) + 1 = 3. \tag{55}$$

The excluded-instrument component contributes $2\kappa^2$, the added w_i component contributes $2(1-\kappa^2)$, and ϵ_i contributes one. Thus, $d_i^*(\kappa) \sim \mathcal{N}(0,3)$ for every κ, making the marginal distribution of $D_i(\kappa)$ invariant across designs. This distribution is not uniform because Φ is applied to a normal index with variance three rather than one. The added term therefore prevents a mechanical change in overall treatment-index dispersion when the excluded-instrument loading is reduced.

Instrument relevance changes because

$$\mathrm{Cov}\big(Z_{1i}, d_i^*(\kappa)\big) = \mathrm{Cov}\big(Z_{2i}, d_i^*(\kappa)\big) = \kappa. \tag{56}$$

By Equations (45) and (46), the components of each observed instrument other than z_{1i} or z_{2i} are independent of $d_i^*(\kappa)$, whereas $\mathrm{Cov}(z_{1i}, d_i^*(\kappa)) = \mathrm{Cov}(z_{2i}, d_i^*(\kappa)) = \kappa$. Thus, as κ declines, so does the covariance between each observed instrument and the latent treatment index. The parameter κ is the structural loading governing instrument relevance, not a complete measure of IVQR identification. Calibration diagnostics later assess the empirical strength separation of the selected designs. At the same time, endogeneity is preserved because u_i is independent of z_{1i}, z_{2i}, and w_i, whereas its covariance with ϵ_i remains fixed:

$$\mathrm{Cov}\big(u_i, d_i^*(\kappa)\big) = \mathrm{Cov}(u_i, \epsilon_i) = 0.3. \tag{57}$$

Because $(u_i, d_i^*(\kappa))$ is jointly Gaussian with zero means and invariant moments $\mathrm{Var}(u_i) = 1$, $\mathrm{Var}(d_i^*(\kappa)) = 3$, and $\mathrm{Cov}(u_i, d_i^*(\kappa)) = 0.3$, its joint distribution is unchanged across κ. The extension also leaves unchanged the observed instruments in Equations (45) and (46), the controls and their distribution, the disturbance distribution, the outcome in Equation (48), and the structural quantile effect in Equation (51). It is therefore designed to isolate weaker excluded-instrument relevance from changes in the marginal treatment distribution, the endogeneity channel, and the structural treatment-effect distribution.

4.3 Simulation Design

The completed Monte Carlo experiment uses the design dimensions

$$\begin{aligned} &n \in \{500, 1000\}, && \tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}, \\ &\kappa \in \{1, 0.50, 0.25, 0.10\}, && R = 500. \end{aligned} \tag{58}$$

The two sample sizes characterize finite-sample behavior at different values of n, while the five quantile indices cover the center and both tails of the structural outcome distribution. The instrument-loading values span the executable baseline at $\kappa = 1$ and progressively lower degrees of excluded-instrument relevance; their empirical calibration is presented later. The Cartesian product contains $2 \times 5 \times 4 = 40$ distinct (n, τ, κ) designs, with $R = 500$ Monte Carlo replications in each design cell.

Each design is evaluated using Oracle-GMM, Full-GMM, and DML-IVQR. Oracle-GMM uses the known low-dimensional set of controls relevant to the DGP and serves as an infeasible benchmark. Full-GMM instead uses the complete high-dimensional control set without the DML regularization strategy. DML-IVQR applies the authors' high-dimensional regularized procedure with an orthogonal score. Chapter 3 provides the formal estimator definitions. Combining the 40 designs with the three estimators yields $40 \times 3 = 120$ design-estimator cells, each containing 500 replications. For every quantile index, the true parameter used in the Monte Carlo evaluation is the value from Equation (51), $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$.

Within a given replication and (n, τ, κ) design, the runner evaluates all three estimators on the same generated sample, thereby holding the simulated data fixed across estimator comparisons. The implemented DML-IVQR estimator uses the BC pivotal-penalty and same-sample nuisance-estimation procedure described in Section 3.4.

4.4 Performance Measures

For a fixed $(n, \tau, \kappa, \text{estimator})$ cell, let $\widehat{\alpha}_r(\tau)$ denote the point estimate in replication r, and let $\alpha_0(\tau)$ be the true value from Equation (51). Replications are indexed by $r = 1, \ldots, R$. Suppressing an estimator superscript, point-estimation performance is summarized by

$$\begin{aligned} \text{Bias}(\tau) &= \frac{1}{R} \sum_{r=1}^{R} \left[\alpha_0(\tau) - \widehat{\alpha}_r(\tau)\right], \\ \text{MAE}(\tau) &= \frac{1}{R} \sum_{r=1}^{R} \left|\widehat{\alpha}_r(\tau) - \alpha_0(\tau)\right|, \\ \text{RMSE}(\tau) &= \left\{\frac{1}{R} \sum_{r=1}^{R} \left[\widehat{\alpha}_r(\tau) - \alpha_0(\tau)\right]^2\right\}^{1/2}. \end{aligned} \tag{59}$$

Every measure is computed separately for each estimator and design cell; aggregation never pools distinct design cells. Under the frozen reporting convention, positive Bias means that estimates lie below the true parameter on average, while negative Bias means that they lie above it. Bias captures signed systematic deviation, MAE measures absolute error, and RMSE places greater weight on large errors.

Coverage is determined by evaluating the test statistic directly at the exact true parameter. With $c_{0.95}$ denoting the critical value defined in Section 4.5, define

$$C_r(\tau) = \mathbf{1}\left\{W_{N,r}(\alpha_0(\tau)) \le c_{0.95}\right\}, \qquad \widehat{\text{Coverage}}(\tau) = \frac{1}{R} \sum_{r=1}^{R} C_r(\tau). \tag{60}$$

Nominal coverage is 0.95. Crucially, coverage evaluates $W_{N,r}$ at $\alpha_0(\tau)$ itself; it is not inferred from whether a numerical confidence-region grid contains the true value and is therefore independent of whether the truth is a grid point. Empirical size is the rejection frequency at the true parameter:

$$\widehat{\text{Size}}(\tau) = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\{W_{N,r}(\alpha_0(\tau)) > c_{0.95}\} = 1 - \widehat{\text{Coverage}}(\tau). \tag{61}$$

The final equality follows exactly from the implemented rejection rule.

Power is evaluated at deliberately false treatment-effect values

$$a_\Delta(\tau) = \alpha_0(\tau) + \Delta, \qquad \Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}. \tag{62}$$

For each offset, empirical power is

$$\widehat{\text{Power}}(\Delta, \tau) = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\{W_{N,r}(a_\Delta(\tau)) > c_{0.95}\}. \tag{63}$$

The offset $\Delta = 0$ is excluded because rejection at the true value measures size rather than power. As with coverage, the statistic is evaluated directly at $a_\Delta(\tau)$, so power does not depend on whether that false value lies on the numerical confidence-region grid.

Inferential informativeness is summarized by the grid-based accepted-set measure within the prespecified primary computational domain. In each replication, candidate values satisfying $W_{N,r}(a) \leq c_{0.95}$ are accepted by the inverted test. The measure totals all accepted components inside the finite domain, including disconnected components; it is neither the distance between the minimum and maximum accepted values nor a claim about the unrestricted theoretical confidence region. Section 4.5 defines the exact numerical grid and trapezoidal construction. The Monte Carlo summary reports the median accepted-set measure across replications. It also reports the all-grid-points acceptance rate, the fraction of replications in which every candidate point in the prespecified primary grid is accepted; this is not acceptance of the entire parameter space. Coverage and size assess empirical calibration at the true parameter, whereas the accepted-set measure and power assess informativeness and discrimination against other parameter values.

4.5 Confidence-Region Construction and Numerical Grid

The Monte Carlo implementation uses the candidate-specific statistic $W_N(a)$ and the uncentered score second-moment matrix defined in Section 3.6. Because the DGP has $d_z = 2$ instrument components, the 95% cutoff is

$$c_{0.95} = \chi^2_{2,0.95} = 5.991464547\ldots. \tag{64}$$

As in Section 3.6, a candidate is accepted when $W_N(a) \leq c_{0.95}$. This asymptotic cutoff is not an exact finite-sample guarantee; finite-sample calibration is evaluated empirically in Chapter 5. The main simulation evaluates the accepted set on the prespecified primary grid

$$\begin{aligned} A_0 &= [-1, 3], & h &= 0.1, \\ G_0 &= \{a_j = -1 + (j-1)h : j = 1, \ldots, 41\}, \\ &= \{-1, -0.9, \ldots, 2.9, 3\}. \end{aligned} \tag{65}$$

The same grid is used across sample sizes, quantiles, instrument-loading values, estimators, and replications. The interval A_0 is solely a computational domain, not the asserted theoretical parameter space. Reaching a boundary of A_0 therefore establishes neither global boundedness nor global unboundedness of the conceptual confidence set.

For replication r, define the binary grid-acceptance indicator, accepted grid, and all-grid-points indicator by

$$\begin{aligned} I_r(a_j) &= \mathbf{1}\left\{W_{N,r}(a_j) \leq c_{0.95}\right\}, \\ C_{r,G_0} &= \left\{a_j \in G_0 : I_r(a_j) = 1\right\}, \\ F_r &= \mathbf{1}\left\{I_r(a_j) = 1 \text{ for every } a_j \in G_0\right\}. \end{aligned} \tag{66}$$

The frozen implementation computes the grid-based accepted-set measure within A_0 as the trapezoidal integral of this binary indicator:

$$M_r(A_0) = \sum_{j=1}^{40} \frac{a_{j+1} - a_j}{2} \left[I_r(a_j) + I_r(a_{j+1})\right]. \tag{67}$$

This quantity approximates the total measure of accepted values inside A_0. Separated accepted regions contribute through their respective local grid intervals, while rejected stretches reduce the measure rather than being filled in by the span between the smallest and largest accepted values. The calculation uses the grid indicators without interpolating critical-value crossings. It lies between zero and four and equals four when all 41 grid points are accepted. It is a finite-domain numerical summary, not an unrestricted confidence-region length. The Monte Carlo mean of F_r is the all-grid-points acceptance rate, which records how often every tested candidate in G_0 is accepted; it does not imply acceptance of the real line or global unboundedness.

The grid supports point-profile evaluation, the accepted-set measure, and the all-grid-points indicator. Coverage and power in Section 4.4 instead evaluate W_N directly at exact true or false candidate values and remain independent of membership in G_0.

4.6 Post-Main Numerical Diagnostics

After the main Monte Carlo design had been frozen and completed, a numerical-domain diagnostic examined whether confidence-set informativeness within the primary domain A_0 was sensitive to its finite endpoints. A finite grid cannot reveal what occurs beyond those endpoints, so acceptance near or at a boundary may cause the measured accepted set to reflect numerical truncation. The investigated domains were

$$A_0 = [-1, 3], \qquad A_1 = [-3, 5], \qquad A_2 = [-5, 7]. \tag{68}$$

The diagnostic used $n = 1000$, $R = 100$ diagnostic replications and primitive samples, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, and Oracle-GMM, Full-GMM, and DML-IVQR-BC. It used $\kappa \in \{1, 0.25, 0.10\}$, a subset selected to represent different levels of instrument relevance. Stage 1 evaluated $W_N(a)$ on the 81-point grid for A_1 with spacing 0.1; its 41-point A_0 summary came from the same profile. Thus, each replication held the primitive sample fixed across domains.

Stage 2 extended the investigated domain to A_2. It reloaded the 100 saved Stage 1 primitive samples, retained the 81 profile values on A_1, and evaluated only the 40 added tail points in $[-5, -3.1] \cup [5.1, 7]$. The resulting A_2 profile contained 121 points at the unchanged spacing of 0.1. Comparisons across domains therefore reflect additional candidate values rather than different Monte Carlo samples. For each domain, the implementation records the grid-based accepted-set measure, the accepted-grid share and count, acceptance at the left, right, either, and both boundaries, and whether all grid points are accepted. It also computes the change in accepted-set measure from A_0 to A_1 and from A_1 to A_2.

These diagnostics distinguish an accepted set numerically localized within a domain from one whose relevant portions continue beyond it. If acceptance reaches a numerical boundary, the warranted conclusion is only that the accepted set is not bounded within the domain examined. It does not establish that the theoretical confidence region is globally unbounded, that the entire real line is accepted, or that identification is completely absent. The diagnostic ended after Stage 2 at A_2; no Stage 3 or open-ended expansion was used. This keeps the exercise bounded rather than repeatedly enlarging the domain until a particular pattern emerges. Main Monte Carlo summaries continue to use A_0. The expansions diagnose the numerical interpretation of accepted-set informativeness, while coverage and power remain direct evaluations at their target values under Section 4.4. Substantive diagnostic findings are reported in Chapter 5.

Table 1. Instrument-strength calibration

n	κ	**Median F**	**F: p10–p90**	**Median partial R^2**	**Partial R^2: p10–p90**
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo strength diagnostics from the preserved 100-replication calibration. The reported dispersion ranges are the 10th and 90th percentiles. These statistics are not formal weak-instrument classification thresholds.

5 Monte Carlo Results

5.1 Instrument-Strength Calibration

Section 4.2 introduced κ as the loading that controls the relevance of the excluded instruments in the latent treatment index. Before turning to the estimators, Table 1 documents how this design parameter maps into two familiar, descriptive measures of instrument relevance. The median first-stage F-statistic declines sharply and monotonically as κ falls. For $n = 500$, it decreases from 101.17 at $\kappa = 1$ to 18.71 at $\kappa = 0.50$ and 1.22 at $\kappa = 0.10$. The same endpoints for $n = 1000$ are 212.82 and 2.04. Thus, larger samples raise the statistic at every fixed value of κ, while the ordering across the four calibration settings remains unchanged.

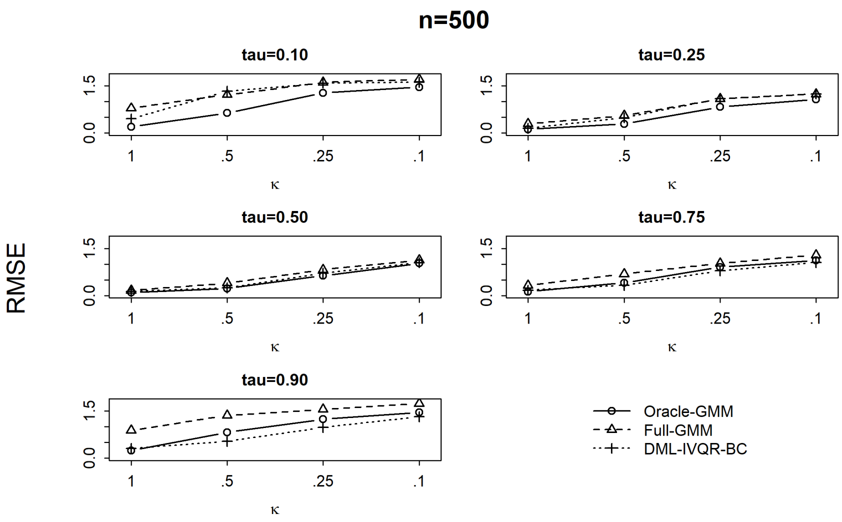
The median partial R^2 provides a complementary measure that is much less sensitive to the change in sample size. At $\kappa = 1$, its medians are 0.294 for $n = 500$ and 0.301 for $n = 1000$, indicating that the excluded instruments explain a substantial share of treatment variation conditional on the controls. At $\kappa = 0.10$, the corresponding values are only 0.005 and 0.004. The close agreement across sample sizes shows that the decline primarily reflects the design's weakening instrument signal, rather than a change in the underlying relevance pattern caused by sample size. By contrast, the first-stage F-statistic increases markedly with n because it reflects both the strength of the relationship and the amount of sample information available for estimating it. Therefore, κ, rather than the raw F-statistic, remains the experimental parameter that defines the strength design.

Taken together, the two diagnostics provide a clear relative ordering of the four strength settings. The case $\kappa = 1$ has the highest relevance and serves as the executable baseline, while $\kappa = 0.50$ is substantially weaker but remains clearly more informative than the two lower settings. The case $\kappa = 0.25$ represents low instrument relevance, and $\kappa = 0.10$ represents very low relevance in this design. These descriptions are comparisons within the simulation and should not be read as formal weak-identification classifications for IVQR. The calibration nevertheless confirms that the chosen values of κ span clear empirical differences in the relevance of the excluded instruments, while preserving the same ranking at both sample sizes. This spread allows the subsequent analysis to examine how estimation and inference change as instrument relevance deteriorates. The next section examines whether this decline in relevance is reflected in point-estimation performance.

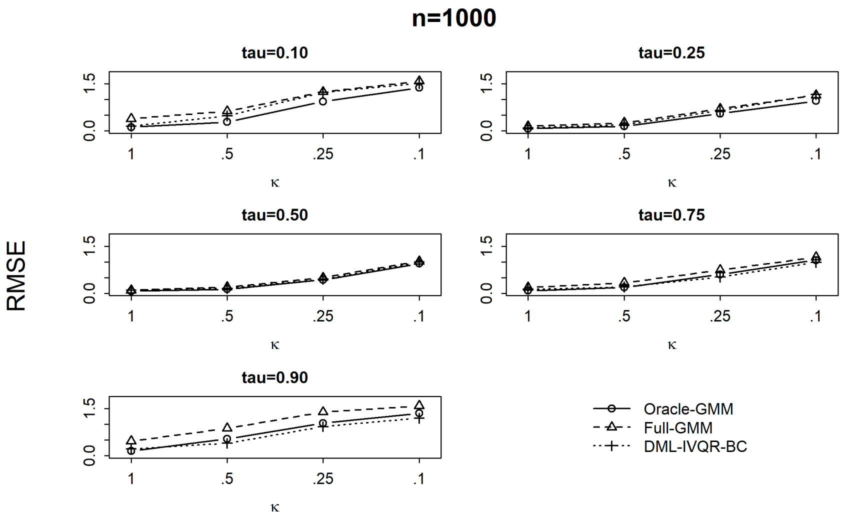
5.2 Point-Estimation Performance

Figure 1 shows the main point-estimation pattern across the complete design. As κ decreases, RMSE rises for Oracle-GMM, Full-GMM, and DML-IVQR-BC at each quantile and sample size. Weaker instrument relevance therefore leads to substantially larger point-estimation errors. When the instruments contain less information about treatment, the GMM criterion also contains less information for locating $\alpha_0(\tau)$. The criterion therefore becomes less informative about which candidate value best matches the moment conditions. This interpretation concerns precision and does not imply that weaker instruments mechanically shift estimates in one direction. The reported estimates are the numerical profile minimizers defined in Sections 3.6 and 4.5, rather than unrestricted continuous optimizers.

Table 2 provides a compact illustration at $\tau = 0.50$ and $n = 1000$. Between $\kappa = 1$ and $\kappa = 0.10$, Oracle-GMM's RMSE rises from 0.072 to 0.945, while the DML value rises from 0.104 to 0.991. Full-GMM shows a similar increase, from 0.100 to 1.018. MAE moves in the same direction: the strongest-to-weakest comparison is 0.047 to 0.735 for Oracle, 0.072 to 0.840 for Full, and 0.079 to 0.779 for DML. Signed bias is smaller than MAE and RMSE over much of this median-quantile comparison. For example, at $\kappa = 0.25$, the biases are 0.045, 0.029, and -0.003, compared with MAEs between 0.281 and 0.372. Under the thesis convention, positive Bias means that estimates are below the truth on average, whereas negative Bias means that they are above it. The comparatively large increases in MAE and RMSE thus cannot be explained only by an average shift in one direction; greater dispersion across replications is an important part of the deterioration. This statement is specific to the median example and is not imposed on the other quantiles.



(a) $n = 500$



(b) $n = 1000$

Figure 1. RMSE across instrument-strength designs

Table 2. Point-estimation performance at the median quantile, $n = 1000$

κ	**Estimator**	**Bias**	**MAE**	**RMSE**
1.00	Oracle-GMM	0.007	0.047	0.072
1.00	Full-GMM	0.007	0.072	0.100
1.00	DML-IVQR-BC	0.070	0.079	0.104
0.50	Oracle-GMM	0.015	0.093	0.134
0.50	Full-GMM	0.014	0.151	0.199
0.50	DML-IVQR-BC	0.059	0.119	0.160
0.25	Oracle-GMM	0.045	0.281	0.431
0.25	Full-GMM	0.029	0.372	0.500
0.25	DML-IVQR-BC	-0.003	0.306	0.448
0.10	Oracle-GMM	-0.044	0.735	0.945
0.10	Full-GMM	-0.012	0.840	1.018
0.10	DML-IVQR-BC	-0.195	0.779	0.991

Note: Each cell uses 500 Monte Carlo replications. The frozen sign convention is Bias $= \alpha_0 - \widehat{\alpha}$.

The full results in Appendix Table A.1 show that Oracle-GMM generally provides the most favorable point-estimation benchmark. Its knowledge of the designated oracle control set avoids the need to estimate the full high-dimensional nuisance model, but its RMSE still increases sharply as relevance weakens. The common deterioration across all three estimators therefore cannot be attributed only to high-dimensional nuisance estimation. Full-GMM, which estimates all controls without regularization, often has higher RMSE than Oracle-GMM and in many designs also has higher RMSE than DML-IVQR-BC. DML is nevertheless neither uniformly better than Full nor uniformly equal to Oracle. Its relative performance is consistent with regularization helping to control the high-dimensional nuisance problem, although the comparison does not isolate that effect because the procedures also differ in details of instrument residualization. Figure 1 also shows lower RMSE at $n = 1000$ than at $n = 500$ for each matched estimator, quantile, and strength setting. The added sample information improves accuracy, especially away from the lowest-relevance design, but it does not remove the pronounced deterioration as κ approaches 0.10 in these finite samples.

DML addresses the difficulty of estimating high-dimensional nuisance parameters; it does not recreate information that weak instruments fail to provide about treatment. The substantial rise in its own RMSE as κ falls therefore illustrates the distinction made in Section 3.3: regularization and orthogonality can limit errors from nuisance estimation, but they do not remove the underlying loss of instrument relevance. The magnitude of the point-estimation deterioration is not identical across quantiles; this heterogeneity is examined directly in Section 5.6. For the present comparison, the central conclusion is that weaker relevance makes the location of the treatment effect increasingly difficult to estimate even for the oracle benchmark, while larger samples improve accuracy and DML-IVQR often records lower error than Full-GMM in many design cells. The next section asks a different question: how much information remains in the test-inverted confidence region.

5.3 Confidence-Region Informativeness

Figures 2 and 3 show that the median grid-based accepted-set measure rises sharply as κ decreases for all three estimators and across the full design. The measure records the total amount of accepted candidate values inside the prespecified domain A_0; a value of four corresponds to acceptance across the entire four-unit numerical domain. This does not mean that the entire real line is accepted. A small measure means that the test rejects most candidate values investigated within A_0, whereas a large measure means that many of those values cannot be rejected. At $\kappa = 1$, the median accepted-set measure is small relative to the domain maximum; the post-main diagnostic below shows that acceptance is also numerically localized within the investigated domains. At $\kappa = 0.10$, every estimator–quantile combination has a median measure of at least 3.8, close to the finite-domain maximum of four. The test therefore loses substantial ability to distinguish among candidate treatment effects as instrument relevance falls.

The median-quantile results for $n = 1000$ provide a representative example. At $\kappa = 1$, the median measures are 0.200 for Oracle-GMM, 0.400 for Full-GMM, and 0.300 for DML-IVQR-BC. At $\kappa = 0.25$, they rise to 1.900, 2.200, and 2.025, and at $\kappa = 0.10$ they reach 3.900, 3.800, and 3.900. Appendix Table E.1 reports the complete results. Oracle often has the smallest measure under the more informative designs, Full is often wider, and DML is closer to Oracle in several cases, but this ordering is not uniform. Near $\kappa = 0.10$, differences among estimators are small relative to their shared loss of informativeness. The oracle's deterioration is especially important because it shows that knowing the designated control set does not overcome weak instrument relevance. Comparing the two panels also shows that increasing the sample size from 500 to 1000 either reduces the median measure or leaves it unchanged in every matched design cell. This finite-sample improvement does not restore a localized accepted set at the weakest setting. The degree of deterioration differs across quantiles, and Section 5.6 examines that heterogeneity directly.

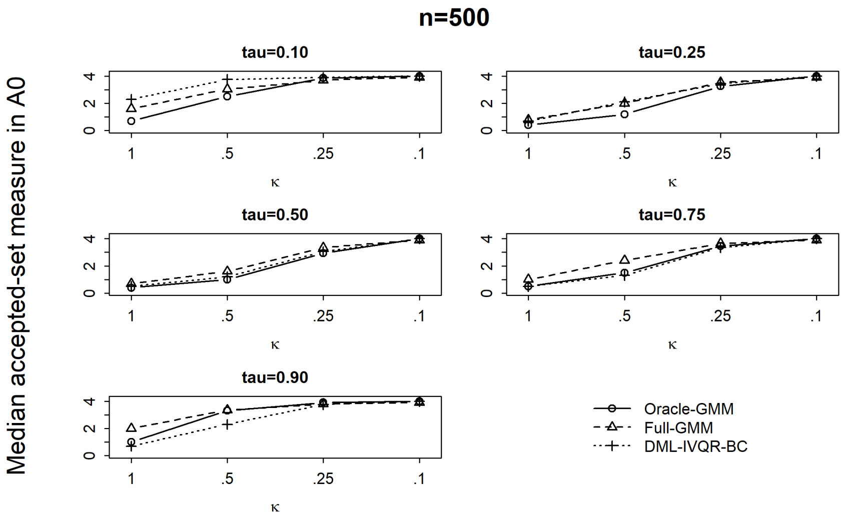


Figure 2. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, $n = 500$

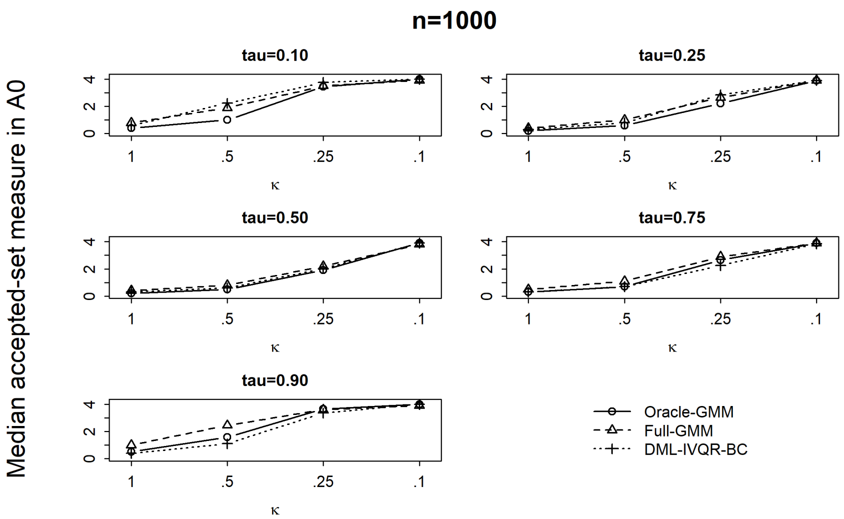


Figure 3. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, $n = 1000$

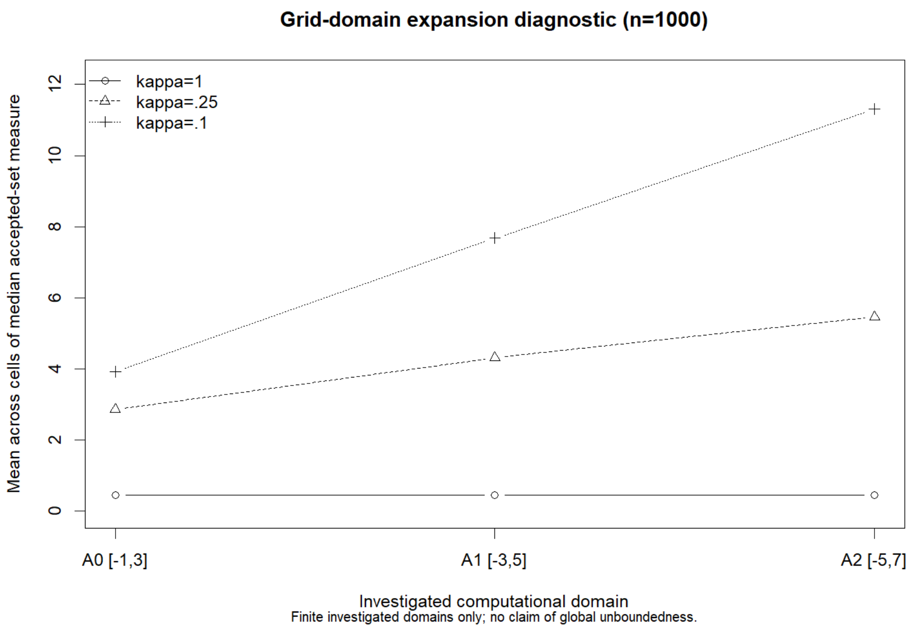


Figure 4. Sensitivity of the accepted set to expansion of the computational domain

Note: The investigated domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. The diagnostic records whether the accepted set stabilizes within these numerical domains; it does not establish global boundedness or unboundedness.

Figure 4 and Appendix Table C.1 show why the boundary of A_0 matters for interpreting the main measure. In the post-main diagnostic, expanding the investigated domain from A_0 to A_1 and then A_2 changes the accepted-set measure very little at $\kappa = 1$; the accepted sets are numerically localized within these domains. At $\kappa = 0.25$, expansion adds substantial accepted measure in some profiles. For example, in the median-quantile DML diagnostic, the median measure rises from 1.950 in A_0 to 2.500 in A_1 and 3.025 in A_2. The primary-domain measure can therefore understate how much non-rejection continues outside A_0. At $\kappa = 0.10$, the increase is much stronger: the corresponding DML sequence is 3.900, 7.500, and 10.325. Across the diagnostic profiles, the aggregate measure does not stabilize within the investigated domains at this weakest setting. The diagnostic does not establish global unboundedness. It shows only that acceptance continues beyond the finite numerical domains that were investigated.

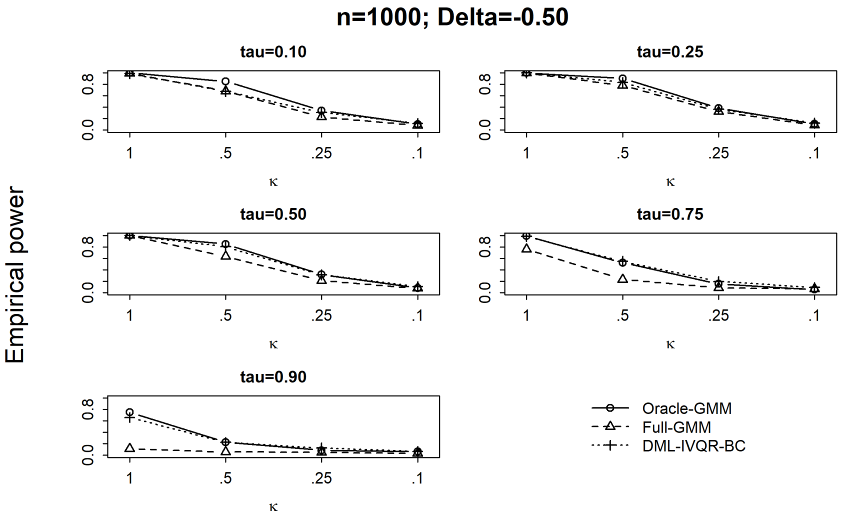
The econometric interpretation follows directly from the inverted test. When the moments are informative, moving a away from the true value tends to push $W_N(a)$ above the cutoff, so the test rejects many false candidates. When the instruments contain little information about treatment, the criterion changes less across candidate values and more of them remain below the cutoff. The resulting accepted set is less informative, but this is not automatically a failure of weak-identification-robust inference. A procedure can be designed to remain valid under weak identification while still providing little information about the parameter. The results here show that the data contain less information for ruling out alternatives; they do not show that identification disappears or that all real values are accepted. The widening accepted sets show that more candidate values survive the test as instrument relevance falls. Section 5.4 examines the same loss of discrimination directly through rejection probabilities at false parameter values.

5.4 Power

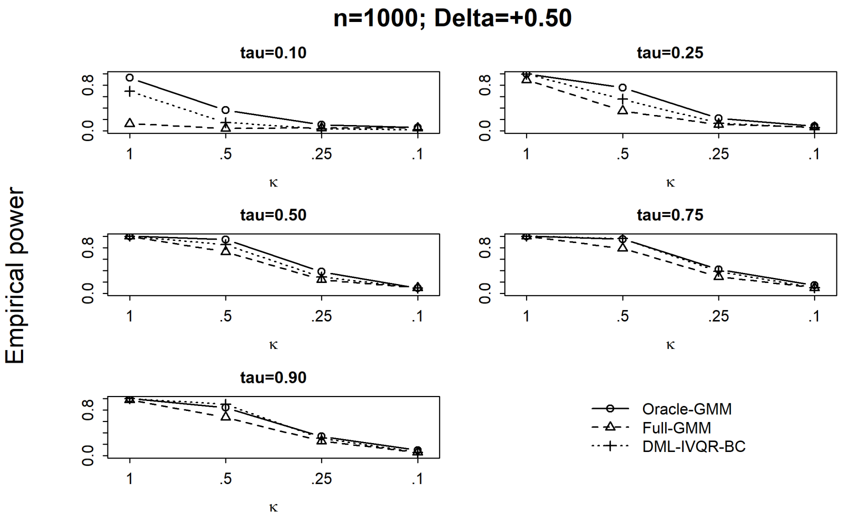
Power measures how often the test rejects a false candidate $a_{\Delta}(\tau) = \alpha_0(\tau) + \Delta$. Figure 5 shows the rejection probabilities for the moderate alternatives $\Delta = -0.50$ and $\Delta = +0.50$ when $n = 1000$. The broad pattern is a sharp decline in power as κ falls for Oracle-GMM, Full-GMM, and DML-IVQR-BC. A few sequences show small reversals when rejection probabilities are already low, so the decline is not mechanically monotone in every cell. The overall direction is nevertheless clear. With less informative instruments, a false treatment-effect value produces moment conditions that are harder to distinguish from those produced at the truth. The test consequently rejects fewer false candidates. This loss of discrimination is a statement about power, not evidence that the testing procedure is invalid.

The median quantile provides a simple numerical example. For $\Delta = -0.50$, Oracle power falls from 1.000 at $\kappa = 1$ to 0.850 at $\kappa = 0.50$, 0.316 at $\kappa = 0.25$, and 0.082 at $\kappa = 0.10$. DML follows a similar path, declining from 1.000 to 0.802, 0.312, and 0.104. Full-GMM falls from 0.998 in the strongest design to 0.076 in the weakest. The positive alternative gives the same main conclusion: Oracle power decreases from 1.000 to 0.092, while the weakest-design rejection probabilities are 0.098 for Full and 0.094 for DML. Thus, moderate false values are rejected with high probability at $\kappa = 1$, rejection remains substantial but lower at $\kappa = 0.50$, and power drops markedly by $\kappa = 0.25$. At $\kappa = 0.10$, probabilities near 0.08–0.10 show that the tests provide little discrimination against these particular false values even with $n = 1000$.

Appendix Table B.1 reports the full set of alternatives and both sample sizes. More distant alternatives are generally easier to reject than alternatives close to the truth, although power is not perfectly monotone in $|\Delta|$ in every finite-sample design. Rejection probabilities can also differ substantially between negative and positive offsets of the same size. Such asymmetry is possible because the candidate-specific IVQR moments and their finite-sample distributions need not be symmetric around the true value. Oracle often has the highest power when the



(a) $\Delta = -0.50$



(b) $\Delta = +0.50$

Figure 5. Rejection probability under moderate false alternatives, $n = 1000$

instruments are informative, and DML often performs competitively with Oracle and exceeds Full in many cells. These rankings vary with τ, Δ, and κ, however, and no estimator is uniformly most powerful. The common decline at low κ, including for Oracle, shows that weak relevance creates a problem beyond high-dimensional nuisance estimation. Power is generally higher at $n = 1000$ than at $n = 500$ for otherwise matched cells, although exceptions remain. Larger samples improve discrimination when there is usable instrument signal, but substantial power loss remains at $\kappa = 0.10$ for the simulated sample sizes.

These findings give the rejection-probability counterpart to Section 5.3. The widening accepted sets showed that more candidates survive test inversion as relevance falls; the power results show directly that false candidates are rejected less often. Both summaries therefore point to declining inferential discrimination. Power must still be separated from empirical size: every probability discussed here is evaluated at a false treatment-effect value, not at $\alpha_0(\tau)$. The size and direction of the power loss also vary across quantiles, and Section 5.6 considers this heterogeneity jointly with the other performance measures. Low power under weak relevance means that false values are difficult to reject; it does not by itself show a calibration failure at the truth. Section 5.5 turns to that separate question by examining whether the tests reject the true parameter at the intended frequency.

5.5 Coverage and Size

Coverage asks whether the true value $\alpha_0(\tau)$ survives the test, while empirical size is the complementary probability that the true value is rejected. The intended coverage is 0.95, corresponding to a 0.05 rejection rate under the implemented weak-inequality rule. The simulations reveal that finite-sample coverage is not uniformly close to this target. Several departures are substantial, including in cells with the strongest loading $\kappa = 1$. This finding is important because the procedure described in Section 3.6 uses an asymptotic chi-square reference distribution. Weak-identification robustness of that asymptotic construction does not imply exact calibration in samples of 500 or 1000 observations. The results instead reveal substantial finite-sample size distortions in some designs.

Table 3 illustrates the problem for $n = 1000$ and $\kappa = 1$. Full-GMM coverage is only 0.502 at $\tau = 0.10$ and 0.530 at $\tau = 0.90$, implying empirical sizes of 0.498 and 0.470. DML-IVQR-BC covers at rates of 0.632 at $\tau = 0.75$ and 0.498 at $\tau = 0.90$, with corresponding sizes of 0.368 and 0.502. Oracle-GMM also undercovers: its coverage is 0.860 at both $\tau = 0.10$ and $\tau = 0.75$. These oracle results show that high-dimensional nuisance estimation is not the

Table 3. Finite-sample coverage under the strongest instrument design

τ	**Estimator**	**Coverage**	**Size**	**MCSE**	$q_{.95}(W_{\text{true}})$
0.10	Oracle-GMM	0.860	0.140	0.016	9.723
0.10	Full-GMM	0.502	0.498	0.022	21.090
0.10	DML-IVQR-BC	0.968	0.032	0.008	5.256
0.25	Oracle-GMM	0.874	0.126	0.015	8.300
0.25	Full-GMM	0.790	0.210	0.018	10.940
0.25	DML-IVQR-BC	0.894	0.106	0.014	7.811
0.50	Oracle-GMM	0.872	0.128	0.015	8.511
0.50	Full-GMM	0.896	0.104	0.014	7.524
0.50	DML-IVQR-BC	0.776	0.224	0.019	11.157
0.75	Oracle-GMM	0.860	0.140	0.016	8.763
0.75	Full-GMM	0.790	0.210	0.018	12.489
0.75	DML-IVQR-BC	0.632	0.368	0.022	16.034
0.90	Oracle-GMM	0.904	0.096	0.013	7.395
0.90	Full-GMM	0.530	0.470	0.022	20.072
0.90	DML-IVQR-BC	0.498	0.502	0.022	18.682

Note: Results use $n = 1000$, $\kappa = 1$, and 500 Monte Carlo replications. The unchanged 95% reference cutoff is $q_{\chi^2_2}(.95) = 5.991$.

whole explanation. The distortions also vary sharply across quantiles. For example, DML coverage is 0.968 at $\tau = 0.10$, slightly above the nominal level, despite its severe undercoverage at upper quantiles. The empirical 95th percentile of $W_N(\alpha_0)$ in Table 3 helps describe the mechanism. The fixed asymptotic cutoff is 5.991. At $\tau = 0.10$, the Full-GMM percentile is 21.090, far above the cutoff, whereas the DML percentile is 5.256. At $\tau = 0.75$, the DML percentile reaches 16.034; at $\tau = 0.90$, the Full and DML values are 20.072 and 18.682. When this finite-sample percentile lies well above 5.991, the fixed cutoff is too low relative to the finite-sample distribution in that design cell. The test then rejects the truth too often and coverage falls below its target. Conversely, the DML percentile of 5.256 at $\tau = 0.10$ is consistent with its slight overcoverage. These empirical percentiles are diagnostics only; they were not used as replacement critical values.

Figures 6 and 7, together with Appendix Table D.1, show that the issue extends beyond the selected cells. Across the 120 design–estimator cells, observed coverage ranges from 0.442 to 0.976. The forensic audit in Appendix D finds that 54 coverage estimates have Monte Carlo 95% intervals that exclude 0.95. The deviations are therefore too widespread to dismiss as isolated simulation noise. The audit also confirms that coverage was evaluated directly at $\alpha_0(\tau)$,

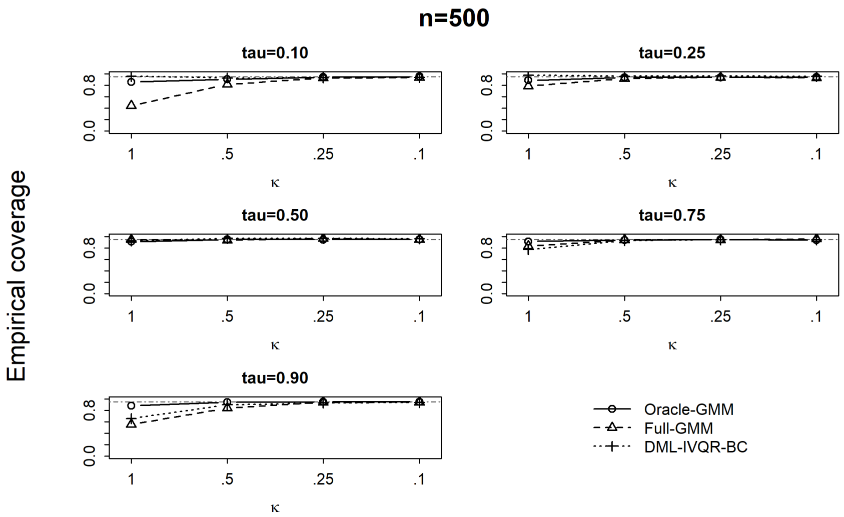


Figure 6. Empirical coverage across instrument-strength designs, $n = 500$

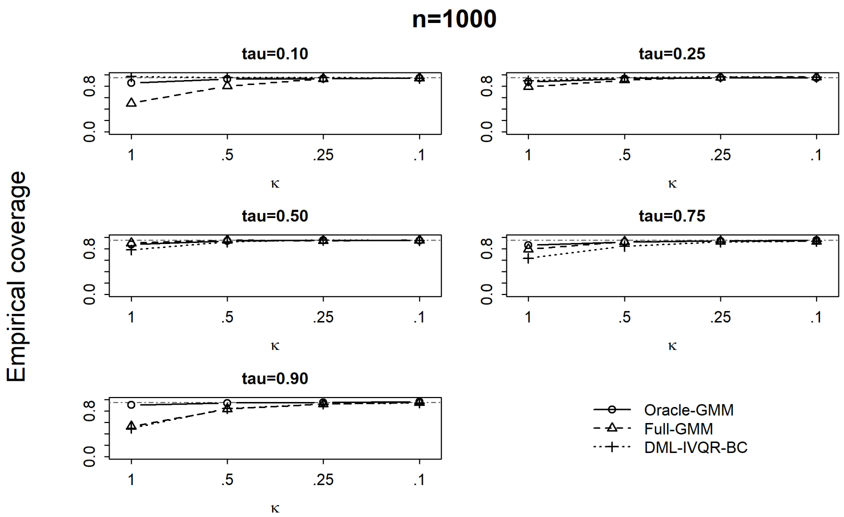


Figure 7. Empirical coverage across instrument-strength designs, $n = 1000$

independently of the confidence-region grid, and that missing or nonfinite statistics do not drive the pattern. Coverage generally moves upward as κ approaches 0.10, sometimes reaching or exceeding the target. Increasing the sample size does not uniformly move coverage toward 0.95 over the two values of n considered. Nor is any estimator uniformly best calibrated: Full-GMM has severe tail distortions in informative designs, DML has strong quantile-specific distortions of its own, and Oracle also undercovers in several cells.

Higher coverage under the weakest designs should not automatically be read as better inference. Coverage records whether the true value survives; informativeness records how many other values survive as well. Section 5.3 showed that accepted sets are often close to the full primary domain when $\kappa = 0.10$, and Section 5.4 showed that false values are then difficult to reject. A confidence region can therefore cover the truth frequently because the test rejects very little, while providing poor discrimination among alternative values. It can also undercover and be poorly calibrated, as several strong-loading cells demonstrate. These possibilities explain why coverage, accepted-set informativeness, and power must be considered separately.

These results do not contradict the asymptotic robustness result; they show that convergence to the reference distribution can be inaccurate at the simulated sample sizes. The calibration distortions are strongly quantile dependent. Section 5.6 therefore brings together point-estimation error, informativeness, power, and coverage to examine how the effect of weakening instruments differs across the conditional outcome distribution.

5.6 Heterogeneity Across Quantiles

The results across Figures 1, 2–3, 5, and 6–7 show that weakening instrument relevance does not affect every quantile in the same way. Point-estimation error, accepted-set informativeness, power, and finite-sample coverage all vary across the conditional outcome distribution. The common direction is that lower relevance makes estimation and inference more difficult, but the severity and shape of that deterioration vary substantially across quantiles. Tail quantiles often present greater difficulties than the median, yet there is no general rule that both tails are always worse for every estimator and metric. In particular, lower- and upper-tail behavior is frequently asymmetric. This heterogeneity remains visible at both sample sizes, although the size of particular differences changes with the design.

Point estimation provides one clear example. Figure 1 and Appendix Table A.1 show that RMSE is often larger in the tails than near the center, especially for Full-GMM. For $n = 1000$ and $\kappa = 1$, Full-GMM's RMSE is 0.100 at the median but 0.465 at the upper-tail quantile; its lower-tail error is also distinctly larger than at the median. This ordering is not identical across estimators or all strength settings, and the relative gaps change as κ falls. The accepted-set results show a related but not identical pattern. Under the strongest loading, the middle quantiles generally have smaller median measures, while the rate at which the measure expands differs across quantiles and estimators. By $\kappa = 0.10$, nearly all measures lie close to the maximum of the primary domain. Cross-quantile differences then become small relative to the common loss of informativeness, although the measures are not exactly equal. Appendix Table E.1 provides the full comparison.

Power makes the lower–upper asymmetry particularly visible. In the DML results for $n = 1000$ and $\kappa = 0.50$, power against the negative moderate alternative is 0.678 at the lower-tail quantile but 0.224 at the upper-tail quantile. For the positive alternative, the ordering reverses: the corresponding values are 0.148 and 0.900. Thus, which quantile retains more power depends partly on the direction of the false alternative; neither tail is uniformly more informative. Appendix Table B.1 shows that such asymmetry also varies across estimators and strength settings. Coverage displays a different form of quantile dependence. In the $n = 1000$, $\kappa = 1$ example from Table 3, DML coverage moves from 0.968 at the lower-tail quantile to 0.776 at the median and 0.498 at the upper-tail quantile. This striking sequence is specific to that design and should not be imposed on every value of κ or n. Full-GMM has severe distortions in both tails in the same table, while Oracle coverage is more stable across these quantiles but still falls below the nominal level in several cases. Finite-sample calibration therefore depends jointly on the quantile and the estimator.

Several features can contribute to these differences. The true structural treatment effect $\alpha_0(\tau)$ changes with τ, the quantile score itself depends on τ, and the effective information available in a finite sample can vary across parts of the outcome distribution. Control and instrument nuisance estimates may also behave differently across quantiles. These features make heterogeneity plausible, but the simulation does not decompose or causally isolate their separate contributions. The main implication is that the effect of weakening instruments cannot be summarized adequately by one average performance number. It depends on where in the conditional outcome distribution estimation and inference are conducted, which is why the design evaluates five quantiles rather than only the median. The next subsection therefore steps back from individual quantiles and summarizes what the three estimator comparisons imply for the thesis as a whole.

5.7 Synthesis Across Estimators

The comparison across estimators points first to a common result. As instrument relevance weakens, point-estimation error rises, accepted sets expand, and power against false values falls for Oracle-GMM, Full-GMM, and DML-IVQR-BC. Oracle-GMM is especially informative as a benchmark because it knows the designated oracle control set and avoids estimating the full high-dimensional nuisance model. Even with this advantage, its accuracy and inferential discrimination deteriorate sharply as κ falls. The main loss of performance therefore cannot be attributed only to variable selection or to the estimation of many nuisance coefficients. It also reflects the reduced information about the treatment effect supplied by the instruments. Larger samples generally improve point accuracy and discrimination, but the differences between the instrument-strength settings remain substantial at both sample sizes examined.

The estimator comparisons still show an important role for regularization. DML-IVQR-BC often records lower point-estimation error than Full-GMM and in several designs remains closer to the oracle benchmark. It also often rejects false candidates more frequently than Full-GMM and produces a smaller accepted-set measure in many cells. These patterns are consistent with regularization helping to control the difficulty of estimating a large nuisance model. They are not uniform: DML does not dominate Full in every design, and it does not reproduce Oracle performance throughout the experiment. Full-GMM is also not uniformly the least favorable procedure, although its use of the full control set without regularization is associated with larger errors, wider accepted sets, or lower power in many comparisons. The rankings depend on the quantile, instrument strength, and performance measure. Moreover, all three procedures deteriorate as relevance falls, and at the weakest setting their differences are often modest relative to the common loss of information. As Section 3.3 emphasizes, orthogonalization protects against nuisance-estimation error; it does not create information about the structural parameter.

Coverage must be interpreted separately from accuracy and informativeness. The implemented tests display substantial finite-sample size distortions in some cells, including designs with the strongest instrument loading. At very weak relevance, coverage often rises, but this occurs alongside accepted sets that cover most of the primary numerical domain and low rejection probabilities at false values. High coverage in such a setting is therefore not, by itself, evidence of informative inference. The results support two distinct conclusions: weaker relevance reduces the ability of every estimator to distinguish among candidate values, while the asymptotic chi-square approximation can also be poorly calibrated in particular finite-sample cells. Quantile heterogeneity further limits simple rankings. Full, DML, and Oracle each behave differently across parts of the conditional outcome distribution, so a comparison based only on the median would miss important tail behavior. No estimator has a uniform advantage across all quantiles and metrics.

Taken together, the Monte Carlo evidence answers the thesis question directly. Weakening instrument relevance has a major practical effect on both point estimation and the informativeness of DML-IVQR inference. The comparison is consistent with regularization reducing part of the additional cost associated with high-dimensional nuisance estimation, but it cannot prevent the loss of information about the structural parameter when the instruments contain little information about treatment. The oracle results make this distinction clear, while the comparison with Full-GMM shows that the way high-dimensional nuisance parameters are estimated can still matter in finite samples. These effects also vary substantially across quantiles. Chapter 6 interprets these findings in relation to weak-identification robustness, the role of DML, and the limitations of the simulation evidence.

6 Discussion

6.1 Main Findings and Weak Identification

Chapter 5 shows a broad deterioration in finite-sample performance as excluded-instrument relevance weakens. As κ falls, point-estimation error increases, accepted sets expand, and power against false values declines for Oracle-GMM, Full-GMM, and DML-IVQR-BC. The size of these changes differs across quantiles, but their common direction is clear. Each measure captures a different feature of performance, yet all three indicate that the moments become less able to distinguish the true treatment effect from alternative candidate values. In this discussion, identifying information means precisely this ability of the moments to separate the true parameter from false alternatives. The parameter κ controls excluded-instrument relevance in the simulation, while the resulting inferential behavior reveals the consequences of reduced identifying information. It is not itself a formal measure of IVQR identification.

This interpretation follows the identification logic of IVQR. Identification of $\alpha_0(\tau)$ depends on whether instrument-induced variation in the endogenous treatment changes the structural quantile restrictions enough to separate the true value from competing values. The robust approach of Chernozhukov and Hansen (2008) evaluates these restrictions at each hypothesized treatment effect. When the instrument variation is informative, moving the candidate a

away from $\alpha_0(\tau)$ tends to move the population moment away from zero, and the corresponding sample criterion $W_N(a)$ tends to reveal that disagreement more clearly. When relevance weakens, nearby and even relatively distant candidates can generate more similar moment behavior. The criterion may then change more slowly or become less discriminating across candidates, although this does not require every realized profile to be literally flat. The same mechanism helps explain three results together: point-estimation error rises, more candidate values remain in the inverted accepted set, and false candidates are rejected less often.

The oracle comparison provides strong evidence within this design that the common deterioration is not only a high-dimensional nuisance-estimation problem. Oracle-GMM knows the designated oracle control set and therefore avoids selecting or estimating the full set of 100 control coefficients. Nevertheless, its point-estimation error rises, its accepted sets expand, and its power falls as κ declines. Variable selection, regularization error, and estimation of the full nuisance vector therefore cannot by themselves explain the shared pattern across Oracle-GMM, Full-GMM, and DML-IVQR-BC. Reduced identifying information is a central driver. This comparison does not perfectly isolate identification because the estimators retain some differences in their nuisance and residualization procedures. It nevertheless places a useful limit on explanations based only on high dimensionality. The construction of Chen et al. (2021) addresses high-dimensional nuisance estimation through regularization and an orthogonal score. The thesis keeps that estimator structure largely fixed while varying relevance through κ, and the deterioration of DML-IVQR-BC shows that addressing nuisance estimation does not by itself make estimation informative when the instruments provide little information about treatment. Section 6.3 develops this distinction further.

The domain diagnostic supports the same interpretation without making a claim about the unrestricted confidence region. At $\kappa = 1$, expanding the numerical domain adds little accepted measure. At $\kappa = 0.25$, expanding the domain adds substantial accepted measure in some profiles, and at $\kappa = 0.10$ the accepted measure continues to grow through the largest investigated domain. As relevance weakens, the test is therefore increasingly unable to exclude candidate values farther from the truth within the ranges examined. The exercise establishes only what occurs inside those finite numerical domains; it does not prove that the theoretical confidence region is globally unbounded. Sample size changes the degree but not the central pattern. Moving from $n = 500$ to $n = 1000$ generally improves point accuracy and inferential discrimination, yet the deterioration across κ remains substantial at both sample sizes. Thus, within the finite samples considered, a larger sample does not eliminate the practical consequences of low relevance. The consequences also vary across quantiles: rates of RMSE deterioration, power asymmetries, accepted-set patterns, and coverage behavior are not uniform in

τ. This variation is consistent with identifying strength and finite-sample information differing across the conditional outcome distribution, but the simulation does not construct a formal quantile-specific measure of identification.

The weakest designs consequently display behavior associated with very limited identifying information, but the numerical exercise does not classify any finite value of κ as formally unidentified. This qualification is consistent with the purpose of weak-identification-robust inference. The procedure developed by Chernozhukov and Hansen (2008) is designed so that its asymptotic validity does not require strong identification. Such robustness does not ensure that the data sharply distinguish among parameter values: test inversion can remain valid while producing a wide or otherwise uninformative confidence region. The simulations provide a finite-sample illustration of this separation between robustness and informativeness. They also leave the calibration of the asymptotic reference distribution as a separate issue. The results therefore distinguish the availability of a weak-identification-robust testing procedure from the amount of information the data provide about the treatment effect. Section 6.2 examines that distinction directly by considering inferential validity, finite-sample calibration, and informativeness.

6.2 Inferential Validity versus Informativeness

Three concepts are needed to interpret the inference results. Asymptotic validity asks whether a test has its intended limiting behavior under the null hypothesis and the maintained assumptions, including when identification is weak. Finite-sample calibration asks whether the implemented test rejects the true parameter at approximately the intended 5% rate in samples of $n = 500$ or $n = 1000$. Informativeness asks how sharply the test distinguishes the true parameter from false alternatives. In this thesis, the accepted-set measure and power provide evidence about that distinction. These concepts are related but not equivalent. A procedure may have a valid limiting reference distribution but approximate it poorly in a moderate sample. It may also be well calibrated while accepting many false values, or it may be informative against false values while rejecting the truth too often. Inferential performance therefore cannot be summarized by any one of these dimensions.

The weak-identification-robust approach of Chernozhukov and Hansen (2008) does not rely on a strongly identified point estimator having an approximately normal distribution. It instead tests the moment restrictions separately at each candidate value and obtains a confidence region by retaining the candidates that are not rejected. Under the maintained assumptions, its asymptotic reference result is designed to remain valid under weak, partial, or failed identification. This is the meaning of weak-identification robustness used here. It does not promise exact coverage in a finite sample, a narrow confidence region, high power, or a precise point estimate. The preserved implementation compares $W_N(a)$ with the asymptotic χ^2_2 critical value of 5.991. Chapter 5 shows that this approximation can be poor at the sample sizes considered. For $n = 1000$ and $\kappa = 1$, Full-GMM coverage is 0.502 at $\tau = 0.10$ and 0.530 at $\tau = 0.90$, while DML-IVQR-BC coverage is 0.632 at $\tau = 0.75$ and 0.498 at $\tau = 0.90$. Oracle-GMM also undercovers at several quantiles. These results indicate that the finite-sample distribution of $W_N(\alpha_0(\tau))$ can differ substantially from its asymptotic reference distribution.

The empirical 95th percentile of $W_N(\alpha_0(\tau))$ helps diagnose this difference. When that percentile is well above 5.991, the asymptotic critical value is low relative to the finite-sample null distribution in the design cell. The test then rejects the true parameter too often and coverage falls below 0.95. This percentile is descriptive; it is not used as a replacement critical value, and the thesis evaluates the preserved procedure without constructing a correction. The forensic audit in Appendix D strengthens the interpretation of the coverage pattern. Coverage is evaluated directly at $\alpha_0(\tau)$ rather than inferred from the finite confidence-region grid, and missing or nonfinite evaluations do not generate the result. Moreover, 54 of the 120 design–estimator cells have Monte Carlo 95% intervals that exclude nominal 0.95 coverage. The distortions therefore cannot reasonably be dismissed as a grid artifact or isolated Monte Carlo noise. The Oracle results also show that high-dimensional nuisance estimation is not a necessary condition for substantial size distortion in this experiment. Oracle avoids the variable-selection problem faced by the high-dimensional procedures but still undercovers in several cells. At the same time, Oracle and the DML-IVQR construction of Chen et al. (2021) are not identical in every nuisance and residualization detail, so their calibration differences should not be assigned to one component alone. The finite-sample evidence therefore concerns the quality of the asymptotic approximation at the simulated sample sizes, not the validity of the limiting result itself. A limiting validity result and the speed at which a finite-sample distribution approaches its limit are different questions.

Calibration must in turn be separated from informativeness. Section 5.3 shows that accepted sets expand sharply as κ falls, while Section 5.4 shows that power against false alternatives declines. At $\kappa = 0.10$, the tests often reject very little. Coverage can then move toward or above 0.95 while many candidate values survive, including the truth. This is not automatically better inference. High coverage combined with a wide accepted set and low power means that the procedure rarely excludes either true or false values. Such a region can cover the truth frequently while providing little useful information about where the parameter lies. The opposite problem is also possible. When coverage is far below 0.95, the true parameter is rejected too often, which is a calibration failure rather than merely a lack of precision. Finite-sample inference can therefore be unsatisfactory through poor calibration, poor informativeness, or both. The coverage number alone does not reveal which problem is present.

Calibration also varies substantially across quantiles. Under the same sample size and instrument loading, DML-IVQR-BC can be close to nominal coverage at one quantile and severely undercovered at another. A single aggregate coverage rate would conceal this variation. More generally, calling a procedure weak-identification robust should not be read as saying that it is always precise or automatically well calibrated in moderate samples. Coverage and size describe treatment of the truth, accepted-set measures describe how many candidates survive, and power describes discrimination against specified false values; all three perspectives are needed. The exact finite-sample conditional pivotal method of Chernozhukov et al. (2009) is conceptually different from the asymptotic chi-square procedure used in this thesis. That method is not implemented here, so the present simulations neither evaluate nor reject it. These findings clarify what weak-identification robustness does and does not guarantee. They also raise a separate question about DML: which parts of observed performance reflect high-dimensional nuisance estimation, and which remain because instrument relevance is low? Section 6.3 turns to that distinction.

6.3 What DML Can and Cannot Address

DML-IVQR addresses the difficulty of estimating high-dimensional nuisance parameters before drawing conclusions about the low-dimensional treatment effect $\alpha_0(\tau)$. In this design, those nuisance parameters include the coefficients in the candidate-specific quantile regression and the coefficients used to residualize the instruments. Estimating many such coefficients freely can be noisy in a finite sample. The procedure of Chen et al. (2021) combines regularized nuisance estimation with a Neyman-orthogonal score. Regularization limits the complexity of the nuisance estimates and can stabilize estimation when many controls are available. Orthogonality then reduces the first-order effect that sufficiently small nuisance-estimation errors have on the population moment at the truth, following the general DML principle in Chernozhukov et al. (2018). This is a local robustness property. It does not mean that the nuisance parameters are estimated without error, that regularization has no cost, or that the score is immune to every source of finite-sample error. It also does not imply strong identification. Instrument validity and the structural restrictions come from the IVQR model, whereas DML concerns how nuisance quantities are estimated around that model.

The comparison with Full-GMM shows why this nuisance-estimation strategy can matter. Full-GMM includes all 100 controls but estimates its nuisance parameters without ℓ_1 regularization. Chapter 5 shows that DML-IVQR-BC often has lower RMSE than Full-GMM, produces smaller accepted-set measures in many designs, and rejects false candidates more frequently in many comparisons. These patterns are consistent with regularization and orthogonal high-dimensional nuisance estimation helping relative to the unregularized full-control benchmark. They do not establish that DML is uniformly preferable: the ordering changes across quantiles, instrument-strength settings, and performance measures. Nor does the experiment isolate regularization as the reason for these differences. DML and Full-GMM differ in other features of nuisance estimation and instrument residualization, including their treatment of the residualization intercept. The comparison therefore does not isolate the contribution of each implementation component. It shows instead that the way a high-dimensional nuisance problem is handled can have important finite-sample consequences.

Oracle-GMM helps delimit this explanation. It knows the designated oracle control set and therefore avoids the main problem of selecting or estimating the full high-dimensional nuisance model. Even with that advantage, Oracle-GMM exhibits rising point-estimation error, expanding accepted sets, and falling power as κ declines. The oracle comparison provides evidence within this design that a source of deterioration is common to all three estimators: weaker excluded-instrument relevance leaves the moments with less information about the treatment effect. Orthogonality changes how nuisance-estimation error enters the score; it does not change how much identifying variation the instruments supply. DML can therefore reduce sensitivity to nuisance-estimation error, but it cannot turn a weakly informative moment condition into a strongly informative one. It does not strengthen the instruments or restore information that the design has removed. This is also why the common deterioration should not be interpreted as evidence against regularization. It reflects a different problem from estimating many nuisance coefficients.

The term DML in this thesis refers to a particular preserved implementation. Its quantile-Lasso step uses the pivotal-penalty route described in Section 3.4, and its nuisance estimates and scores are computed on the same sample. It does not use sample splitting or cross-fitting, unlike many implementations of the general DML framework. This fact helps define the scope of the comparison without assigning a particular result to the penalty choice or the absence of cross-fitting. Relative performance also varies with τ. DML is close to Oracle or clearly better than Full-GMM in some quantile–design combinations, while in others the gaps differ or its finite-sample calibration is poor. There is consequently no universal ranking among DML, Full, and Oracle. As Section 6.2 explains, orthogonal nuisance robustness also does not guarantee exact finite-sample chi-square calibration. The observed DML coverage distortions at some quantiles reinforce that nuisance robustness and calibration are separate properties.

The simulation extension therefore highlights a boundary of what the DML component is designed to address. Chen et al. (2021) combine IVQR with high-dimensional nuisance estimation, orthogonal and debiased moments, and weak-identification-robust inference. The thesis preserves that basic estimator structure while deliberately reducing instrument relevance through κ. The resulting evidence complements their framework: robustness to nuisance estimation and asymptotic robustness to weak identification do not ensure informative inference when the instruments contain little information about treatment. In a high-dimensional IVQR setting, managing nuisance complexity and having adequate identifying information are separate requirements. A technically sophisticated estimator may still produce wide accepted regions and low power when the underlying moments are weakly informative. This distinction also clarifies the scope of the evidence. The simulation holds many features of the original design fixed and studies one specific DML-IVQR implementation. Section 6.4 therefore discusses the limitations of these conclusions and the dimensions that remain for future work.

6.4 Limitations

The conclusions come from one high-dimensional IVQR Monte Carlo design based on Chen et al. (2021). The extension changes excluded-instrument relevance through κ while holding the remaining structural features of the DGP fixed. This restriction is useful for internal interpretation because differences across κ are not accompanied by changes in the marginal treatment distribution, the endogeneity channel, or the structural quantile effect. It also limits external generalization. The results have not been established for other treatment equations, disturbance distributions, degrees of endogeneity, instrument dimensions, control-dependence structures, sparsity patterns, or structural quantile functions. In addition, κ is a simulation parameter controlling the loading on the excluded-instrument primitives, not a universal weak-instrument index or a general statistic for IVQR identification. The first-stage F-statistic and partial R^2 document the relative strength separation generated by the design, but they are descriptive calibrations. Labels such as high, low, and very low relevance therefore apply within this experiment and should not be translated into universal thresholds.

The simulation also studies a finite collection of sample sizes, quantiles, and alternatives. The values $n \in \{500, 1000\}$ reveal substantial finite-sample differences, but they do not characterize very small samples, much larger samples, or rates of convergence as n grows. Similarly, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$ is broad enough to display cross-quantile heterogeneity within the design, but it does not map behavior at every quantile or in more extreme tails. Power is evaluated only at the prespecified offsets $\Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}$. The observed asymmetries characterize those evaluated alternatives, but they do not describe the complete continuous power function. Finally, the main experiment uses 500 Monte Carlo replications per design cell. The resulting summaries remain subject to Monte Carlo sampling uncertainty and should not be read as population quantities. Appendix D accounts for this uncertainty explicitly when assessing coverage through Monte Carlo confidence intervals. That uncertainty does not erase the large and repeated patterns documented in Chapter 5.

The results characterize the specific preserved DML-IVQR-BC implementation used in the thesis rather than every possible DML-IVQR procedure. Its nuisance parameters and scores are evaluated on the same sample without cross-fitting; it uses the preserved pivotal penalty, the fitted-normal density proxy, and the particular regularized instrument-residualization step described in Chapter 3. These are implementation choices that define the experiment, not errors. Other nuisance learners or DML implementations could be compared in future work, but the current design does not establish how they would perform and does not imply that cross-fitting would necessarily improve the results. The estimator comparison also is not a clean decomposition of individual components. Oracle-GMM, Full-GMM, and DML-IVQR-BC share the structural target and instruments, but they differ in regularization, their residualization procedures, and their treatment of the residualization intercept. Comparisons between DML and Full therefore cannot isolate the effect of regularization alone. They support conclusions about the procedures as implemented, not a causal attribution to one computational choice.

The accepted-set analysis has a separate numerical limitation. The main results evaluate $W_N(a)$ on the prespecified finite domain A_0 with grid spacing $h = 0.1$. The accepted-set measure is consequently a grid-based approximation within that domain, not the unrestricted length of a theoretical confidence region. Because the calculation does not interpolate critical-value crossings, fine features of the criterion between adjacent grid points may not be represented exactly. Earlier numerical checks supported retaining this resolution for the main analysis, but they do not change the approximate nature of the measure. The post-main diagnostic extends the domain through A_1 and A_2, yet those domains also remain finite. It can therefore establish neither global boundedness nor global unboundedness, and it cannot show acceptance over the entire real line. The diagnostic is also deliberately narrower than the main experiment: it uses $n = 1000$, 100 diagnostic replications, $\kappa \in \{1, 0.25, 0.10\}$, all five quantiles, and all three estimators. It excludes $n = 500$ and $\kappa = 0.50$ because its purpose is to diagnose sensitivity to the numerical domain, not to repeat the complete Monte Carlo design.

The coverage conclusions are specific to the preserved asymptotic chi-square procedure. The thesis documents substantial finite-sample size distortion in some cells, but it does not evaluate bootstrap critical values, other finite-sample corrections, or the exact conditional pivotal method of Chernozhukov et al. (2009). Those approaches are outside the current design, and the results do not establish whether any of them would remove the observed distortions. The thesis also contains no empirical application. A Monte Carlo setting permits controlled variation in instrument relevance and makes the true $\alpha_0(\tau)$ known, allowing direct evaluation of error, coverage, and power. It does not show how frequently the same patterns arise in a particular real-world setting. Future work could consider alternative DGPs, additional sample sizes, different instrument structures, other nuisance-learning methods, alternative DML implementations, finite-sample inference procedures, or empirical applications without presuming that any extension would improve performance. These boundaries define the scope of the conclusions. Within the preserved DGP and implementation, reducing instrument relevance clearly produces larger point-estimation errors, less informative accepted sets, lower power, and substantial quantile heterogeneity, while finite-sample calibration of the asymptotic test is imperfect in some design cells.

7 Conclusion

This thesis asks how systematically weakening excluded-instrument relevance affects the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how these effects differ across quantiles. The Monte Carlo experiment extends the framework of Chen et al. (2021) by changing instrument relevance through κ while preserving the main structural features of their design. The marginal treatment distribution, endogeneity channel, outcome equation, structural quantile effect, controls, and observed instrument construction remain fixed across relevance settings. Within this controlled comparison, the answer is clear: as excluded-instrument relevance weakens, point-estimation error increases substantially, inverted accepted sets become less informative, and power against false treatment-effect values declines. These patterns occur for Oracle-GMM, Full-GMM, and DML-IVQR-BC, although their magnitude differs across quantiles and estimators.

The point-estimation evidence shows that RMSE rises as κ falls at every quantile and sample size for all three estimators. Because Oracle-GMM knows the designated oracle control set and avoids estimating the full high-dimensional nuisance model, its deterioration is especially informative. High-dimensional nuisance estimation cannot by itself explain a pattern shared by the oracle and both full-control procedures; reduced information supplied by the instruments is a central part of the problem. The inference profiles tell the same story from a different perspective. As relevance weakens, more candidate values survive the inverted test, the grid-based accepted-set measure within the primary computational domain grows, and false alternatives are rejected less frequently. At $\kappa = 0.10$, every estimator–quantile combination has a median accepted-set measure of at least 3.8 within a domain whose maximum measure is four. The domain-expansion diagnostic further shows that acceptance continues beyond the primary domain and that the aggregate measure does not stabilize within the largest investigated domain at this setting. The warranted conclusion is that the accepted set is not numerically localized within the investigated domains in many weakest-relevance profiles, not that the theoretical confidence region is globally unbounded or that identification necessarily fails. Power declines sharply in the broad comparison, with some small finite-sample reversals when rejection probabilities are already low.

Finite-sample calibration provides a separate qualification. The candidate-specific procedure follows the weak-identification-robust logic of Chernozhukov and Hansen (2008), but the implemented asymptotic chi-square test is not uniformly well calibrated in the simulated samples. Coverage ranges from 0.442 to 0.976 across the 120 design–estimator cells, and the forensic audit finds that 54 of the 120 coverage estimates have Monte Carlo 95% intervals excluding nominal 0.95. Substantial undercoverage occurs in some cells even at $\kappa = 1$, and Oracle-GMM also undercovers in several designs. These calibration problems therefore cannot be attributed only to regularized or high-dimensional nuisance estimation. They concern how accurately the finite-sample distribution is approximated by its asymptotic reference distribution; they do not invalidate the limiting weak-identification-robust theory. At very weak relevance, coverage often moves upward, but this is not automatically better inference. Frequent inclusion of the truth can coexist with accepted sets covering most of the investigated domain and with little power against false values. Calibration and informativeness must consequently be evaluated separately.

The estimator comparison refines this interpretation without changing it. Oracle-GMM generally supplies the strongest benchmark, but it is not an upper bound and also deteriorates sharply as relevance falls. DML-IVQR-BC often records lower RMSE, smaller accepted-set measures, and higher power than Full-GMM, but it does not dominate Full-GMM uniformly. These differences cannot be assigned entirely to regularization because the procedures also differ in nuisance estimation and instrument residualization. The defensible conclusion is that the treatment of high-dimensional nuisance parameters matters in finite samples, while DML cannot replace identifying information that the instruments do not supply. Increasing the sample size from 500 to 1000 generally improves point accuracy, inferential discrimination, and accepted-set localization, but it does not eliminate the deterioration associated with very low relevance at the sample sizes examined. Performance is also heterogeneous across τ: lower-tail, median, and upper-tail results differ in RMSE, power, accepted-set behavior, and coverage, and power can be asymmetric between positive and negative alternatives. No tail is uniformly the most difficult. Evaluating several quantiles is therefore necessary to characterize weak-relevance behavior in this design.

The contribution is systematic finite-sample evidence from a controlled instrument-relevance experiment. The thesis jointly evaluates point accuracy, accepted-set informativeness, power, coverage and size, and heterogeneity across quantiles for an existing methodology. It does not propose a new structural IVQR model, DML estimator, orthogonal score, identification theorem, or weak-identification-robust test. Its scope is correspondingly limited. The evidence comes from one preserved Monte Carlo DGP, two sample sizes, and one same-sample DML-IVQR implementation without cross-fitting. Accepted-set summaries are evaluated on finite numerical domains, inference uses the asymptotic chi-square reference distribution, and the thesis contains no empirical application. Future research could examine alternative DGPs and instrument structures, larger samples, other nuisance learners, cross-fitted DML-IVQR implementations, bootstrap or other finite-sample inference procedures, and empirical applications. Such comparisons would clarify how broadly the documented patterns extend, without presuming that any particular alternative must improve performance.

The central implication is that robustness has boundaries that should be stated plainly. DML addresses sensitivity to high-dimensional nuisance estimation, while candidate-specific weak-identification-robust testing addresses asymptotic inference without requiring strong identification. Neither creates identifying information. When instruments contain little information about treatment, an orthogonal and identification-robust procedure may still estimate imprecisely and exclude few false structural quantile effects. A robust method should reveal that lack of information rather than manufacture precision. At the same time, finite-sample calibration of an asymptotic procedure must be assessed rather than assumed. High-dimensional nuisance robustness and weak-identification robustness are valuable properties, but neither is a substitute for instrument relevance.

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Appendix A Full Monte Carlo Results

Table A.1. Full point-estimation performance

n	τ	κ	Estimator	Bias	MAE	RMSE
500	0.10	1.00	Oracle-GMM	-0.115	0.155	0.200
500	0.10	1.00	Full-GMM	-0.684	0.693	0.789
500	0.10	1.00	DML-IVQR-BC	-0.204	0.296	0.453
500	0.10	0.50	Oracle-GMM	-0.285	0.394	0.642
500	0.10	0.50	Full-GMM	-0.901	0.960	1.219
500	0.10	0.50	DML-IVQR-BC	-0.827	0.926	1.335
500	0.10	0.25	Oracle-GMM	-0.705	0.895	1.276
500	0.10	0.25	Full-GMM	-1.155	1.283	1.612
500	0.10	0.25	DML-IVQR-BC	-1.059	1.207	1.585
500	0.10	0.10	Oracle-GMM	-0.847	1.097	1.457
500	0.10	0.10	Full-GMM	-1.272	1.396	1.707
500	0.10	0.10	DML-IVQR-BC	-1.065	1.258	1.618
500	0.25	1.00	Oracle-GMM	-0.048	0.097	0.123
500	0.25	1.00	Full-GMM	-0.218	0.246	0.295
500	0.25	1.00	DML-IVQR-BC	0.006	0.118	0.150
500	0.25	0.50	Oracle-GMM	-0.083	0.213	0.294
500	0.25	0.50	Full-GMM	-0.272	0.423	0.553
500	0.25	0.50	DML-IVQR-BC	-0.156	0.311	0.486
500	0.25	0.25	Oracle-GMM	-0.264	0.579	0.831
500	0.25	0.25	Full-GMM	-0.502	0.844	1.088
500	0.25	0.25	DML-IVQR-BC	-0.576	0.801	1.092
500	0.25	0.10	Oracle-GMM	-0.312	0.833	1.071
500	0.25	0.10	Full-GMM	-0.549	1.006	1.241
500	0.25	0.10	DML-IVQR-BC	-0.601	0.987	1.245
500	0.50	1.00	Oracle-GMM	-0.010	0.075	0.102
500	0.50	1.00	Full-GMM	0.009	0.131	0.169
500	0.50	1.00	DML-IVQR-BC	0.062	0.100	0.131
500	0.50	0.50	Oracle-GMM	-0.007	0.164	0.228
500	0.50	0.50	Full-GMM	0.011	0.303	0.400
500	0.50	0.50	DML-IVQR-BC	0.010	0.183	0.247
500	0.50	0.25	Oracle-GMM	-0.033	0.442	0.634
500	0.50	0.25	Full-GMM	0.008	0.640	0.827
500	0.50	0.25	DML-IVQR-BC	-0.199	0.499	0.715
500	0.50	0.10	Oracle-GMM	-0.060	0.818	1.027
500	0.50	0.10	Full-GMM	0.036	0.949	1.129
500	0.50	0.10	DML-IVQR-BC	-0.268	0.848	1.046

Continued on next page

TableA.1. Full point-estimation performance (Continued)

n	τ	κ	**Estimator**	**Bias**	**MAE**	**RMSE**
500	0.75	1.00	Oracle-GMM	0.034	0.097	0.126
500	0.75	1.00	Full-GMM	0.231	0.266	0.327
500	0.75	1.00	DML-IVQR-BC	0.125	0.144	0.175
500	0.75	0.50	Oracle-GMM	0.117	0.251	0.410
500	0.75	0.50	Full-GMM	0.325	0.499	0.686
500	0.75	0.50	DML-IVQR-BC	0.118	0.238	0.329
500	0.75	0.25	Oracle-GMM	0.300	0.613	0.903
500	0.75	0.25	Full-GMM	0.417	0.798	1.030
500	0.75	0.25	DML-IVQR-BC	0.106	0.562	0.793
500	0.75	0.10	Oracle-GMM	0.260	0.868	1.123
500	0.75	0.10	Full-GMM	0.488	1.048	1.288
500	0.75	0.10	DML-IVQR-BC	0.159	0.828	1.055
500	0.90	1.00	Oracle-GMM	0.111	0.179	0.246
500	0.90	1.00	Full-GMM	0.736	0.751	0.880
500	0.90	1.00	DML-IVQR-BC	0.244	0.254	0.306
500	0.90	0.50	Oracle-GMM	0.348	0.510	0.822
500	0.90	0.50	Full-GMM	1.005	1.087	1.362
500	0.90	0.50	DML-IVQR-BC	0.286	0.373	0.539
500	0.90	0.25	Oracle-GMM	0.594	0.842	1.237
500	0.90	0.25	Full-GMM	1.071	1.230	1.551
500	0.90	0.25	DML-IVQR-BC	0.378	0.663	0.975
500	0.90	0.10	Oracle-GMM	0.812	1.097	1.456
500	0.90	0.10	Full-GMM	1.266	1.416	1.730
500	0.90	0.10	DML-IVQR-BC	0.648	0.968	1.322
1000	0.10	1.00	Oracle-GMM	-0.054	0.093	0.121
1000	0.10	1.00	Full-GMM	-0.329	0.337	0.387
1000	0.10	1.00	DML-IVQR-BC	-0.005	0.117	0.150
1000	0.10	0.50	Oracle-GMM	-0.101	0.194	0.287
1000	0.10	0.50	Full-GMM	-0.407	0.465	0.630
1000	0.10	0.50	DML-IVQR-BC	-0.159	0.306	0.482
1000	0.10	0.25	Oracle-GMM	-0.384	0.566	0.934
1000	0.10	0.25	Full-GMM	-0.773	0.914	1.242
1000	0.10	0.25	DML-IVQR-BC	-0.681	0.844	1.229
1000	0.10	0.10	Oracle-GMM	-0.747	0.975	1.376
1000	0.10	0.10	Full-GMM	-1.074	1.251	1.582
1000	0.10	0.10	DML-IVQR-BC	-0.932	1.146	1.514
1000	0.25	1.00	Oracle-GMM	-0.018	0.063	0.077
1000	0.25	1.00	Full-GMM	-0.098	0.126	0.153
1000	0.25	1.00	DML-IVQR-BC	0.044	0.078	0.097
1000	0.25	0.50	Oracle-GMM	-0.029	0.113	0.147

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TableA.1. Full point-estimation performance (Continued)

n	τ	κ	**Estimator**	**Bias**	**MAE**	**RMSE**
1000	0.25	0.50	Full-GMM	-0.114	0.201	0.259
1000	0.25	0.50	DML-IVQR-BC	-0.000	0.145	0.198
1000	0.25	0.25	Oracle-GMM	-0.131	0.332	0.550
1000	0.25	0.25	Full-GMM	-0.232	0.486	0.706
1000	0.25	0.25	DML-IVQR-BC	-0.199	0.405	0.647
1000	0.25	0.10	Oracle-GMM	-0.235	0.696	0.954
1000	0.25	0.10	Full-GMM	-0.410	0.878	1.136
1000	0.25	0.10	DML-IVQR-BC	-0.490	0.862	1.157
1000	0.50	1.00	Oracle-GMM	0.007	0.047	0.072
1000	0.50	1.00	Full-GMM	0.007	0.072	0.100
1000	0.50	1.00	DML-IVQR-BC	0.070	0.079	0.104
1000	0.50	0.50	Oracle-GMM	0.015	0.093	0.134
1000	0.50	0.50	Full-GMM	0.014	0.151	0.199
1000	0.50	0.50	DML-IVQR-BC	0.059	0.119	0.160
1000	0.50	0.25	Oracle-GMM	0.045	0.281	0.431
1000	0.50	0.25	Full-GMM	0.029	0.372	0.500
1000	0.50	0.25	DML-IVQR-BC	-0.003	0.306	0.448
1000	0.50	0.10	Oracle-GMM	-0.044	0.735	0.945
1000	0.50	0.10	Full-GMM	-0.012	0.840	1.018
1000	0.50	0.10	DML-IVQR-BC	-0.195	0.779	0.991
1000	0.75	1.00	Oracle-GMM	0.025	0.068	0.085
1000	0.75	1.00	Full-GMM	0.126	0.151	0.186
1000	0.75	1.00	DML-IVQR-BC	0.105	0.113	0.133
1000	0.75	0.50	Oracle-GMM	0.055	0.134	0.189
1000	0.75	0.50	Full-GMM	0.175	0.244	0.332
1000	0.75	0.50	DML-IVQR-BC	0.111	0.149	0.203
1000	0.75	0.25	Oracle-GMM	0.186	0.381	0.604
1000	0.75	0.25	Full-GMM	0.304	0.517	0.745
1000	0.75	0.25	DML-IVQR-BC	0.132	0.336	0.516
1000	0.75	0.10	Oracle-GMM	0.256	0.788	1.058
1000	0.75	0.10	Full-GMM	0.422	0.907	1.147
1000	0.75	0.10	DML-IVQR-BC	0.136	0.753	0.982
1000	0.90	1.00	Oracle-GMM	0.072	0.110	0.148
1000	0.90	1.00	Full-GMM	0.387	0.397	0.465
1000	0.90	1.00	DML-IVQR-BC	0.183	0.186	0.215
1000	0.90	0.50	Oracle-GMM	0.211	0.295	0.540
1000	0.90	0.50	Full-GMM	0.584	0.629	0.871
1000	0.90	0.50	DML-IVQR-BC	0.226	0.260	0.396
1000	0.90	0.25	Oracle-GMM	0.491	0.658	1.038
1000	0.90	0.25	Full-GMM	0.901	1.058	1.389

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TableA.1. Full point-estimation performance (Continued)

n	τ	κ	Estimator	Bias	MAE	RMSE
1000	0.90	0.25	DML-IVQR-BC	0.417	0.580	0.927
1000	0.90	0.10	Oracle-GMM	0.653	0.948	1.350
1000	0.90	0.10	Full-GMM	1.050	1.236	1.583
1000	0.90	0.10	DML-IVQR-BC	0.519	0.831	1.191

Note: All 120 Monte Carlo design cells. Bias follows $\alpha_0 - \widehat{\alpha}$; each cell uses 500 replications.

Appendix B Detailed Power Results

Table B.1. Full power results

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.10	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
500	0.10	1.00	Oracle-GMM	-0.50	0.968	0.032	500	0
500	0.10	1.00	Oracle-GMM	-0.25	0.694	0.306	500	0
500	0.10	1.00	Oracle-GMM	0.25	0.120	0.880	500	0
500	0.10	1.00	Oracle-GMM	0.50	0.482	0.518	500	0
500	0.10	1.00	Oracle-GMM	1.00	0.930	0.070	500	0
500	0.10	1.00	Full-GMM	-1.00	0.970	0.030	500	0
500	0.10	1.00	Full-GMM	-0.50	0.898	0.102	500	0
500	0.10	1.00	Full-GMM	-0.25	0.770	0.230	500	0
500	0.10	1.00	Full-GMM	0.25	0.248	0.752	500	0
500	0.10	1.00	Full-GMM	0.50	0.050	0.950	500	0
500	0.10	1.00	Full-GMM	1.00	0.134	0.866	500	0
500	0.10	1.00	DML-IVQR-BC	-1.00	0.998	0.002	500	0
500	0.10	1.00	DML-IVQR-BC	-0.50	0.744	0.256	500	0
500	0.10	1.00	DML-IVQR-BC	-0.25	0.262	0.738	500	0
500	0.10	1.00	DML-IVQR-BC	0.25	0.028	0.972	500	0
500	0.10	1.00	DML-IVQR-BC	0.50	0.058	0.942	500	0
500	0.10	1.00	DML-IVQR-BC	1.00	0.184	0.816	500	0
500	0.10	0.50	Oracle-GMM	-1.00	0.794	0.206	500	0
500	0.10	0.50	Oracle-GMM	-0.50	0.532	0.468	500	0
500	0.10	0.50	Oracle-GMM	-0.25	0.292	0.708	500	0
500	0.10	0.50	Oracle-GMM	0.25	0.090	0.910	500	0
500	0.10	0.50	Oracle-GMM	0.50	0.144	0.856	500	0
500	0.10	0.50	Oracle-GMM	1.00	0.308	0.692	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.10	0.50	Full-GMM	-1.00	0.558	0.442	500	0
500	0.10	0.50	Full-GMM	-0.50	0.438	0.562	500	0
500	0.10	0.50	Full-GMM	-0.25	0.310	0.690	500	0
500	0.10	0.50	Full-GMM	0.25	0.108	0.892	500	0
500	0.10	0.50	Full-GMM	0.50	0.068	0.932	500	0
500	0.10	0.50	Full-GMM	1.00	0.082	0.918	500	0
500	0.10	0.50	DML-IVQR-BC	-1.00	0.834	0.166	500	0
500	0.10	0.50	DML-IVQR-BC	-0.50	0.406	0.594	500	0
500	0.10	0.50	DML-IVQR-BC	-0.25	0.174	0.826	500	0
500	0.10	0.50	DML-IVQR-BC	0.25	0.038	0.962	500	0
500	0.10	0.50	DML-IVQR-BC	0.50	0.042	0.958	500	0
500	0.10	0.50	DML-IVQR-BC	1.00	0.032	0.968	500	0
500	0.10	0.25	Oracle-GMM	-1.00	0.246	0.754	500	0
500	0.10	0.25	Oracle-GMM	-0.50	0.188	0.812	500	0
500	0.10	0.25	Oracle-GMM	-0.25	0.112	0.888	500	0
500	0.10	0.25	Oracle-GMM	0.25	0.050	0.950	500	0
500	0.10	0.25	Oracle-GMM	0.50	0.074	0.926	500	0
500	0.10	0.25	Oracle-GMM	1.00	0.104	0.896	500	0
500	0.10	0.25	Full-GMM	-1.00	0.168	0.832	500	0
500	0.10	0.25	Full-GMM	-0.50	0.134	0.866	500	0
500	0.10	0.25	Full-GMM	-0.25	0.106	0.894	500	0
500	0.10	0.25	Full-GMM	0.25	0.052	0.948	500	0
500	0.10	0.25	Full-GMM	0.50	0.050	0.950	500	0
500	0.10	0.25	Full-GMM	1.00	0.026	0.974	500	0
500	0.10	0.25	DML-IVQR-BC	-1.00	0.432	0.568	500	0
500	0.10	0.25	DML-IVQR-BC	-0.50	0.224	0.776	500	0
500	0.10	0.25	DML-IVQR-BC	-0.25	0.116	0.884	500	0
500	0.10	0.25	DML-IVQR-BC	0.25	0.046	0.954	500	0
500	0.10	0.25	DML-IVQR-BC	0.50	0.036	0.964	500	0
500	0.10	0.25	DML-IVQR-BC	1.00	0.036	0.964	500	0
500	0.10	0.10	Oracle-GMM	-1.00	0.062	0.938	500	0
500	0.10	0.10	Oracle-GMM	-0.50	0.090	0.910	500	0
500	0.10	0.10	Oracle-GMM	-0.25	0.062	0.938	500	0
500	0.10	0.10	Oracle-GMM	0.25	0.052	0.948	500	0
500	0.10	0.10	Oracle-GMM	0.50	0.070	0.930	500	0
500	0.10	0.10	Oracle-GMM	1.00	0.056	0.944	500	0
500	0.10	0.10	Full-GMM	-1.00	0.056	0.944	500	0
500	0.10	0.10	Full-GMM	-0.50	0.052	0.948	500	0
500	0.10	0.10	Full-GMM	-0.25	0.072	0.928	500	0
500	0.10	0.10	Full-GMM	0.25	0.070	0.930	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.10	0.10	Full-GMM	0.50	0.058	0.942	500	0
500	0.10	0.10	Full-GMM	1.00	0.038	0.962	500	0
500	0.10	0.10	DML-IVQR-BC	-1.00	0.194	0.806	500	0
500	0.10	0.10	DML-IVQR-BC	-0.50	0.136	0.864	500	0
500	0.10	0.10	DML-IVQR-BC	-0.25	0.100	0.900	500	0
500	0.10	0.10	DML-IVQR-BC	0.25	0.058	0.942	500	0
500	0.10	0.10	DML-IVQR-BC	0.50	0.062	0.938	500	0
500	0.10	0.10	DML-IVQR-BC	1.00	0.054	0.946	500	0
500	0.25	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
500	0.25	1.00	Oracle-GMM	-0.50	0.994	0.006	500	0
500	0.25	1.00	Oracle-GMM	-0.25	0.818	0.182	500	0
500	0.25	1.00	Oracle-GMM	0.25	0.450	0.550	500	0
500	0.25	1.00	Oracle-GMM	0.50	0.930	0.070	500	0
500	0.25	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
500	0.25	1.00	Full-GMM	-1.00	1.000	0.000	500	0
500	0.25	1.00	Full-GMM	-0.50	0.928	0.072	500	0
500	0.25	1.00	Full-GMM	-0.25	0.692	0.308	500	0
500	0.25	1.00	Full-GMM	0.25	0.052	0.948	500	0
500	0.25	1.00	Full-GMM	0.50	0.242	0.758	500	0
500	0.25	1.00	Full-GMM	1.00	0.908	0.092	500	0
500	0.25	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
500	0.25	1.00	DML-IVQR-BC	-0.50	0.940	0.060	500	0
500	0.25	1.00	DML-IVQR-BC	-0.25	0.402	0.598	500	0
500	0.25	1.00	DML-IVQR-BC	0.25	0.248	0.752	500	0
500	0.25	1.00	DML-IVQR-BC	0.50	0.668	0.332	500	0
500	0.25	1.00	DML-IVQR-BC	1.00	0.980	0.020	500	0
500	0.25	0.50	Oracle-GMM	-1.00	0.906	0.094	500	0
500	0.25	0.50	Oracle-GMM	-0.50	0.618	0.382	500	0
500	0.25	0.50	Oracle-GMM	-0.25	0.330	0.670	500	0
500	0.25	0.50	Oracle-GMM	0.25	0.138	0.862	500	0
500	0.25	0.50	Oracle-GMM	0.50	0.330	0.670	500	0
500	0.25	0.50	Oracle-GMM	1.00	0.654	0.346	500	0
500	0.25	0.50	Full-GMM	-1.00	0.762	0.238	500	0
500	0.25	0.50	Full-GMM	-0.50	0.448	0.552	500	0
500	0.25	0.50	Full-GMM	-0.25	0.242	0.758	500	0
500	0.25	0.50	Full-GMM	0.25	0.054	0.946	500	0
500	0.25	0.50	Full-GMM	0.50	0.092	0.908	500	0
500	0.25	0.50	Full-GMM	1.00	0.308	0.692	500	0
500	0.25	0.50	DML-IVQR-BC	-1.00	0.930	0.070	500	0
500	0.25	0.50	DML-IVQR-BC	-0.50	0.530	0.470	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.25	0.50	DML-IVQR-BC	-0.25	0.170	0.830	500	0
500	0.25	0.50	DML-IVQR-BC	0.25	0.042	0.958	500	0
500	0.25	0.50	DML-IVQR-BC	0.50	0.144	0.856	500	0
500	0.25	0.50	DML-IVQR-BC	1.00	0.316	0.684	500	0
500	0.25	0.25	Oracle-GMM	-1.00	0.370	0.630	500	0
500	0.25	0.25	Oracle-GMM	-0.50	0.224	0.776	500	0
500	0.25	0.25	Oracle-GMM	-0.25	0.106	0.894	500	0
500	0.25	0.25	Oracle-GMM	0.25	0.072	0.928	500	0
500	0.25	0.25	Oracle-GMM	0.50	0.100	0.900	500	0
500	0.25	0.25	Oracle-GMM	1.00	0.178	0.822	500	0
500	0.25	0.25	Full-GMM	-1.00	0.292	0.708	500	0
500	0.25	0.25	Full-GMM	-0.50	0.184	0.816	500	0
500	0.25	0.25	Full-GMM	-0.25	0.120	0.880	500	0
500	0.25	0.25	Full-GMM	0.25	0.054	0.946	500	0
500	0.25	0.25	Full-GMM	0.50	0.070	0.930	500	0
500	0.25	0.25	Full-GMM	1.00	0.122	0.878	500	0
500	0.25	0.25	DML-IVQR-BC	-1.00	0.456	0.544	500	0
500	0.25	0.25	DML-IVQR-BC	-0.50	0.214	0.786	500	0
500	0.25	0.25	DML-IVQR-BC	-0.25	0.098	0.902	500	0
500	0.25	0.25	DML-IVQR-BC	0.25	0.030	0.970	500	0
500	0.25	0.25	DML-IVQR-BC	0.50	0.052	0.948	500	0
500	0.25	0.25	DML-IVQR-BC	1.00	0.064	0.936	500	0
500	0.25	0.10	Oracle-GMM	-1.00	0.104	0.896	500	0
500	0.25	0.10	Oracle-GMM	-0.50	0.078	0.922	500	0
500	0.25	0.10	Oracle-GMM	-0.25	0.070	0.930	500	0
500	0.25	0.10	Oracle-GMM	0.25	0.066	0.934	500	0
500	0.25	0.10	Oracle-GMM	0.50	0.050	0.950	500	0
500	0.25	0.10	Oracle-GMM	1.00	0.064	0.936	500	0
500	0.25	0.10	Full-GMM	-1.00	0.108	0.892	500	0
500	0.25	0.10	Full-GMM	-0.50	0.068	0.932	500	0
500	0.25	0.10	Full-GMM	-0.25	0.078	0.922	500	0
500	0.25	0.10	Full-GMM	0.25	0.044	0.956	500	0
500	0.25	0.10	Full-GMM	0.50	0.050	0.950	500	0
500	0.25	0.10	Full-GMM	1.00	0.072	0.928	500	0
500	0.25	0.10	DML-IVQR-BC	-1.00	0.168	0.832	500	0
500	0.25	0.10	DML-IVQR-BC	-0.50	0.114	0.886	500	0
500	0.25	0.10	DML-IVQR-BC	-0.25	0.072	0.928	500	0
500	0.25	0.10	DML-IVQR-BC	0.25	0.038	0.962	500	0
500	0.25	0.10	DML-IVQR-BC	0.50	0.034	0.966	500	0
500	0.25	0.10	DML-IVQR-BC	1.00	0.044	0.956	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.50	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
500	0.50	1.00	Oracle-GMM	-0.50	0.990	0.010	500	0
500	0.50	1.00	Oracle-GMM	-0.25	0.730	0.270	500	0
500	0.50	1.00	Oracle-GMM	0.25	0.734	0.266	500	0
500	0.50	1.00	Oracle-GMM	0.50	0.994	0.006	500	0
500	0.50	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
500	0.50	1.00	Full-GMM	-1.00	1.000	0.000	500	0
500	0.50	1.00	Full-GMM	-0.50	0.790	0.210	500	0
500	0.50	1.00	Full-GMM	-0.25	0.312	0.688	500	0
500	0.50	1.00	Full-GMM	0.25	0.324	0.676	500	0
500	0.50	1.00	Full-GMM	0.50	0.854	0.146	500	0
500	0.50	1.00	Full-GMM	1.00	1.000	0.000	500	0
500	0.50	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
500	0.50	1.00	DML-IVQR-BC	-0.50	0.962	0.038	500	0
500	0.50	1.00	DML-IVQR-BC	-0.25	0.398	0.602	500	0
500	0.50	1.00	DML-IVQR-BC	0.25	0.716	0.284	500	0
500	0.50	1.00	DML-IVQR-BC	0.50	0.984	0.016	500	0
500	0.50	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
500	0.50	0.50	Oracle-GMM	-1.00	0.838	0.162	500	0
500	0.50	0.50	Oracle-GMM	-0.50	0.552	0.448	500	0
500	0.50	0.50	Oracle-GMM	-0.25	0.230	0.770	500	0
500	0.50	0.50	Oracle-GMM	0.25	0.220	0.780	500	0
500	0.50	0.50	Oracle-GMM	0.50	0.560	0.440	500	0
500	0.50	0.50	Oracle-GMM	1.00	0.924	0.076	500	0
500	0.50	0.50	Full-GMM	-1.00	0.630	0.370	500	0
500	0.50	0.50	Full-GMM	-0.50	0.260	0.740	500	0
500	0.50	0.50	Full-GMM	-0.25	0.122	0.878	500	0
500	0.50	0.50	Full-GMM	0.25	0.116	0.884	500	0
500	0.50	0.50	Full-GMM	0.50	0.276	0.724	500	0
500	0.50	0.50	Full-GMM	1.00	0.704	0.296	500	0
500	0.50	0.50	DML-IVQR-BC	-1.00	0.894	0.106	500	0
500	0.50	0.50	DML-IVQR-BC	-0.50	0.466	0.534	500	0
500	0.50	0.50	DML-IVQR-BC	-0.25	0.156	0.844	500	0
500	0.50	0.50	DML-IVQR-BC	0.25	0.144	0.856	500	0
500	0.50	0.50	DML-IVQR-BC	0.50	0.392	0.608	500	0
500	0.50	0.50	DML-IVQR-BC	1.00	0.754	0.246	500	0
500	0.50	0.25	Oracle-GMM	-1.00	0.294	0.706	500	0
500	0.50	0.25	Oracle-GMM	-0.50	0.144	0.856	500	0
500	0.50	0.25	Oracle-GMM	-0.25	0.096	0.904	500	0
500	0.50	0.25	Oracle-GMM	0.25	0.090	0.910	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.50	0.25	Oracle-GMM	0.50	0.164	0.836	500	0
500	0.50	0.25	Oracle-GMM	1.00	0.330	0.670	500	0
500	0.50	0.25	Full-GMM	-1.00	0.202	0.798	500	0
500	0.50	0.25	Full-GMM	-0.50	0.094	0.906	500	0
500	0.50	0.25	Full-GMM	-0.25	0.080	0.920	500	0
500	0.50	0.25	Full-GMM	0.25	0.052	0.948	500	0
500	0.50	0.25	Full-GMM	0.50	0.082	0.918	500	0
500	0.50	0.25	Full-GMM	1.00	0.216	0.784	500	0
500	0.50	0.25	DML-IVQR-BC	-1.00	0.354	0.646	500	0
500	0.50	0.25	DML-IVQR-BC	-0.50	0.168	0.832	500	0
500	0.50	0.25	DML-IVQR-BC	-0.25	0.094	0.906	500	0
500	0.50	0.25	DML-IVQR-BC	0.25	0.054	0.946	500	0
500	0.50	0.25	DML-IVQR-BC	0.50	0.072	0.928	500	0
500	0.50	0.25	DML-IVQR-BC	1.00	0.178	0.822	500	0
500	0.50	0.10	Oracle-GMM	-1.00	0.102	0.898	500	0
500	0.50	0.10	Oracle-GMM	-0.50	0.074	0.926	500	0
500	0.50	0.10	Oracle-GMM	-0.25	0.068	0.932	500	0
500	0.50	0.10	Oracle-GMM	0.25	0.066	0.934	500	0
500	0.50	0.10	Oracle-GMM	0.50	0.062	0.938	500	0
500	0.50	0.10	Oracle-GMM	1.00	0.062	0.938	500	0
500	0.50	0.10	Full-GMM	-1.00	0.098	0.902	500	0
500	0.50	0.10	Full-GMM	-0.50	0.088	0.912	500	0
500	0.50	0.10	Full-GMM	-0.25	0.064	0.936	500	0
500	0.50	0.10	Full-GMM	0.25	0.044	0.956	500	0
500	0.50	0.10	Full-GMM	0.50	0.060	0.940	500	0
500	0.50	0.10	Full-GMM	1.00	0.088	0.912	500	0
500	0.50	0.10	DML-IVQR-BC	-1.00	0.116	0.884	500	0
500	0.50	0.10	DML-IVQR-BC	-0.50	0.104	0.896	500	0
500	0.50	0.10	DML-IVQR-BC	-0.25	0.066	0.934	500	0
500	0.50	0.10	DML-IVQR-BC	0.25	0.032	0.968	500	0
500	0.50	0.10	DML-IVQR-BC	0.50	0.030	0.970	500	0
500	0.50	0.10	DML-IVQR-BC	1.00	0.056	0.944	500	0
500	0.75	1.00	Oracle-GMM	-1.00	0.990	0.010	500	0
500	0.75	1.00	Oracle-GMM	-0.50	0.832	0.168	500	0
500	0.75	1.00	Oracle-GMM	-0.25	0.366	0.634	500	0
500	0.75	1.00	Oracle-GMM	0.25	0.772	0.228	500	0
500	0.75	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
500	0.75	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
500	0.75	1.00	Full-GMM	-1.00	0.790	0.210	500	0
500	0.75	1.00	Full-GMM	-0.50	0.202	0.798	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.75	1.00	Full-GMM	-0.25	0.034	0.966	500	0
500	0.75	1.00	Full-GMM	0.25	0.608	0.392	500	0
500	0.75	1.00	Full-GMM	0.50	0.922	0.078	500	0
500	0.75	1.00	Full-GMM	1.00	1.000	0.000	500	0
500	0.75	1.00	DML-IVQR-BC	-1.00	0.998	0.002	500	0
500	0.75	1.00	DML-IVQR-BC	-0.50	0.754	0.246	500	0
500	0.75	1.00	DML-IVQR-BC	-0.25	0.160	0.840	500	0
500	0.75	1.00	DML-IVQR-BC	0.25	0.896	0.104	500	0
500	0.75	1.00	DML-IVQR-BC	0.50	0.998	0.002	500	0
500	0.75	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
500	0.75	0.50	Oracle-GMM	-1.00	0.510	0.490	500	0
500	0.75	0.50	Oracle-GMM	-0.50	0.252	0.748	500	0
500	0.75	0.50	Oracle-GMM	-0.25	0.124	0.876	500	0
500	0.75	0.50	Oracle-GMM	0.25	0.286	0.714	500	0
500	0.75	0.50	Oracle-GMM	0.50	0.646	0.354	500	0
500	0.75	0.50	Oracle-GMM	1.00	0.932	0.068	500	0
500	0.75	0.50	Full-GMM	-1.00	0.234	0.766	500	0
500	0.75	0.50	Full-GMM	-0.50	0.058	0.942	500	0
500	0.75	0.50	Full-GMM	-0.25	0.040	0.960	500	0
500	0.75	0.50	Full-GMM	0.25	0.200	0.800	500	0
500	0.75	0.50	Full-GMM	0.50	0.410	0.590	500	0
500	0.75	0.50	Full-GMM	1.00	0.740	0.260	500	0
500	0.75	0.50	DML-IVQR-BC	-1.00	0.614	0.386	500	0
500	0.75	0.50	DML-IVQR-BC	-0.50	0.284	0.716	500	0
500	0.75	0.50	DML-IVQR-BC	-0.25	0.088	0.912	500	0
500	0.75	0.50	DML-IVQR-BC	0.25	0.304	0.696	500	0
500	0.75	0.50	DML-IVQR-BC	0.50	0.574	0.426	500	0
500	0.75	0.50	DML-IVQR-BC	1.00	0.844	0.156	500	0
500	0.75	0.25	Oracle-GMM	-1.00	0.150	0.850	500	0
500	0.75	0.25	Oracle-GMM	-0.50	0.086	0.914	500	0
500	0.75	0.25	Oracle-GMM	-0.25	0.060	0.940	500	0
500	0.75	0.25	Oracle-GMM	0.25	0.112	0.888	500	0
500	0.75	0.25	Oracle-GMM	0.50	0.184	0.816	500	0
500	0.75	0.25	Oracle-GMM	1.00	0.358	0.642	500	0
500	0.75	0.25	Full-GMM	-1.00	0.092	0.908	500	0
500	0.75	0.25	Full-GMM	-0.50	0.036	0.964	500	0
500	0.75	0.25	Full-GMM	-0.25	0.040	0.960	500	0
500	0.75	0.25	Full-GMM	0.25	0.074	0.926	500	0
500	0.75	0.25	Full-GMM	0.50	0.106	0.894	500	0
500	0.75	0.25	Full-GMM	1.00	0.248	0.752	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.75	0.25	DML-IVQR-BC	-1.00	0.212	0.788	500	0
500	0.75	0.25	DML-IVQR-BC	-0.50	0.112	0.888	500	0
500	0.75	0.25	DML-IVQR-BC	-0.25	0.070	0.930	500	0
500	0.75	0.25	DML-IVQR-BC	0.25	0.084	0.916	500	0
500	0.75	0.25	DML-IVQR-BC	0.50	0.158	0.842	500	0
500	0.75	0.25	DML-IVQR-BC	1.00	0.230	0.770	500	0
500	0.75	0.10	Oracle-GMM	-1.00	0.062	0.938	500	0
500	0.75	0.10	Oracle-GMM	-0.50	0.056	0.944	500	0
500	0.75	0.10	Oracle-GMM	-0.25	0.050	0.950	500	0
500	0.75	0.10	Oracle-GMM	0.25	0.072	0.928	500	0
500	0.75	0.10	Oracle-GMM	0.50	0.076	0.924	500	0
500	0.75	0.10	Oracle-GMM	1.00	0.104	0.896	500	0
500	0.75	0.10	Full-GMM	-1.00	0.050	0.950	500	0
500	0.75	0.10	Full-GMM	-0.50	0.052	0.948	500	0
500	0.75	0.10	Full-GMM	-0.25	0.042	0.958	500	0
500	0.75	0.10	Full-GMM	0.25	0.042	0.958	500	0
500	0.75	0.10	Full-GMM	0.50	0.068	0.932	500	0
500	0.75	0.10	Full-GMM	1.00	0.068	0.932	500	0
500	0.75	0.10	DML-IVQR-BC	-1.00	0.102	0.898	500	0
500	0.75	0.10	DML-IVQR-BC	-0.50	0.076	0.924	500	0
500	0.75	0.10	DML-IVQR-BC	-0.25	0.056	0.944	500	0
500	0.75	0.10	DML-IVQR-BC	0.25	0.050	0.950	500	0
500	0.75	0.10	DML-IVQR-BC	0.50	0.044	0.956	500	0
500	0.75	0.10	DML-IVQR-BC	1.00	0.058	0.942	500	0
500	0.90	1.00	Oracle-GMM	-1.00	0.658	0.342	500	0
500	0.90	1.00	Oracle-GMM	-0.50	0.278	0.722	500	0
500	0.90	1.00	Oracle-GMM	-0.25	0.096	0.904	500	0
500	0.90	1.00	Oracle-GMM	0.25	0.584	0.416	500	0
500	0.90	1.00	Oracle-GMM	0.50	0.954	0.046	500	0
500	0.90	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
500	0.90	1.00	Full-GMM	-1.00	0.098	0.902	500	0
500	0.90	1.00	Full-GMM	-0.50	0.068	0.932	500	0
500	0.90	1.00	Full-GMM	-0.25	0.186	0.814	500	0
500	0.90	1.00	Full-GMM	0.25	0.724	0.276	500	0
500	0.90	1.00	Full-GMM	0.50	0.834	0.166	500	0
500	0.90	1.00	Full-GMM	1.00	0.960	0.040	500	0
500	0.90	1.00	DML-IVQR-BC	-1.00	0.832	0.168	500	0
500	0.90	1.00	DML-IVQR-BC	-0.50	0.262	0.738	500	0
500	0.90	1.00	DML-IVQR-BC	-0.25	0.072	0.928	500	0
500	0.90	1.00	DML-IVQR-BC	0.25	0.882	0.118	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.90	1.00	DML-IVQR-BC	0.50	0.998	0.002	500	0
500	0.90	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
500	0.90	0.50	Oracle-GMM	-1.00	0.166	0.834	500	0
500	0.90	0.50	Oracle-GMM	-0.50	0.088	0.912	500	0
500	0.90	0.50	Oracle-GMM	-0.25	0.052	0.948	500	0
500	0.90	0.50	Oracle-GMM	0.25	0.212	0.788	500	0
500	0.90	0.50	Oracle-GMM	0.50	0.446	0.554	500	0
500	0.90	0.50	Oracle-GMM	1.00	0.794	0.206	500	0
500	0.90	0.50	Full-GMM	-1.00	0.038	0.962	500	0
500	0.90	0.50	Full-GMM	-0.50	0.048	0.952	500	0
500	0.90	0.50	Full-GMM	-0.25	0.072	0.928	500	0
500	0.90	0.50	Full-GMM	0.25	0.210	0.790	500	0
500	0.90	0.50	Full-GMM	0.50	0.324	0.676	500	0
500	0.90	0.50	Full-GMM	1.00	0.524	0.476	500	0
500	0.90	0.50	DML-IVQR-BC	-1.00	0.300	0.700	500	0
500	0.90	0.50	DML-IVQR-BC	-0.50	0.102	0.898	500	0
500	0.90	0.50	DML-IVQR-BC	-0.25	0.054	0.946	500	0
500	0.90	0.50	DML-IVQR-BC	0.25	0.314	0.686	500	0
500	0.90	0.50	DML-IVQR-BC	0.50	0.530	0.470	500	0
500	0.90	0.50	DML-IVQR-BC	1.00	0.766	0.234	500	0
500	0.90	0.25	Oracle-GMM	-1.00	0.078	0.922	500	0
500	0.90	0.25	Oracle-GMM	-0.50	0.044	0.956	500	0
500	0.90	0.25	Oracle-GMM	-0.25	0.042	0.958	500	0
500	0.90	0.25	Oracle-GMM	0.25	0.102	0.898	500	0
500	0.90	0.25	Oracle-GMM	0.50	0.162	0.838	500	0
500	0.90	0.25	Oracle-GMM	1.00	0.260	0.740	500	0
500	0.90	0.25	Full-GMM	-1.00	0.058	0.942	500	0
500	0.90	0.25	Full-GMM	-0.50	0.038	0.962	500	0
500	0.90	0.25	Full-GMM	-0.25	0.050	0.950	500	0
500	0.90	0.25	Full-GMM	0.25	0.086	0.914	500	0
500	0.90	0.25	Full-GMM	0.50	0.116	0.884	500	0
500	0.90	0.25	Full-GMM	1.00	0.172	0.828	500	0
500	0.90	0.25	DML-IVQR-BC	-1.00	0.110	0.890	500	0
500	0.90	0.25	DML-IVQR-BC	-0.50	0.086	0.914	500	0
500	0.90	0.25	DML-IVQR-BC	-0.25	0.064	0.936	500	0
500	0.90	0.25	DML-IVQR-BC	0.25	0.100	0.900	500	0
500	0.90	0.25	DML-IVQR-BC	0.50	0.154	0.846	500	0
500	0.90	0.25	DML-IVQR-BC	1.00	0.174	0.826	500	0
500	0.90	0.10	Oracle-GMM	-1.00	0.062	0.938	500	0
500	0.90	0.10	Oracle-GMM	-0.50	0.046	0.954	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.90	0.10	Oracle-GMM	-0.25	0.032	0.968	500	0
500	0.90	0.10	Oracle-GMM	0.25	0.058	0.942	500	0
500	0.90	0.10	Oracle-GMM	0.50	0.068	0.932	500	0
500	0.90	0.10	Oracle-GMM	1.00	0.096	0.904	500	0
500	0.90	0.10	Full-GMM	-1.00	0.026	0.974	500	0
500	0.90	0.10	Full-GMM	-0.50	0.032	0.968	500	0
500	0.90	0.10	Full-GMM	-0.25	0.042	0.958	500	0
500	0.90	0.10	Full-GMM	0.25	0.058	0.942	500	0
500	0.90	0.10	Full-GMM	0.50	0.052	0.948	500	0
500	0.90	0.10	Full-GMM	1.00	0.058	0.942	500	0
500	0.90	0.10	DML-IVQR-BC	-1.00	0.050	0.950	500	0
500	0.90	0.10	DML-IVQR-BC	-0.50	0.064	0.936	500	0
500	0.90	0.10	DML-IVQR-BC	-0.25	0.056	0.944	500	0
500	0.90	0.10	DML-IVQR-BC	0.25	0.060	0.940	500	0
500	0.90	0.10	DML-IVQR-BC	0.50	0.068	0.932	500	0
500	0.90	0.10	DML-IVQR-BC	1.00	0.040	0.960	500	0
1000	0.10	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
1000	0.10	1.00	Oracle-GMM	-0.50	1.000	0.000	500	0
1000	0.10	1.00	Oracle-GMM	-0.25	0.938	0.062	500	0
1000	0.10	1.00	Oracle-GMM	0.25	0.462	0.538	500	0
1000	0.10	1.00	Oracle-GMM	0.50	0.930	0.070	500	0
1000	0.10	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.10	1.00	Full-GMM	-1.00	1.000	0.000	500	0
1000	0.10	1.00	Full-GMM	-0.50	0.994	0.006	500	0
1000	0.10	1.00	Full-GMM	-0.25	0.882	0.118	500	0
1000	0.10	1.00	Full-GMM	0.25	0.100	0.900	500	0
1000	0.10	1.00	Full-GMM	0.50	0.126	0.874	500	0
1000	0.10	1.00	Full-GMM	1.00	0.822	0.178	500	0
1000	0.10	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
1000	0.10	1.00	DML-IVQR-BC	-0.50	0.988	0.012	500	0
1000	0.10	1.00	DML-IVQR-BC	-0.25	0.482	0.518	500	0
1000	0.10	1.00	DML-IVQR-BC	0.25	0.296	0.704	500	0
1000	0.10	1.00	DML-IVQR-BC	0.50	0.696	0.304	500	0
1000	0.10	1.00	DML-IVQR-BC	1.00	0.966	0.034	500	0
1000	0.10	0.50	Oracle-GMM	-1.00	0.988	0.012	500	0
1000	0.10	0.50	Oracle-GMM	-0.50	0.852	0.148	500	0
1000	0.10	0.50	Oracle-GMM	-0.25	0.438	0.562	500	0
1000	0.10	0.50	Oracle-GMM	0.25	0.156	0.844	500	0
1000	0.10	0.50	Oracle-GMM	0.50	0.364	0.636	500	0
1000	0.10	0.50	Oracle-GMM	1.00	0.662	0.338	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
1000	0.10	0.50	Full-GMM	-1.00	0.914	0.086	500	0
1000	0.10	0.50	Full-GMM	-0.50	0.682	0.318	500	0
1000	0.10	0.50	Full-GMM	-0.25	0.440	0.560	500	0
1000	0.10	0.50	Full-GMM	0.25	0.052	0.948	500	0
1000	0.10	0.50	Full-GMM	0.50	0.046	0.954	500	0
1000	0.10	0.50	Full-GMM	1.00	0.274	0.726	500	0
1000	0.10	0.50	DML-IVQR-BC	-1.00	0.976	0.024	500	0
1000	0.10	0.50	DML-IVQR-BC	-0.50	0.678	0.322	500	0
1000	0.10	0.50	DML-IVQR-BC	-0.25	0.202	0.798	500	0
1000	0.10	0.50	DML-IVQR-BC	0.25	0.066	0.934	500	0
1000	0.10	0.50	DML-IVQR-BC	0.50	0.148	0.852	500	0
1000	0.10	0.50	DML-IVQR-BC	1.00	0.336	0.664	500	0
1000	0.10	0.25	Oracle-GMM	-1.00	0.526	0.474	500	0
1000	0.10	0.25	Oracle-GMM	-0.50	0.342	0.658	500	0
1000	0.10	0.25	Oracle-GMM	-0.25	0.154	0.846	500	0
1000	0.10	0.25	Oracle-GMM	0.25	0.078	0.922	500	0
1000	0.10	0.25	Oracle-GMM	0.50	0.110	0.890	500	0
1000	0.10	0.25	Oracle-GMM	1.00	0.194	0.806	500	0
1000	0.10	0.25	Full-GMM	-1.00	0.392	0.608	500	0
1000	0.10	0.25	Full-GMM	-0.50	0.230	0.770	500	0
1000	0.10	0.25	Full-GMM	-0.25	0.144	0.856	500	0
1000	0.10	0.25	Full-GMM	0.25	0.044	0.956	500	0
1000	0.10	0.25	Full-GMM	0.50	0.050	0.950	500	0
1000	0.10	0.25	Full-GMM	1.00	0.096	0.904	500	0
1000	0.10	0.25	DML-IVQR-BC	-1.00	0.608	0.392	500	0
1000	0.10	0.25	DML-IVQR-BC	-0.50	0.306	0.694	500	0
1000	0.10	0.25	DML-IVQR-BC	-0.25	0.098	0.902	500	0
1000	0.10	0.25	DML-IVQR-BC	0.25	0.028	0.972	500	0
1000	0.10	0.25	DML-IVQR-BC	0.50	0.036	0.964	500	0
1000	0.10	0.25	DML-IVQR-BC	1.00	0.058	0.942	500	0
1000	0.10	0.10	Oracle-GMM	-1.00	0.132	0.868	500	0
1000	0.10	0.10	Oracle-GMM	-0.50	0.104	0.896	500	0
1000	0.10	0.10	Oracle-GMM	-0.25	0.078	0.922	500	0
1000	0.10	0.10	Oracle-GMM	0.25	0.056	0.944	500	0
1000	0.10	0.10	Oracle-GMM	0.50	0.058	0.942	500	0
1000	0.10	0.10	Oracle-GMM	1.00	0.052	0.948	500	0
1000	0.10	0.10	Full-GMM	-1.00	0.096	0.904	500	0
1000	0.10	0.10	Full-GMM	-0.50	0.080	0.920	500	0
1000	0.10	0.10	Full-GMM	-0.25	0.054	0.946	500	0
1000	0.10	0.10	Full-GMM	0.25	0.056	0.944	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.10	0.10	Full-GMM	0.50	0.050	0.950	500	0
1000	0.10	0.10	Full-GMM	1.00	0.046	0.954	500	0
1000	0.10	0.10	DML-IVQR-BC	-1.00	0.242	0.758	500	0
1000	0.10	0.10	DML-IVQR-BC	-0.50	0.118	0.882	500	0
1000	0.10	0.10	DML-IVQR-BC	-0.25	0.076	0.924	500	0
1000	0.10	0.10	DML-IVQR-BC	0.25	0.038	0.962	500	0
1000	0.10	0.10	DML-IVQR-BC	0.50	0.022	0.978	500	0
1000	0.10	0.10	DML-IVQR-BC	1.00	0.024	0.976	500	0
1000	0.25	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
1000	0.25	1.00	Oracle-GMM	-0.50	1.000	0.000	500	0
1000	0.25	1.00	Oracle-GMM	-0.25	0.980	0.020	500	0
1000	0.25	1.00	Oracle-GMM	0.25	0.898	0.102	500	0
1000	0.25	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
1000	0.25	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.25	1.00	Full-GMM	-1.00	1.000	0.000	500	0
1000	0.25	1.00	Full-GMM	-0.50	0.998	0.002	500	0
1000	0.25	1.00	Full-GMM	-0.25	0.904	0.096	500	0
1000	0.25	1.00	Full-GMM	0.25	0.282	0.718	500	0
1000	0.25	1.00	Full-GMM	0.50	0.894	0.106	500	0
1000	0.25	1.00	Full-GMM	1.00	1.000	0.000	500	0
1000	0.25	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
1000	0.25	1.00	DML-IVQR-BC	-0.50	1.000	0.000	500	0
1000	0.25	1.00	DML-IVQR-BC	-0.25	0.746	0.254	500	0
1000	0.25	1.00	DML-IVQR-BC	0.25	0.838	0.162	500	0
1000	0.25	1.00	DML-IVQR-BC	0.50	0.996	0.004	500	0
1000	0.25	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
1000	0.25	0.50	Oracle-GMM	-1.00	0.998	0.002	500	0
1000	0.25	0.50	Oracle-GMM	-0.50	0.902	0.098	500	0
1000	0.25	0.50	Oracle-GMM	-0.25	0.542	0.458	500	0
1000	0.25	0.50	Oracle-GMM	0.25	0.318	0.682	500	0
1000	0.25	0.50	Oracle-GMM	0.50	0.758	0.242	500	0
1000	0.25	0.50	Oracle-GMM	1.00	0.968	0.032	500	0
1000	0.25	0.50	Full-GMM	-1.00	0.986	0.014	500	0
1000	0.25	0.50	Full-GMM	-0.50	0.782	0.218	500	0
1000	0.25	0.50	Full-GMM	-0.25	0.418	0.582	500	0
1000	0.25	0.50	Full-GMM	0.25	0.082	0.918	500	0
1000	0.25	0.50	Full-GMM	0.50	0.350	0.650	500	0
1000	0.25	0.50	Full-GMM	1.00	0.824	0.176	500	0
1000	0.25	0.50	DML-IVQR-BC	-1.00	0.998	0.002	500	0
1000	0.25	0.50	DML-IVQR-BC	-0.50	0.832	0.168	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.25	0.50	DML-IVQR-BC	-0.25	0.310	0.690	500	0
1000	0.25	0.50	DML-IVQR-BC	0.25	0.262	0.738	500	0
1000	0.25	0.50	DML-IVQR-BC	0.50	0.556	0.444	500	0
1000	0.25	0.50	DML-IVQR-BC	1.00	0.848	0.152	500	0
1000	0.25	0.25	Oracle-GMM	-1.00	0.610	0.390	500	0
1000	0.25	0.25	Oracle-GMM	-0.50	0.386	0.614	500	0
1000	0.25	0.25	Oracle-GMM	-0.25	0.188	0.812	500	0
1000	0.25	0.25	Oracle-GMM	0.25	0.124	0.876	500	0
1000	0.25	0.25	Oracle-GMM	0.50	0.224	0.776	500	0
1000	0.25	0.25	Oracle-GMM	1.00	0.416	0.584	500	0
1000	0.25	0.25	Full-GMM	-1.00	0.512	0.488	500	0
1000	0.25	0.25	Full-GMM	-0.50	0.322	0.678	500	0
1000	0.25	0.25	Full-GMM	-0.25	0.128	0.872	500	0
1000	0.25	0.25	Full-GMM	0.25	0.052	0.948	500	0
1000	0.25	0.25	Full-GMM	0.50	0.112	0.888	500	0
1000	0.25	0.25	Full-GMM	1.00	0.308	0.692	500	0
1000	0.25	0.25	DML-IVQR-BC	-1.00	0.688	0.312	500	0
1000	0.25	0.25	DML-IVQR-BC	-0.50	0.362	0.638	500	0
1000	0.25	0.25	DML-IVQR-BC	-0.25	0.156	0.844	500	0
1000	0.25	0.25	DML-IVQR-BC	0.25	0.084	0.916	500	0
1000	0.25	0.25	DML-IVQR-BC	0.50	0.136	0.864	500	0
1000	0.25	0.25	DML-IVQR-BC	1.00	0.260	0.740	500	0
1000	0.25	0.10	Oracle-GMM	-1.00	0.170	0.830	500	0
1000	0.25	0.10	Oracle-GMM	-0.50	0.106	0.894	500	0
1000	0.25	0.10	Oracle-GMM	-0.25	0.086	0.914	500	0
1000	0.25	0.10	Oracle-GMM	0.25	0.064	0.936	500	0
1000	0.25	0.10	Oracle-GMM	0.50	0.086	0.914	500	0
1000	0.25	0.10	Oracle-GMM	1.00	0.094	0.906	500	0
1000	0.25	0.10	Full-GMM	-1.00	0.126	0.874	500	0
1000	0.25	0.10	Full-GMM	-0.50	0.088	0.912	500	0
1000	0.25	0.10	Full-GMM	-0.25	0.056	0.944	500	0
1000	0.25	0.10	Full-GMM	0.25	0.052	0.948	500	0
1000	0.25	0.10	Full-GMM	0.50	0.066	0.934	500	0
1000	0.25	0.10	Full-GMM	1.00	0.096	0.904	500	0
1000	0.25	0.10	DML-IVQR-BC	-1.00	0.220	0.780	500	0
1000	0.25	0.10	DML-IVQR-BC	-0.50	0.120	0.880	500	0
1000	0.25	0.10	DML-IVQR-BC	-0.25	0.090	0.910	500	0
1000	0.25	0.10	DML-IVQR-BC	0.25	0.044	0.956	500	0
1000	0.25	0.10	DML-IVQR-BC	0.50	0.060	0.940	500	0
1000	0.25	0.10	DML-IVQR-BC	1.00	0.056	0.944	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
1000	0.50	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
1000	0.50	1.00	Oracle-GMM	-0.50	1.000	0.000	500	0
1000	0.50	1.00	Oracle-GMM	-0.25	0.948	0.052	500	0
1000	0.50	1.00	Oracle-GMM	0.25	0.980	0.020	500	0
1000	0.50	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
1000	0.50	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.50	1.00	Full-GMM	-1.00	1.000	0.000	500	0
1000	0.50	1.00	Full-GMM	-0.50	0.998	0.002	500	0
1000	0.50	1.00	Full-GMM	-0.25	0.692	0.308	500	0
1000	0.50	1.00	Full-GMM	0.25	0.754	0.246	500	0
1000	0.50	1.00	Full-GMM	0.50	0.998	0.002	500	0
1000	0.50	1.00	Full-GMM	1.00	1.000	0.000	500	0
1000	0.50	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
1000	0.50	1.00	DML-IVQR-BC	-0.50	1.000	0.000	500	0
1000	0.50	1.00	DML-IVQR-BC	-0.25	0.774	0.226	500	0
1000	0.50	1.00	DML-IVQR-BC	0.25	0.988	0.012	500	0
1000	0.50	1.00	DML-IVQR-BC	0.50	1.000	0.000	500	0
1000	0.50	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
1000	0.50	0.50	Oracle-GMM	-1.00	0.986	0.014	500	0
1000	0.50	0.50	Oracle-GMM	-0.50	0.850	0.150	500	0
1000	0.50	0.50	Oracle-GMM	-0.25	0.426	0.574	500	0
1000	0.50	0.50	Oracle-GMM	0.25	0.540	0.460	500	0
1000	0.50	0.50	Oracle-GMM	0.50	0.940	0.060	500	0
1000	0.50	0.50	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.50	0.50	Full-GMM	-1.00	0.952	0.048	500	0
1000	0.50	0.50	Full-GMM	-0.50	0.634	0.366	500	0
1000	0.50	0.50	Full-GMM	-0.25	0.242	0.758	500	0
1000	0.50	0.50	Full-GMM	0.25	0.298	0.702	500	0
1000	0.50	0.50	Full-GMM	0.50	0.726	0.274	500	0
1000	0.50	0.50	Full-GMM	1.00	0.988	0.012	500	0
1000	0.50	0.50	DML-IVQR-BC	-1.00	0.994	0.006	500	0
1000	0.50	0.50	DML-IVQR-BC	-0.50	0.802	0.198	500	0
1000	0.50	0.50	DML-IVQR-BC	-0.25	0.284	0.716	500	0
1000	0.50	0.50	DML-IVQR-BC	0.25	0.494	0.506	500	0
1000	0.50	0.50	DML-IVQR-BC	0.50	0.852	0.148	500	0
1000	0.50	0.50	DML-IVQR-BC	1.00	0.994	0.006	500	0
1000	0.50	0.25	Oracle-GMM	-1.00	0.540	0.460	500	0
1000	0.50	0.25	Oracle-GMM	-0.50	0.316	0.684	500	0
1000	0.50	0.25	Oracle-GMM	-0.25	0.124	0.876	500	0
1000	0.50	0.25	Oracle-GMM	0.25	0.190	0.810	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.50	0.25	Oracle-GMM	0.50	0.378	0.622	500	0
1000	0.50	0.25	Oracle-GMM	1.00	0.626	0.374	500	0
1000	0.50	0.25	Full-GMM	-1.00	0.430	0.570	500	0
1000	0.50	0.25	Full-GMM	-0.50	0.212	0.788	500	0
1000	0.50	0.25	Full-GMM	-0.25	0.092	0.908	500	0
1000	0.50	0.25	Full-GMM	0.25	0.102	0.898	500	0
1000	0.50	0.25	Full-GMM	0.50	0.238	0.762	500	0
1000	0.50	0.25	Full-GMM	1.00	0.524	0.476	500	0
1000	0.50	0.25	DML-IVQR-BC	-1.00	0.632	0.368	500	0
1000	0.50	0.25	DML-IVQR-BC	-0.50	0.312	0.688	500	0
1000	0.50	0.25	DML-IVQR-BC	-0.25	0.108	0.892	500	0
1000	0.50	0.25	DML-IVQR-BC	0.25	0.152	0.848	500	0
1000	0.50	0.25	DML-IVQR-BC	0.50	0.286	0.714	500	0
1000	0.50	0.25	DML-IVQR-BC	1.00	0.486	0.514	500	0
1000	0.50	0.10	Oracle-GMM	-1.00	0.120	0.880	500	0
1000	0.50	0.10	Oracle-GMM	-0.50	0.082	0.918	500	0
1000	0.50	0.10	Oracle-GMM	-0.25	0.064	0.936	500	0
1000	0.50	0.10	Oracle-GMM	0.25	0.084	0.916	500	0
1000	0.50	0.10	Oracle-GMM	0.50	0.092	0.908	500	0
1000	0.50	0.10	Oracle-GMM	1.00	0.138	0.862	500	0
1000	0.50	0.10	Full-GMM	-1.00	0.112	0.888	500	0
1000	0.50	0.10	Full-GMM	-0.50	0.076	0.924	500	0
1000	0.50	0.10	Full-GMM	-0.25	0.054	0.946	500	0
1000	0.50	0.10	Full-GMM	0.25	0.066	0.934	500	0
1000	0.50	0.10	Full-GMM	0.50	0.098	0.902	500	0
1000	0.50	0.10	Full-GMM	1.00	0.160	0.840	500	0
1000	0.50	0.10	DML-IVQR-BC	-1.00	0.154	0.846	500	0
1000	0.50	0.10	DML-IVQR-BC	-0.50	0.104	0.896	500	0
1000	0.50	0.10	DML-IVQR-BC	-0.25	0.060	0.940	500	0
1000	0.50	0.10	DML-IVQR-BC	0.25	0.070	0.930	500	0
1000	0.50	0.10	DML-IVQR-BC	0.50	0.094	0.906	500	0
1000	0.50	0.10	DML-IVQR-BC	1.00	0.080	0.920	500	0
1000	0.75	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
1000	0.75	1.00	Oracle-GMM	-0.50	0.994	0.006	500	0
1000	0.75	1.00	Oracle-GMM	-0.25	0.746	0.254	500	0
1000	0.75	1.00	Oracle-GMM	0.25	0.976	0.024	500	0
1000	0.75	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
1000	0.75	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.75	1.00	Full-GMM	-1.00	1.000	0.000	500	0
1000	0.75	1.00	Full-GMM	-0.50	0.760	0.240	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.75	1.00	Full-GMM	-0.25	0.162	0.838	500	0
1000	0.75	1.00	Full-GMM	0.25	0.906	0.094	500	0
1000	0.75	1.00	Full-GMM	0.50	1.000	0.000	500	0
1000	0.75	1.00	Full-GMM	1.00	1.000	0.000	500	0
1000	0.75	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
1000	0.75	1.00	DML-IVQR-BC	-0.50	0.984	0.016	500	0
1000	0.75	1.00	DML-IVQR-BC	-0.25	0.510	0.490	500	0
1000	0.75	1.00	DML-IVQR-BC	0.25	0.996	0.004	500	0
1000	0.75	1.00	DML-IVQR-BC	0.50	1.000	0.000	500	0
1000	0.75	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
1000	0.75	0.50	Oracle-GMM	-1.00	0.828	0.172	500	0
1000	0.75	0.50	Oracle-GMM	-0.50	0.526	0.474	500	0
1000	0.75	0.50	Oracle-GMM	-0.25	0.260	0.740	500	0
1000	0.75	0.50	Oracle-GMM	0.25	0.554	0.446	500	0
1000	0.75	0.50	Oracle-GMM	0.50	0.950	0.050	500	0
1000	0.75	0.50	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.75	0.50	Full-GMM	-1.00	0.654	0.346	500	0
1000	0.75	0.50	Full-GMM	-0.50	0.230	0.770	500	0
1000	0.75	0.50	Full-GMM	-0.25	0.078	0.922	500	0
1000	0.75	0.50	Full-GMM	0.25	0.414	0.586	500	0
1000	0.75	0.50	Full-GMM	0.50	0.786	0.214	500	0
1000	0.75	0.50	Full-GMM	1.00	0.996	0.004	500	0
1000	0.75	0.50	DML-IVQR-BC	-1.00	0.906	0.094	500	0
1000	0.75	0.50	DML-IVQR-BC	-0.50	0.546	0.454	500	0
1000	0.75	0.50	DML-IVQR-BC	-0.25	0.184	0.816	500	0
1000	0.75	0.50	DML-IVQR-BC	0.25	0.640	0.360	500	0
1000	0.75	0.50	DML-IVQR-BC	0.50	0.958	0.042	500	0
1000	0.75	0.50	DML-IVQR-BC	1.00	0.998	0.002	500	0
1000	0.75	0.25	Oracle-GMM	-1.00	0.286	0.714	500	0
1000	0.75	0.25	Oracle-GMM	-0.50	0.154	0.846	500	0
1000	0.75	0.25	Oracle-GMM	-0.25	0.084	0.916	500	0
1000	0.75	0.25	Oracle-GMM	0.25	0.220	0.780	500	0
1000	0.75	0.25	Oracle-GMM	0.50	0.418	0.582	500	0
1000	0.75	0.25	Oracle-GMM	1.00	0.712	0.288	500	0
1000	0.75	0.25	Full-GMM	-1.00	0.192	0.808	500	0
1000	0.75	0.25	Full-GMM	-0.50	0.084	0.916	500	0
1000	0.75	0.25	Full-GMM	-0.25	0.040	0.960	500	0
1000	0.75	0.25	Full-GMM	0.25	0.152	0.848	500	0
1000	0.75	0.25	Full-GMM	0.50	0.292	0.708	500	0
1000	0.75	0.25	Full-GMM	1.00	0.548	0.452	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.75	0.25	DML-IVQR-BC	-1.00	0.398	0.602	500	0
1000	0.75	0.25	DML-IVQR-BC	-0.50	0.194	0.806	500	0
1000	0.75	0.25	DML-IVQR-BC	-0.25	0.100	0.900	500	0
1000	0.75	0.25	DML-IVQR-BC	0.25	0.212	0.788	500	0
1000	0.75	0.25	DML-IVQR-BC	0.50	0.384	0.616	500	0
1000	0.75	0.25	DML-IVQR-BC	1.00	0.610	0.390	500	0
1000	0.75	0.10	Oracle-GMM	-1.00	0.088	0.912	500	0
1000	0.75	0.10	Oracle-GMM	-0.50	0.054	0.946	500	0
1000	0.75	0.10	Oracle-GMM	-0.25	0.046	0.954	500	0
1000	0.75	0.10	Oracle-GMM	0.25	0.070	0.930	500	0
1000	0.75	0.10	Oracle-GMM	0.50	0.142	0.858	500	0
1000	0.75	0.10	Oracle-GMM	1.00	0.166	0.834	500	0
1000	0.75	0.10	Full-GMM	-1.00	0.070	0.930	500	0
1000	0.75	0.10	Full-GMM	-0.50	0.064	0.936	500	0
1000	0.75	0.10	Full-GMM	-0.25	0.070	0.930	500	0
1000	0.75	0.10	Full-GMM	0.25	0.074	0.926	500	0
1000	0.75	0.10	Full-GMM	0.50	0.096	0.904	500	0
1000	0.75	0.10	Full-GMM	1.00	0.122	0.878	500	0
1000	0.75	0.10	DML-IVQR-BC	-1.00	0.102	0.898	500	0
1000	0.75	0.10	DML-IVQR-BC	-0.50	0.090	0.910	500	0
1000	0.75	0.10	DML-IVQR-BC	-0.25	0.080	0.920	500	0
1000	0.75	0.10	DML-IVQR-BC	0.25	0.068	0.932	500	0
1000	0.75	0.10	DML-IVQR-BC	0.50	0.100	0.900	500	0
1000	0.75	0.10	DML-IVQR-BC	1.00	0.118	0.882	500	0
1000	0.90	1.00	Oracle-GMM	-1.00	0.972	0.028	500	0
1000	0.90	1.00	Oracle-GMM	-0.50	0.748	0.252	500	0
1000	0.90	1.00	Oracle-GMM	-0.25	0.322	0.678	500	0
1000	0.90	1.00	Oracle-GMM	0.25	0.928	0.072	500	0
1000	0.90	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
1000	0.90	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.90	1.00	Full-GMM	-1.00	0.544	0.456	500	0
1000	0.90	1.00	Full-GMM	-0.50	0.112	0.888	500	0
1000	0.90	1.00	Full-GMM	-0.25	0.114	0.886	500	0
1000	0.90	1.00	Full-GMM	0.25	0.840	0.160	500	0
1000	0.90	1.00	Full-GMM	0.50	0.982	0.018	500	0
1000	0.90	1.00	Full-GMM	1.00	1.000	0.000	500	0
1000	0.90	1.00	DML-IVQR-BC	-1.00	0.990	0.010	500	0
1000	0.90	1.00	DML-IVQR-BC	-0.50	0.660	0.340	500	0
1000	0.90	1.00	DML-IVQR-BC	-0.25	0.128	0.872	500	0
1000	0.90	1.00	DML-IVQR-BC	0.25	0.996	0.004	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.90	1.00	DML-IVQR-BC	0.50	1.000	0.000	500	0
1000	0.90	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
1000	0.90	0.50	Oracle-GMM	-1.00	0.426	0.574	500	0
1000	0.90	0.50	Oracle-GMM	-0.50	0.228	0.772	500	0
1000	0.90	0.50	Oracle-GMM	-0.25	0.070	0.930	500	0
1000	0.90	0.50	Oracle-GMM	0.25	0.440	0.560	500	0
1000	0.90	0.50	Oracle-GMM	0.50	0.844	0.156	500	0
1000	0.90	0.50	Oracle-GMM	1.00	0.998	0.002	500	0
1000	0.90	0.50	Full-GMM	-1.00	0.172	0.828	500	0
1000	0.90	0.50	Full-GMM	-0.50	0.056	0.944	500	0
1000	0.90	0.50	Full-GMM	-0.25	0.074	0.926	500	0
1000	0.90	0.50	Full-GMM	0.25	0.400	0.600	500	0
1000	0.90	0.50	Full-GMM	0.50	0.672	0.328	500	0
1000	0.90	0.50	Full-GMM	1.00	0.936	0.064	500	0
1000	0.90	0.50	DML-IVQR-BC	-1.00	0.578	0.422	500	0
1000	0.90	0.50	DML-IVQR-BC	-0.50	0.224	0.776	500	0
1000	0.90	0.50	DML-IVQR-BC	-0.25	0.070	0.930	500	0
1000	0.90	0.50	DML-IVQR-BC	0.25	0.616	0.384	500	0
1000	0.90	0.50	DML-IVQR-BC	0.50	0.900	0.100	500	0
1000	0.90	0.50	DML-IVQR-BC	1.00	0.996	0.004	500	0
1000	0.90	0.25	Oracle-GMM	-1.00	0.144	0.856	500	0
1000	0.90	0.25	Oracle-GMM	-0.50	0.086	0.914	500	0
1000	0.90	0.25	Oracle-GMM	-0.25	0.052	0.948	500	0
1000	0.90	0.25	Oracle-GMM	0.25	0.166	0.834	500	0
1000	0.90	0.25	Oracle-GMM	0.50	0.338	0.662	500	0
1000	0.90	0.25	Oracle-GMM	1.00	0.558	0.442	500	0
1000	0.90	0.25	Full-GMM	-1.00	0.066	0.934	500	0
1000	0.90	0.25	Full-GMM	-0.50	0.050	0.950	500	0
1000	0.90	0.25	Full-GMM	-0.25	0.050	0.950	500	0
1000	0.90	0.25	Full-GMM	0.25	0.126	0.874	500	0
1000	0.90	0.25	Full-GMM	0.50	0.258	0.742	500	0
1000	0.90	0.25	Full-GMM	1.00	0.416	0.584	500	0
1000	0.90	0.25	DML-IVQR-BC	-1.00	0.208	0.792	500	0
1000	0.90	0.25	DML-IVQR-BC	-0.50	0.126	0.874	500	0
1000	0.90	0.25	DML-IVQR-BC	-0.25	0.062	0.938	500	0
1000	0.90	0.25	DML-IVQR-BC	0.25	0.194	0.806	500	0
1000	0.90	0.25	DML-IVQR-BC	0.50	0.306	0.694	500	0
1000	0.90	0.25	DML-IVQR-BC	1.00	0.490	0.510	500	0
1000	0.90	0.10	Oracle-GMM	-1.00	0.080	0.920	500	0
1000	0.90	0.10	Oracle-GMM	-0.50	0.062	0.938	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
1000	0.90	0.10	Oracle-GMM	-0.25	0.038	0.962	500	0
1000	0.90	0.10	Oracle-GMM	0.25	0.092	0.908	500	0
1000	0.90	0.10	Oracle-GMM	0.50	0.096	0.904	500	0
1000	0.90	0.10	Oracle-GMM	1.00	0.144	0.856	500	0
1000	0.90	0.10	Full-GMM	-1.00	0.052	0.948	500	0
1000	0.90	0.10	Full-GMM	-0.50	0.034	0.966	500	0
1000	0.90	0.10	Full-GMM	-0.25	0.042	0.958	500	0
1000	0.90	0.10	Full-GMM	0.25	0.066	0.934	500	0
1000	0.90	0.10	Full-GMM	0.50	0.060	0.940	500	0
1000	0.90	0.10	Full-GMM	1.00	0.088	0.912	500	0
1000	0.90	0.10	DML-IVQR-BC	-1.00	0.096	0.904	500	0
1000	0.90	0.10	DML-IVQR-BC	-0.50	0.064	0.936	500	0
1000	0.90	0.10	DML-IVQR-BC	-0.25	0.054	0.946	500	0
1000	0.90	0.10	DML-IVQR-BC	0.25	0.082	0.918	500	0
1000	0.90	0.10	DML-IVQR-BC	0.50	0.058	0.942	500	0
1000	0.90	0.10	DML-IVQR-BC	1.00	0.100	0.900	500	0

Note: Rejection probabilities at false values $a = \alpha_0(\tau) + \Delta$. These probabilities are power, not empirical size.

Appendix C Boundary and Grid-Domain Diagnostic

Table C.1. Grid-domain diagnostic: accepted-set measures

n	τ	κ	**Estimator**	**Med.** A_0	**Med.** A_1	**Med.** A_2	**Mean** A_0	**Mean** A_1	**Mean** A_2
1000	0.10	1.00	Oracle-GMM	0.400	0.400	0.400	0.386	0.386	0.386
1000	0.10	1.00	Full-GMM	0.800	0.800	0.800	0.757	0.757	0.757
1000	0.10	1.00	DML-IVQR-BC	0.600	0.600	0.600	0.705	0.705	0.705
1000	0.10	0.25	Oracle-GMM	3.400	5.325	7.250	3.019	4.994	6.838
1000	0.10	0.25	Full-GMM	3.375	5.950	8.200	3.304	5.774	8.037
1000	0.10	0.25	DML-IVQR-BC	3.750	5.950	7.950	3.607	6.040	8.245
1000	0.10	0.10	Oracle-GMM	4.000	7.900	11.900	3.745	7.381	11.002
1000	0.10	0.10	Full-GMM	3.900	7.800	11.600	3.785	7.394	10.975
1000	0.10	0.10	DML-IVQR-BC	4.000	7.575	10.775	3.845	7.219	10.339
1000	0.25	1.00	Oracle-GMM	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.25	1.00	Full-GMM	0.400	0.400	0.400	0.429	0.429	0.429
1000	0.25	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.346	0.346	0.346

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TableC.1. Grid-domain diagnostic: accepted-set measures (Continued)

n	τ	κ	**Estimator**	**Med.** A_0	**Med.** A_1	**Med.** A_2	**Mean** A_0	**Mean** A_1	**Mean** A_2
1000	0.25	0.25	Oracle-GMM	2.175	2.800	2.850	2.224	3.158	3.964
1000	0.25	0.25	Full-GMM	2.450	3.500	4.075	2.517	3.768	4.745
1000	0.25	0.25	DML-IVQR-BC	2.750	4.650	6.225	2.630	4.130	5.712
1000	0.25	0.10	Oracle-GMM	3.900	7.700	11.500	3.492	6.699	9.872
1000	0.25	0.10	Full-GMM	3.825	7.600	11.100	3.595	6.891	10.083
1000	0.25	0.10	DML-IVQR-BC	4.000	7.700	11.125	3.651	6.926	9.984
1000	0.50	1.00	Oracle-GMM	0.200	0.200	0.200	0.235	0.235	0.235
1000	0.50	1.00	Full-GMM	0.400	0.400	0.400	0.373	0.373	0.373
1000	0.50	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.50	0.25	Oracle-GMM	1.875	2.200	2.200	1.960	2.682	3.229
1000	0.50	0.25	Full-GMM	2.250	2.400	2.550	2.368	3.171	3.802
1000	0.50	0.25	DML-IVQR-BC	1.950	2.500	3.025	2.124	3.020	3.942
1000	0.50	0.10	Oracle-GMM	3.825	7.600	11.300	3.521	6.640	9.687
1000	0.50	0.10	Full-GMM	3.900	7.625	11.450	3.536	6.718	9.817
1000	0.50	0.10	DML-IVQR-BC	3.900	7.500	10.325	3.538	6.671	9.585
1000	0.75	1.00	Oracle-GMM	0.300	0.300	0.300	0.290	0.290	0.290
1000	0.75	1.00	Full-GMM	0.500	0.500	0.500	0.512	0.512	0.512
1000	0.75	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.279	0.279	0.279
1000	0.75	0.25	Oracle-GMM	2.825	4.200	4.850	2.628	3.945	4.990
1000	0.75	0.25	Full-GMM	2.950	4.450	5.150	2.827	4.267	5.478
1000	0.75	0.25	DML-IVQR-BC	2.325	3.300	3.900	2.402	3.595	4.788
1000	0.75	0.10	Oracle-GMM	3.975	7.750	11.700	3.670	6.999	10.357
1000	0.75	0.10	Full-GMM	3.900	7.750	11.500	3.661	7.023	10.267
1000	0.75	0.10	DML-IVQR-BC	3.900	7.350	10.375	3.554	6.691	9.736
1000	0.90	1.00	Oracle-GMM	0.500	0.500	0.500	0.519	0.519	0.519
1000	0.90	1.00	Full-GMM	1.000	1.000	1.000	1.026	1.031	1.031
1000	0.90	1.00	DML-IVQR-BC	0.400	0.400	0.400	0.454	0.454	0.454
1000	0.90	0.25	Oracle-GMM	3.650	5.850	7.850	3.381	5.625	7.675
1000	0.90	0.25	Full-GMM	3.675	6.150	8.575	3.555	6.123	8.432
1000	0.90	0.25	DML-IVQR-BC	3.450	5.400	7.300	3.114	5.249	7.374
1000	0.90	0.10	Oracle-GMM	4.000	7.900	11.800	3.712	7.241	10.771
1000	0.90	0.10	Full-GMM	3.900	7.700	11.500	3.817	7.448	11.031
1000	0.90	0.10	DML-IVQR-BC	4.000	7.700	11.600	3.644	7.042	10.395

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table C.2. Grid-domain diagnostic: added accepted measure

n	τ	κ	**Estimator**	**Med.** $A_0 \to A_1$	**Mean** $A_0 \to A_1$	**Med.** $A_1 \to A_2$	**Mean** $A_1 \to A_2$
1000	0.10	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.10	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.10	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.10	0.25	Oracle-GMM	2.000	1.974	2.000	1.844
1000	0.10	0.25	Full-GMM	2.500	2.470	2.200	2.263
1000	0.10	0.25	DML-IVQR-BC	2.150	2.433	2.000	2.205
1000	0.10	0.10	Oracle-GMM	4.000	3.636	4.000	3.621
1000	0.10	0.10	Full-GMM	3.850	3.610	3.875	3.580
1000	0.10	0.10	DML-IVQR-BC	3.700	3.374	3.450	3.120
1000	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.25	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.25	0.25	Oracle-GMM	0.275	0.933	0.000	0.806
1000	0.25	0.25	Full-GMM	1.025	1.251	0.600	0.978
1000	0.25	0.25	DML-IVQR-BC	1.750	1.500	2.000	1.582
1000	0.25	0.10	Oracle-GMM	3.900	3.207	3.900	3.174
1000	0.25	0.10	Full-GMM	3.800	3.296	3.800	3.192
1000	0.25	0.10	DML-IVQR-BC	3.950	3.275	3.625	3.058
1000	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.50	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.50	0.25	Oracle-GMM	0.000	0.722	0.000	0.547
1000	0.50	0.25	Full-GMM	0.100	0.803	0.000	0.630
1000	0.50	0.25	DML-IVQR-BC	0.250	0.895	0.400	0.921
1000	0.50	0.10	Oracle-GMM	3.900	3.119	4.000	3.047
1000	0.50	0.10	Full-GMM	3.900	3.182	3.900	3.099
1000	0.50	0.10	DML-IVQR-BC	3.850	3.134	2.850	2.914
1000	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.75	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.75	0.25	Oracle-GMM	1.325	1.317	0.400	1.045
1000	0.75	0.25	Full-GMM	1.375	1.440	0.900	1.211
1000	0.75	0.25	DML-IVQR-BC	0.525	1.193	1.200	1.192
1000	0.75	0.10	Oracle-GMM	4.000	3.329	4.000	3.358
1000	0.75	0.10	Full-GMM	3.900	3.362	3.800	3.244
1000	0.75	0.10	DML-IVQR-BC	3.575	3.136	3.600	3.045
1000	0.90	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.90	1.00	Full-GMM	0.000	0.005	0.000	0.000
1000	0.90	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.90	0.25	Oracle-GMM	2.050	2.244	2.000	2.050

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TableC.2. Grid-domain diagnostic: added accepted measure (Continued)

n	τ	κ	**Estimator**	**Med.** $A_0 \to A_1$	**Mean** $A_0 \to A_1$	**Med.** $A_1 \to A_2$	**Mean** $A_1 \to A_2$
1000	0.90	0.25	Full-GMM	2.600	2.568	2.200	2.309
1000	0.90	0.25	DML-IVQR-BC	2.000	2.135	2.000	2.125
1000	0.90	0.10	Oracle-GMM	4.000	3.529	4.000	3.530
1000	0.90	0.10	Full-GMM	3.800	3.631	3.825	3.583
1000	0.90	0.10	DML-IVQR-BC	4.000	3.399	4.000	3.353

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table C.3. Grid-domain diagnostic: boundary and full-grid acceptance

n	τ	κ	**Estimator**	**Boundary** A_0	**Boundary** A_1	**Boundary** A_2	**All** A_0	**All** A_1	**All** A_2
1000	0.10	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	0.25	Oracle-GMM	0.880	0.770	0.740	0.150	0.070	0.050
1000	0.10	0.25	Full-GMM	0.920	0.840	0.750	0.090	0.030	0.020
1000	0.10	0.25	DML-IVQR-BC	0.950	0.980	0.980	0.240	0.060	0.050
1000	0.10	0.10	Oracle-GMM	1.000	1.000	1.000	0.580	0.430	0.390
1000	0.10	0.10	Full-GMM	0.990	0.980	0.990	0.430	0.210	0.150
1000	0.10	0.10	DML-IVQR-BC	1.000	0.990	0.990	0.560	0.260	0.200
1000	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	0.25	Oracle-GMM	0.400	0.360	0.270	0.070	0.050	0.030
1000	0.25	0.25	Full-GMM	0.560	0.390	0.390	0.060	0.030	0.010
1000	0.25	0.25	DML-IVQR-BC	0.610	0.740	0.740	0.090	0.050	0.020
1000	0.25	0.10	Oracle-GMM	0.940	0.930	0.940	0.470	0.380	0.320
1000	0.25	0.10	Full-GMM	0.960	0.930	0.900	0.320	0.250	0.210
1000	0.25	0.10	DML-IVQR-BC	0.980	0.990	0.990	0.550	0.400	0.290
1000	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.25	Oracle-GMM	0.390	0.200	0.150	0.050	0.040	0.020
1000	0.50	0.25	Full-GMM	0.400	0.240	0.260	0.050	0.020	0.020
1000	0.50	0.25	DML-IVQR-BC	0.440	0.340	0.500	0.060	0.020	0.020
1000	0.50	0.10	Oracle-GMM	0.910	0.890	0.830	0.420	0.350	0.320
1000	0.50	0.10	Full-GMM	0.940	0.890	0.850	0.450	0.320	0.280
1000	0.50	0.10	DML-IVQR-BC	0.900	0.930	0.960	0.450	0.360	0.300
1000	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	0.25	Oracle-GMM	0.620	0.470	0.370	0.070	0.050	0.040
1000	0.75	0.25	Full-GMM	0.650	0.550	0.530	0.080	0.010	0.000
1000	0.75	0.25	DML-IVQR-BC	0.550	0.450	0.550	0.090	0.060	0.030
1000	0.75	0.10	Oracle-GMM	0.940	0.960	0.930	0.500	0.400	0.380
1000	0.75	0.10	Full-GMM	0.980	0.920	0.920	0.420	0.270	0.220
1000	0.75	0.10	DML-IVQR-BC	0.950	0.970	0.980	0.470	0.360	0.310
1000	0.90	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	0.25	Oracle-GMM	0.850	0.760	0.760	0.250	0.090	0.060

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TableC.3. Grid-domain diagnostic: boundary and full-grid acceptance (Continued)

n	τ	κ	**Estimator**	**Boundary** A_0	**Boundary** A_1	**Boundary** A_2	**All** A_0	**All** A_1	**All** A_2
1000	0.90	0.25	Full-GMM	0.960	0.800	0.820	0.110	0.060	0.030
1000	0.90	0.25	DML-IVQR-BC	0.820	0.770	0.790	0.210	0.100	0.060
1000	0.90	0.10	Oracle-GMM	0.980	1.000	1.000	0.620	0.450	0.430
1000	0.90	0.10	Full-GMM	1.000	0.990	0.990	0.380	0.110	0.060
1000	0.90	0.10	DML-IVQR-BC	0.990	0.980	0.990	0.510	0.394	0.343

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table C.4. Grid-domain diagnostic: profiles and numerical errors

n	τ	κ	**Estimator**	R	**Succ.** A_0	**Succ.** A_1	**Succ.** A_2	**Fail** A_0	**Fail** A_1	**Fail** A_2	**Stage 1**	**Stage 2**
1000	0.10	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.90	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0

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TableC.4. Grid-domain diagnostic: profiles and numerical errors (Continued)

n	τ	κ	**Estimator**	R	**Succ.** A_0	**Succ.** A_1	**Succ.** A_2	**Fail** A_0	**Fail** A_1	**Fail** A_2	**Stage 1**	**Stage 2**
1000	0.90	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	DML-IVQR-BC	100	100	99	99	0	1	1	1	0

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table C.5. Grid-domain diagnostic: stabilization within investigated domains

n	τ	κ	**Estimator**	**Investigated-domain status**
1000	0.10	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.10	1.00	Full-GMM	near-stable within investigated domains
1000	0.10	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.10	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.10	0.25	Full-GMM	not stabilized within investigated domains
1000	0.10	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.10	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.10	0.10	Full-GMM	not stabilized within investigated domains
1000	0.10	0.10	DML-IVQR-BC	not stabilized within investigated domains
1000	0.25	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.25	1.00	Full-GMM	near-stable within investigated domains
1000	0.25	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.25	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.25	0.25	Full-GMM	not stabilized within investigated domains
1000	0.25	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.25	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.25	0.10	Full-GMM	not stabilized within investigated domains
1000	0.25	0.10	DML-IVQR-BC	not stabilized within investigated domains
1000	0.50	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.50	1.00	Full-GMM	near-stable within investigated domains
1000	0.50	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.50	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.50	0.25	Full-GMM	not stabilized within investigated domains
1000	0.50	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.50	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.50	0.10	Full-GMM	not stabilized within investigated domains
1000	0.50	0.10	DML-IVQR-BC	not stabilized within investigated domains
1000	0.75	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.75	1.00	Full-GMM	near-stable within investigated domains
1000	0.75	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.75	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.75	0.25	Full-GMM	not stabilized within investigated domains
1000	0.75	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.75	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.75	0.10	Full-GMM	not stabilized within investigated domains
1000	0.75	0.10	DML-IVQR-BC	not stabilized within investigated domains
1000	0.90	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.90	1.00	Full-GMM	near-stable within investigated domains
1000	0.90	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.90	0.25	Oracle-GMM	not stabilized within investigated domains

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TableC.5. Grid-domain diagnostic: stabilization within investigated domains (Continued)

n	τ	κ	**Estimator**	**Investigated-domain status**
1000	0.90	0.25	Full-GMM	not stabilized within investigated domains
1000	0.90	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.90	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.90	0.10	Full-GMM	not stabilized within investigated domains
1000	0.90	0.10	DML-IVQR-BC	not stabilized within investigated domains

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Appendix D Coverage Forensic Audit

Table D.1. Full empirical coverage and size

n	τ	κ	**Estimator**	R	**Coverage**	**Size**	**MCSE**	**MC 95% interval**	**.95 inside**
500	0.10	1.00	Oracle-GMM	500	0.858	0.142	0.016	[0.825, 0.886]	No
500	0.10	1.00	Full-GMM	500	0.442	0.558	0.022	[0.399, 0.486]	No
500	0.10	1.00	DML-IVQR-BC	500	0.958	0.042	0.009	[0.937, 0.972]	Yes
500	0.10	0.50	Oracle-GMM	500	0.904	0.096	0.013	[0.875, 0.927]	No
500	0.10	0.50	Full-GMM	500	0.816	0.184	0.017	[0.780, 0.848]	No
500	0.10	0.50	DML-IVQR-BC	500	0.928	0.072	0.012	[0.902, 0.948]	No
500	0.10	0.25	Oracle-GMM	500	0.944	0.056	0.010	[0.920, 0.961]	Yes
500	0.10	0.25	Full-GMM	500	0.928	0.072	0.012	[0.902, 0.948]	No
500	0.10	0.25	DML-IVQR-BC	500	0.930	0.070	0.011	[0.904, 0.949]	No
500	0.10	0.10	Oracle-GMM	500	0.954	0.046	0.009	[0.932, 0.969]	Yes
500	0.10	0.10	Full-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
500	0.10	0.10	DML-IVQR-BC	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
500	0.25	1.00	Oracle-GMM	500	0.886	0.114	0.014	[0.855, 0.911]	No
500	0.25	1.00	Full-GMM	500	0.788	0.212	0.018	[0.750, 0.822]	No
500	0.25	1.00	DML-IVQR-BC	500	0.976	0.024	0.007	[0.959, 0.986]	No
500	0.25	0.50	Oracle-GMM	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
500	0.25	0.50	Full-GMM	500	0.920	0.080	0.012	[0.893, 0.941]	No
500	0.25	0.50	DML-IVQR-BC	500	0.960	0.040	0.009	[0.939, 0.974]	Yes
500	0.25	0.25	Oracle-GMM	500	0.938	0.062	0.011	[0.913, 0.956]	Yes
500	0.25	0.25	Full-GMM	500	0.938	0.062	0.011	[0.913, 0.956]	Yes
500	0.25	0.25	DML-IVQR-BC	500	0.962	0.038	0.009	[0.941, 0.976]	Yes
500	0.25	0.10	Oracle-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes

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TableD.1. Full empirical coverage and size (Continued)

n	τ	κ	Estimator	R	Coverage	Size	MCSE	MC 95% interval	.95 inside
500	0.25	0.10	Full-GMM	500	0.930	0.070	0.011	[0.904, 0.949]	No
500	0.25	0.10	DML-IVQR-BC	500	0.958	0.042	0.009	[0.937, 0.972]	Yes
500	0.50	1.00	Oracle-GMM	500	0.908	0.092	0.013	[0.879, 0.930]	No
500	0.50	1.00	Full-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
500	0.50	1.00	DML-IVQR-BC	500	0.922	0.078	0.012	[0.895, 0.942]	No
500	0.50	0.50	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.50	0.50	Full-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
500	0.50	0.50	DML-IVQR-BC	500	0.966	0.034	0.008	[0.946, 0.979]	Yes
500	0.50	0.25	Oracle-GMM	500	0.944	0.056	0.010	[0.920, 0.961]	Yes
500	0.50	0.25	Full-GMM	500	0.962	0.038	0.009	[0.941, 0.976]	Yes
500	0.50	0.25	DML-IVQR-BC	500	0.966	0.034	0.008	[0.946, 0.979]	Yes
500	0.50	0.10	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.50	0.10	Full-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.50	0.10	DML-IVQR-BC	500	0.958	0.042	0.009	[0.937, 0.972]	Yes
500	0.75	1.00	Oracle-GMM	500	0.912	0.088	0.013	[0.884, 0.934]	No
500	0.75	1.00	Full-GMM	500	0.832	0.168	0.017	[0.797, 0.862]	No
500	0.75	1.00	DML-IVQR-BC	500	0.770	0.230	0.019	[0.731, 0.805]	No
500	0.75	0.50	Oracle-GMM	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
500	0.75	0.50	Full-GMM	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
500	0.75	0.50	DML-IVQR-BC	500	0.926	0.074	0.012	[0.900, 0.946]	No
500	0.75	0.25	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.75	0.25	Full-GMM	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
500	0.75	0.25	DML-IVQR-BC	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
500	0.75	0.10	Oracle-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
500	0.75	0.10	Full-GMM	500	0.956	0.044	0.009	[0.934, 0.971]	Yes
500	0.75	0.10	DML-IVQR-BC	500	0.938	0.062	0.011	[0.913, 0.956]	Yes
500	0.90	1.00	Oracle-GMM	500	0.882	0.118	0.014	[0.851, 0.907]	No
500	0.90	1.00	Full-GMM	500	0.556	0.444	0.022	[0.512, 0.599]	No
500	0.90	1.00	DML-IVQR-BC	500	0.658	0.342	0.021	[0.615, 0.698]	No
500	0.90	0.50	Oracle-GMM	500	0.944	0.056	0.010	[0.920, 0.961]	Yes
500	0.90	0.50	Full-GMM	500	0.840	0.160	0.016	[0.805, 0.870]	No
500	0.90	0.50	DML-IVQR-BC	500	0.898	0.102	0.014	[0.868, 0.922]	No
500	0.90	0.25	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.90	0.25	Full-GMM	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
500	0.90	0.25	DML-IVQR-BC	500	0.930	0.070	0.011	[0.904, 0.949]	No
500	0.90	0.10	Oracle-GMM	500	0.950	0.050	0.010	[0.927, 0.966]	Yes
500	0.90	0.10	Full-GMM	500	0.944	0.056	0.010	[0.920, 0.961]	Yes
500	0.90	0.10	DML-IVQR-BC	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
1000	0.10	1.00	Oracle-GMM	500	0.860	0.140	0.016	[0.827, 0.888]	No
1000	0.10	1.00	Full-GMM	500	0.502	0.498	0.022	[0.458, 0.546]	No

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TableD.1. Full empirical coverage and size (Continued)

n	τ	κ	**Estimator**	R	**Coverage**	**Size**	**MCSE**	**MC 95% interval**	**.95 inside**
1000	0.10	1.00	DML-IVQR-BC	500	0.968	0.032	0.008	[0.949, 0.980]	Yes
1000	0.10	0.50	Oracle-GMM	500	0.926	0.074	0.012	[0.900, 0.946]	No
1000	0.10	0.50	Full-GMM	500	0.804	0.196	0.018	[0.767, 0.836]	No
1000	0.10	0.50	DML-IVQR-BC	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.10	0.25	Oracle-GMM	500	0.930	0.070	0.011	[0.904, 0.949]	No
1000	0.10	0.25	Full-GMM	500	0.932	0.068	0.011	[0.906, 0.951]	Yes
1000	0.10	0.25	DML-IVQR-BC	500	0.950	0.050	0.010	[0.927, 0.966]	Yes
1000	0.10	0.10	Oracle-GMM	500	0.946	0.054	0.010	[0.923, 0.963]	Yes
1000	0.10	0.10	Full-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
1000	0.10	0.10	DML-IVQR-BC	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
1000	0.25	1.00	Oracle-GMM	500	0.874	0.126	0.015	[0.842, 0.900]	No
1000	0.25	1.00	Full-GMM	500	0.790	0.210	0.018	[0.752, 0.823]	No
1000	0.25	1.00	DML-IVQR-BC	500	0.894	0.106	0.014	[0.864, 0.918]	No
1000	0.25	0.50	Oracle-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
1000	0.25	0.50	Full-GMM	500	0.908	0.092	0.013	[0.879, 0.930]	No
1000	0.25	0.50	DML-IVQR-BC	500	0.952	0.048	0.010	[0.930, 0.968]	Yes
1000	0.25	0.25	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.25	0.25	Full-GMM	500	0.956	0.044	0.009	[0.934, 0.971]	Yes
1000	0.25	0.25	DML-IVQR-BC	500	0.962	0.038	0.009	[0.941, 0.976]	Yes
1000	0.25	0.10	Oracle-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
1000	0.25	0.10	Full-GMM	500	0.962	0.038	0.009	[0.941, 0.976]	Yes
1000	0.25	0.10	DML-IVQR-BC	500	0.954	0.046	0.009	[0.932, 0.969]	Yes
1000	0.50	1.00	Oracle-GMM	500	0.872	0.128	0.015	[0.840, 0.898]	No
1000	0.50	1.00	Full-GMM	500	0.896	0.104	0.014	[0.866, 0.920]	No
1000	0.50	1.00	DML-IVQR-BC	500	0.776	0.224	0.019	[0.737, 0.810]	No
1000	0.50	0.50	Oracle-GMM	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
1000	0.50	0.50	Full-GMM	500	0.946	0.054	0.010	[0.923, 0.963]	Yes
1000	0.50	0.50	DML-IVQR-BC	500	0.918	0.082	0.012	[0.891, 0.939]	No
1000	0.50	0.25	Oracle-GMM	500	0.946	0.054	0.010	[0.923, 0.963]	Yes
1000	0.50	0.25	Full-GMM	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
1000	0.50	0.25	DML-IVQR-BC	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.50	0.10	Oracle-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
1000	0.50	0.10	Full-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.50	0.10	DML-IVQR-BC	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
1000	0.75	1.00	Oracle-GMM	500	0.860	0.140	0.016	[0.827, 0.888]	No
1000	0.75	1.00	Full-GMM	500	0.790	0.210	0.018	[0.752, 0.823]	No
1000	0.75	1.00	DML-IVQR-BC	500	0.632	0.368	0.022	[0.589, 0.673]	No
1000	0.75	0.50	Oracle-GMM	500	0.916	0.084	0.012	[0.888, 0.937]	No
1000	0.75	0.50	Full-GMM	500	0.922	0.078	0.012	[0.895, 0.942]	No
1000	0.75	0.50	DML-IVQR-BC	500	0.842	0.158	0.016	[0.807, 0.871]	No

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TableD.1. Full empirical coverage and size (Continued)

n	τ	κ	**Estimator**	R	**Coverage**	**Size**	**MCSE**	**MC 95% interval**	**.95 inside**
1000	0.75	0.25	Oracle-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
1000	0.75	0.25	Full-GMM	500	0.928	0.072	0.012	[0.902, 0.948]	No
1000	0.75	0.25	DML-IVQR-BC	500	0.916	0.084	0.012	[0.888, 0.937]	No
1000	0.75	0.10	Oracle-GMM	500	0.946	0.054	0.010	[0.923, 0.963]	Yes
1000	0.75	0.10	Full-GMM	500	0.926	0.074	0.012	[0.900, 0.946]	No
1000	0.75	0.10	DML-IVQR-BC	500	0.930	0.070	0.011	[0.904, 0.949]	No
1000	0.90	1.00	Oracle-GMM	500	0.904	0.096	0.013	[0.875, 0.927]	No
1000	0.90	1.00	Full-GMM	500	0.530	0.470	0.022	[0.486, 0.573]	No
1000	0.90	1.00	DML-IVQR-BC	500	0.498	0.502	0.022	[0.454, 0.542]	No
1000	0.90	0.50	Oracle-GMM	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
1000	0.90	0.50	Full-GMM	500	0.834	0.166	0.017	[0.799, 0.864]	No
1000	0.90	0.50	DML-IVQR-BC	500	0.844	0.156	0.016	[0.810, 0.873]	No
1000	0.90	0.25	Oracle-GMM	500	0.952	0.048	0.010	[0.930, 0.968]	Yes
1000	0.90	0.25	Full-GMM	500	0.920	0.080	0.012	[0.893, 0.941]	No
1000	0.90	0.25	DML-IVQR-BC	500	0.922	0.078	0.012	[0.895, 0.942]	No
1000	0.90	0.10	Oracle-GMM	500	0.964	0.036	0.008	[0.944, 0.977]	Yes
1000	0.90	0.10	Full-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.90	0.10	DML-IVQR-BC	500	0.940	0.060	0.011	[0.916, 0.958]	Yes

Note: Coverage is evaluated directly at $\alpha_0(\tau)$ using the unchanged χ^2_2 cutoff. Intervals are Wilson 95% Monte Carlo intervals.

Appendix E Additional $W_N(a)$ Profiles

Table E.1. Full accepted-set informativeness: measure and grid share

n	τ	κ	**Estimator**	**Med. measure**	**Mean measure**	**Med. share**	**Mean share**	**Mean points**
500	0.10	1.00	Oracle-GMM	0.700	0.736	0.171	0.180	7.37
500	0.10	1.00	Full-GMM	1.600	1.615	0.390	0.395	16.21
500	0.10	1.00	DML-IVQR-BC	2.300	2.303	0.561	0.564	23.13
500	0.10	0.50	Oracle-GMM	2.500	2.437	0.610	0.603	24.70
500	0.10	0.50	Full-GMM	3.050	3.005	0.756	0.745	30.55
500	0.10	0.50	DML-IVQR-BC	3.750	3.576	0.927	0.888	36.43
500	0.10	0.25	Oracle-GMM	3.850	3.497	0.951	0.873	35.78
500	0.10	0.25	Full-GMM	3.700	3.620	0.927	0.904	37.06

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TableE.1. Full accepted-set informativeness: measure and grid share (Continued)

n	τ	κ	Estimator	Med. measure	Mean measure	Med. share	Mean share	Mean points
500	0.10	0.25	DML-IVQR-BC	3.900	3.755	0.976	0.936	38.39
500	0.10	0.10	Oracle-GMM	4.000	3.747	1.000	0.936	38.39
500	0.10	0.10	Full-GMM	3.900	3.766	0.976	0.941	38.60
500	0.10	0.10	DML-IVQR-BC	4.000	3.757	1.000	0.938	38.47
500	0.25	1.00	Oracle-GMM	0.400	0.423	0.098	0.103	4.23
500	0.25	1.00	Full-GMM	0.800	0.824	0.195	0.201	8.24
500	0.25	1.00	DML-IVQR-BC	0.700	0.702	0.171	0.171	7.02
500	0.25	0.50	Oracle-GMM	1.200	1.370	0.293	0.335	13.75
500	0.25	0.50	Full-GMM	2.000	2.120	0.488	0.521	21.36
500	0.25	0.50	DML-IVQR-BC	2.100	2.132	0.512	0.525	21.51
500	0.25	0.25	Oracle-GMM	3.250	3.094	0.805	0.770	31.56
500	0.25	0.25	Full-GMM	3.550	3.342	0.878	0.833	34.13
500	0.25	0.25	DML-IVQR-BC	3.450	3.387	0.854	0.842	34.53
500	0.25	0.10	Oracle-GMM	4.000	3.695	1.000	0.923	37.86
500	0.25	0.10	Full-GMM	3.900	3.708	0.976	0.926	37.97
500	0.25	0.10	DML-IVQR-BC	4.000	3.726	1.000	0.930	38.15
500	0.50	1.00	Oracle-GMM	0.400	0.375	0.098	0.092	3.75
500	0.50	1.00	Full-GMM	0.700	0.676	0.171	0.165	6.76
500	0.50	1.00	DML-IVQR-BC	0.500	0.473	0.122	0.115	4.73
500	0.50	0.50	Oracle-GMM	1.000	1.112	0.244	0.272	11.15
500	0.50	0.50	Full-GMM	1.600	1.715	0.390	0.420	17.22
500	0.50	0.50	DML-IVQR-BC	1.200	1.339	0.293	0.327	13.43
500	0.50	0.25	Oracle-GMM	2.925	2.863	0.732	0.711	29.16
500	0.50	0.25	Full-GMM	3.350	3.185	0.829	0.792	32.47
500	0.50	0.25	DML-IVQR-BC	3.050	2.997	0.756	0.745	30.53
500	0.50	0.10	Oracle-GMM	4.000	3.656	1.000	0.913	37.45
500	0.50	0.10	Full-GMM	3.900	3.670	0.976	0.917	37.59
500	0.50	0.10	DML-IVQR-BC	4.000	3.652	1.000	0.912	37.39
500	0.75	1.00	Oracle-GMM	0.500	0.497	0.122	0.121	4.97
500	0.75	1.00	Full-GMM	1.000	0.978	0.244	0.239	9.78

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TableE.1. Full accepted-set informativeness: measure and grid share (Continued)

n	τ	κ	Estimator	Med. measure	Mean measure	Med. share	Mean share	Mean points
500	0.75	1.00	DML-IVQR-BC	0.500	0.495	0.122	0.121	4.95
500	0.75	0.50	Oracle-GMM	1.500	1.676	0.366	0.411	16.86
500	0.75	0.50	Full-GMM	2.400	2.397	0.585	0.590	24.20
500	0.75	0.50	DML-IVQR-BC	1.300	1.489	0.317	0.365	14.98
500	0.75	0.25	Oracle-GMM	3.450	3.241	0.854	0.806	33.06
500	0.75	0.25	Full-GMM	3.650	3.473	0.902	0.865	35.46
500	0.75	0.25	DML-IVQR-BC	3.350	3.089	0.829	0.769	31.52
500	0.75	0.10	Oracle-GMM	4.000	3.731	1.000	0.932	38.21
500	0.75	0.10	Full-GMM	3.900	3.760	0.976	0.939	38.52
500	0.75	0.10	DML-IVQR-BC	4.000	3.661	1.000	0.915	37.50
500	0.90	1.00	Oracle-GMM	1.000	1.131	0.244	0.276	11.31
500	0.90	1.00	Full-GMM	2.000	2.013	0.488	0.494	20.26
500	0.90	1.00	DML-IVQR-BC	0.700	0.793	0.171	0.193	7.93
500	0.90	0.50	Oracle-GMM	3.300	2.944	0.817	0.729	29.89
500	0.90	0.50	Full-GMM	3.350	3.293	0.829	0.818	33.54
500	0.90	0.50	DML-IVQR-BC	2.300	2.347	0.561	0.580	23.78
500	0.90	0.25	Oracle-GMM	3.900	3.617	0.976	0.903	37.00
500	0.90	0.25	Full-GMM	3.800	3.715	0.951	0.928	38.04
500	0.90	0.25	DML-IVQR-BC	3.750	3.401	0.927	0.849	34.80
500	0.90	0.10	Oracle-GMM	4.000	3.779	1.000	0.944	38.72
500	0.90	0.10	Full-GMM	3.900	3.816	0.976	0.954	39.11
500	0.90	0.10	DML-IVQR-BC	4.000	3.704	1.000	0.926	37.95
1000	0.10	1.00	Oracle-GMM	0.400	0.379	0.098	0.092	3.79
1000	0.10	1.00	Full-GMM	0.800	0.766	0.195	0.187	7.66
1000	0.10	1.00	DML-IVQR-BC	0.600	0.668	0.146	0.163	6.68
1000	0.10	0.50	Oracle-GMM	1.000	1.291	0.244	0.316	12.97
1000	0.10	0.50	Full-GMM	1.900	1.941	0.463	0.477	19.56
1000	0.10	0.50	DML-IVQR-BC	2.225	2.283	0.549	0.562	23.04
1000	0.10	0.25	Oracle-GMM	3.450	3.077	0.854	0.765	31.36
1000	0.10	0.25	Full-GMM	3.500	3.318	0.878	0.826	33.87

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TableE.1. Full accepted-set informativeness: measure and grid share (Continued)

n	τ	κ	Estimator	Med. measure	Mean measure	Med. share	Mean share	Mean points
1000	0.10	0.25	DML-IVQR-BC	3.750	3.577	0.927	0.890	36.49
1000	0.10	0.10	Oracle-GMM	4.000	3.740	1.000	0.934	38.30
1000	0.10	0.10	Full-GMM	3.900	3.781	0.976	0.945	38.73
1000	0.10	0.10	DML-IVQR-BC	4.000	3.826	1.000	0.955	39.15
1000	0.25	1.00	Oracle-GMM	0.200	0.249	0.049	0.061	2.49
1000	0.25	1.00	Full-GMM	0.400	0.433	0.098	0.106	4.33
1000	0.25	1.00	DML-IVQR-BC	0.300	0.341	0.073	0.083	3.41
1000	0.25	0.50	Oracle-GMM	0.600	0.659	0.146	0.161	6.59
1000	0.25	0.50	Full-GMM	1.000	1.020	0.244	0.249	10.20
1000	0.25	0.50	DML-IVQR-BC	0.800	0.915	0.195	0.223	9.16
1000	0.25	0.25	Oracle-GMM	2.200	2.271	0.537	0.561	23.02
1000	0.25	0.25	Full-GMM	2.650	2.611	0.659	0.647	26.51
1000	0.25	0.25	DML-IVQR-BC	2.850	2.669	0.707	0.661	27.09
1000	0.25	0.10	Oracle-GMM	3.900	3.544	0.976	0.885	36.27
1000	0.25	0.10	Full-GMM	3.900	3.627	0.976	0.906	37.14
1000	0.25	0.10	DML-IVQR-BC	3.900	3.637	0.976	0.908	37.22
1000	0.50	1.00	Oracle-GMM	0.200	0.228	0.049	0.056	2.28
1000	0.50	1.00	Full-GMM	0.400	0.369	0.098	0.090	3.69
1000	0.50	1.00	DML-IVQR-BC	0.300	0.261	0.073	0.064	2.61
1000	0.50	0.50	Oracle-GMM	0.500	0.562	0.122	0.137	5.62
1000	0.50	0.50	Full-GMM	0.800	0.833	0.195	0.203	8.33
1000	0.50	0.50	DML-IVQR-BC	0.600	0.641	0.146	0.156	6.41
1000	0.50	0.25	Oracle-GMM	1.900	1.946	0.463	0.480	19.67
1000	0.50	0.25	Full-GMM	2.200	2.277	0.537	0.562	23.03
1000	0.50	0.25	DML-IVQR-BC	2.025	2.068	0.500	0.510	20.93
1000	0.50	0.10	Oracle-GMM	3.900	3.501	0.976	0.874	35.83
1000	0.50	0.10	Full-GMM	3.800	3.520	0.951	0.879	36.02
1000	0.50	0.10	DML-IVQR-BC	3.900	3.514	0.976	0.877	35.96
1000	0.75	1.00	Oracle-GMM	0.300	0.280	0.073	0.068	2.80
1000	0.75	1.00	Full-GMM	0.500	0.518	0.122	0.126	5.18

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TableE.1. Full accepted-set informativeness: measure and grid share (Continued)

n	τ	κ	Estimator	Med. measure	Mean measure	Med. share	Mean share	Mean points
1000	0.75	1.00	DML-IVQR-BC	0.300	0.283	0.073	0.069	2.83
1000	0.75	0.50	Oracle-GMM	0.700	0.830	0.171	0.203	8.31
1000	0.75	0.50	Full-GMM	1.100	1.217	0.268	0.297	12.18
1000	0.75	0.50	DML-IVQR-BC	0.700	0.737	0.171	0.180	7.37
1000	0.75	0.25	Oracle-GMM	2.650	2.488	0.659	0.615	25.22
1000	0.75	0.25	Full-GMM	2.900	2.791	0.707	0.691	28.32
1000	0.75	0.25	DML-IVQR-BC	2.250	2.266	0.561	0.560	22.98
1000	0.75	0.10	Oracle-GMM	3.900	3.609	0.976	0.901	36.92
1000	0.75	0.10	Full-GMM	3.900	3.647	0.976	0.911	37.35
1000	0.75	0.10	DML-IVQR-BC	3.850	3.519	0.951	0.878	36.01
1000	0.90	1.00	Oracle-GMM	0.500	0.515	0.122	0.126	5.15
1000	0.90	1.00	Full-GMM	1.000	1.035	0.244	0.252	10.35
1000	0.90	1.00	DML-IVQR-BC	0.400	0.446	0.098	0.109	4.46
1000	0.90	0.50	Oracle-GMM	1.575	1.864	0.390	0.458	18.78
1000	0.90	0.50	Full-GMM	2.450	2.417	0.610	0.595	24.39
1000	0.90	0.50	DML-IVQR-BC	1.100	1.414	0.268	0.346	14.21
1000	0.90	0.25	Oracle-GMM	3.650	3.279	0.902	0.815	33.42
1000	0.90	0.25	Full-GMM	3.600	3.462	0.902	0.862	35.35
1000	0.90	0.25	DML-IVQR-BC	3.350	2.986	0.829	0.742	30.44
1000	0.90	0.10	Oracle-GMM	4.000	3.690	1.000	0.922	37.80
1000	0.90	0.10	Full-GMM	3.900	3.767	0.976	0.941	38.59
1000	0.90	0.10	DML-IVQR-BC	4.000	3.604	1.000	0.900	36.92

Note: Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$. The measure is finite-domain and is not an unrestricted confidence-region length.

Table E.2. Full accepted-set informativeness: boundary diagnostics, $n = 500$

n	τ	κ	**Estimator**	**Left**	**Right**	**Either**	**Both**	**All**
500	0.10	1.00	Oracle-GMM	0.004	0.000	0.004	0.000	0.000
500	0.10	1.00	Full-GMM	0.058	0.056	0.106	0.008	0.000
500	0.10	1.00	DML-IVQR-BC	0.036	0.180	0.212	0.004	0.000
500	0.10	0.50	Oracle-GMM	0.288	0.376	0.552	0.112	0.038
500	0.10	0.50	Full-GMM	0.482	0.514	0.740	0.256	0.006
500	0.10	0.50	DML-IVQR-BC	0.402	0.926	0.954	0.374	0.216
500	0.10	0.25	Oracle-GMM	0.788	0.824	0.962	0.650	0.370
500	0.10	0.25	Full-GMM	0.838	0.864	0.972	0.730	0.146
500	0.10	0.25	DML-IVQR-BC	0.702	0.980	0.996	0.686	0.440
500	0.10	0.10	Oracle-GMM	0.928	0.926	0.994	0.860	0.564
500	0.10	0.10	Full-GMM	0.936	0.940	0.990	0.886	0.292
500	0.10	0.10	DML-IVQR-BC	0.840	0.952	0.992	0.800	0.514
500	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
500	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
500	0.25	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
500	0.25	0.50	Oracle-GMM	0.048	0.060	0.102	0.006	0.000
500	0.25	0.50	Full-GMM	0.168	0.158	0.298	0.028	0.000
500	0.25	0.50	DML-IVQR-BC	0.020	0.354	0.364	0.010	0.004
500	0.25	0.25	Oracle-GMM	0.592	0.652	0.836	0.408	0.190
500	0.25	0.25	Full-GMM	0.684	0.734	0.906	0.512	0.172
500	0.25	0.25	DML-IVQR-BC	0.428	0.906	0.946	0.388	0.204
500	0.25	0.10	Oracle-GMM	0.880	0.924	0.996	0.808	0.514
500	0.25	0.10	Full-GMM	0.876	0.916	0.988	0.804	0.364
500	0.25	0.10	DML-IVQR-BC	0.806	0.966	0.990	0.782	0.502
500	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
500	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
500	0.50	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
500	0.50	0.50	Oracle-GMM	0.042	0.018	0.056	0.004	0.000
500	0.50	0.50	Full-GMM	0.096	0.044	0.132	0.008	0.000
500	0.50	0.50	DML-IVQR-BC	0.010	0.072	0.078	0.004	0.000
500	0.50	0.25	Oracle-GMM	0.560	0.490	0.706	0.344	0.168
500	0.50	0.25	Full-GMM	0.644	0.602	0.798	0.448	0.150
500	0.50	0.25	DML-IVQR-BC	0.380	0.742	0.810	0.312	0.150
500	0.50	0.10	Oracle-GMM	0.898	0.868	0.968	0.798	0.540
500	0.50	0.10	Full-GMM	0.884	0.882	0.974	0.792	0.404
500	0.50	0.10	DML-IVQR-BC	0.818	0.936	0.984	0.770	0.540
500	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
500	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
500	0.75	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
500	0.75	0.50	Oracle-GMM	0.172	0.030	0.196	0.006	0.000

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TableE.2. Full accepted-set informativeness: boundary diagnostics, n = 500 (Continued)

n	τ	κ	**Estimator**	**Left**	**Right**	**Either**	**Both**	**All**
500	0.75	0.50	Full-GMM	0.300	0.164	0.394	0.070	0.010
500	0.75	0.50	DML-IVQR-BC	0.076	0.098	0.164	0.010	0.000
500	0.75	0.25	Oracle-GMM	0.750	0.552	0.872	0.430	0.224
500	0.75	0.25	Full-GMM	0.766	0.702	0.916	0.552	0.176
500	0.75	0.25	DML-IVQR-BC	0.552	0.714	0.854	0.412	0.226
500	0.75	0.10	Oracle-GMM	0.906	0.896	0.988	0.814	0.564
500	0.75	0.10	Full-GMM	0.928	0.912	0.992	0.848	0.426
500	0.75	0.10	DML-IVQR-BC	0.842	0.946	0.994	0.794	0.524
500	0.90	1.00	Oracle-GMM	0.012	0.004	0.016	0.000	0.000
500	0.90	1.00	Full-GMM	0.160	0.102	0.246	0.016	0.000
500	0.90	1.00	DML-IVQR-BC	0.002	0.000	0.002	0.000	0.000
500	0.90	0.50	Oracle-GMM	0.558	0.346	0.698	0.206	0.098
500	0.90	0.50	Full-GMM	0.640	0.584	0.834	0.390	0.022
500	0.90	0.50	DML-IVQR-BC	0.294	0.338	0.538	0.094	0.028
500	0.90	0.25	Oracle-GMM	0.872	0.790	0.978	0.684	0.446
500	0.90	0.25	Full-GMM	0.898	0.878	0.996	0.780	0.190
500	0.90	0.25	DML-IVQR-BC	0.738	0.838	0.978	0.598	0.326
500	0.90	0.10	Oracle-GMM	0.938	0.914	0.996	0.856	0.616
500	0.90	0.10	Full-GMM	0.936	0.960	0.996	0.900	0.292
500	0.90	0.10	DML-IVQR-BC	0.884	0.946	0.998	0.832	0.558

Note: Boundary rates refer to the endpoints of $A_0 = [-1, 3]$; ‘All’ is the rate at which all 41 primary-grid points are accepted.

Table E.3. Full accepted-set informativeness: boundary diagnostics, n = 1000

n	τ	κ	**Estimator**	**Left**	**Right**	**Either**	**Both**	**All**
1000	0.10	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.10	0.50	Oracle-GMM	0.050	0.072	0.122	0.000	0.000
1000	0.10	0.50	Full-GMM	0.176	0.132	0.292	0.016	0.002
1000	0.10	0.50	DML-IVQR-BC	0.088	0.332	0.398	0.022	0.006
1000	0.10	0.25	Oracle-GMM	0.556	0.626	0.836	0.346	0.166
1000	0.10	0.25	Full-GMM	0.670	0.714	0.908	0.476	0.092
1000	0.10	0.25	DML-IVQR-BC	0.536	0.904	0.956	0.484	0.260
1000	0.10	0.10	Oracle-GMM	0.862	0.936	0.986	0.812	0.548
1000	0.10	0.10	Full-GMM	0.910	0.934	0.998	0.846	0.358
1000	0.10	0.10	DML-IVQR-BC	0.814	0.968	0.992	0.790	0.528
1000	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.25	0.50	Oracle-GMM	0.000	0.004	0.004	0.000	0.000
1000	0.25	0.50	Full-GMM	0.002	0.008	0.010	0.000	0.000
1000	0.25	0.50	DML-IVQR-BC	0.000	0.014	0.014	0.000	0.000
1000	0.25	0.25	Oracle-GMM	0.268	0.340	0.500	0.108	0.052
1000	0.25	0.25	Full-GMM	0.382	0.408	0.606	0.184	0.046
1000	0.25	0.25	DML-IVQR-BC	0.206	0.594	0.652	0.148	0.062
1000	0.25	0.10	Oracle-GMM	0.818	0.840	0.964	0.694	0.454
1000	0.25	0.10	Full-GMM	0.868	0.860	0.974	0.754	0.378
1000	0.25	0.10	DML-IVQR-BC	0.756	0.938	0.978	0.716	0.488
1000	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.50	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.50	Full-GMM	0.002	0.000	0.002	0.000	0.000
1000	0.50	0.50	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.25	Oracle-GMM	0.238	0.182	0.342	0.078	0.036
1000	0.50	0.25	Full-GMM	0.286	0.234	0.420	0.100	0.030
1000	0.50	0.25	DML-IVQR-BC	0.146	0.352	0.408	0.090	0.040
1000	0.50	0.10	Oracle-GMM	0.828	0.812	0.916	0.724	0.418
1000	0.50	0.10	Full-GMM	0.824	0.832	0.934	0.722	0.366
1000	0.50	0.10	DML-IVQR-BC	0.740	0.882	0.926	0.696	0.428
1000	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.75	0.50	Oracle-GMM	0.008	0.000	0.008	0.000	0.000

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TableE.3. Full accepted-set informativeness: boundary diagnostics, $n = 1000$ (Continued)

n	τ	κ	**Estimator**	**Left**	**Right**	**Either**	**Both**	**All**
1000	0.75	0.50	Full-GMM	0.010	0.000	0.010	0.000	0.000
1000	0.75	0.50	DML-IVQR-BC	0.004	0.000	0.004	0.000	0.000
1000	0.75	0.25	Oracle-GMM	0.472	0.222	0.568	0.126	0.058
1000	0.75	0.25	Full-GMM	0.516	0.308	0.628	0.196	0.046
1000	0.75	0.25	DML-IVQR-BC	0.314	0.320	0.498	0.136	0.054
1000	0.75	0.10	Oracle-GMM	0.856	0.818	0.956	0.718	0.458
1000	0.75	0.10	Full-GMM	0.904	0.862	0.974	0.792	0.380
1000	0.75	0.10	DML-IVQR-BC	0.770	0.872	0.960	0.682	0.412
1000	0.90	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.90	1.00	Full-GMM	0.002	0.000	0.002	0.000	0.000
1000	0.90	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.90	0.50	Oracle-GMM	0.236	0.044	0.270	0.010	0.004
1000	0.90	0.50	Full-GMM	0.274	0.148	0.374	0.048	0.000
1000	0.90	0.50	DML-IVQR-BC	0.106	0.030	0.134	0.002	0.000
1000	0.90	0.25	Oracle-GMM	0.728	0.534	0.864	0.398	0.232
1000	0.90	0.25	Full-GMM	0.796	0.666	0.940	0.522	0.142
1000	0.90	0.25	DML-IVQR-BC	0.574	0.576	0.808	0.342	0.174
1000	0.90	0.10	Oracle-GMM	0.906	0.886	0.980	0.812	0.580
1000	0.90	0.10	Full-GMM	0.924	0.906	0.994	0.836	0.360
1000	0.90	0.10	DML-IVQR-BC	0.838	0.914	0.986	0.766	0.506

Note: Boundary rates refer to the endpoints of $A_0 = [-1, 3]$; 'All' is the rate at which all 41 primary-grid points are accepted.

Appendix F Implementation and Reproducibility

Declaration

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

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